

Infineon Wi-Fi CLM regulatory manual

About this document

Scope and purpose

AN225347 introduces you to the Infineon Wi-Fi regulatory compliance process and steps required to submit a regulatory configuration to Infineon.

This application note provides the background on various Wi-Fi requirements and technical details for regulatory certification. At the end of this application note, you will learn how to test your Infineon Wi-Fi system to confirm output power targets, transfer confirmed power targets to the Infineon submission template, and integrate the resulting output into your final product. In addition, this application note provides guidance related to SKU planning for manufacturing your Wi-Fi-enabled product.

Intended audience

This document is intended for anyone planning to get regulatory approval for their designs based on Infineon AIROC™ connectivity devices.

Table of contents

Table of contents

About this document.....	1
Table of contents.....	2
1 Introduction	6
2 More information	7
2.1 Infineon Wi-Fi + Bluetooth® combo solutions	7
2.2 Infineon Wi-Fi solutions	7
2.3 Infineon Bluetooth® and Bluetooth® module solutions	7
2.4 Infineon AIROC™ Bluetooth® community.....	7
2.5 Technical support.....	7
3 Wi-Fi basics.....	8
3.1 Wi-Fi bands, channels, and data rates.....	8
3.2 2.4-GHz channels and mapping	9
3.2.1 40-MHz bandwidth in the 2.4-GHz spectrum	10
3.3 5-GHz channels and mapping.....	11
3.3.1 5-GHz, 20-MHz bandwidth	11
3.3.2 40-MHz and 80-MHz bandwidths	11
3.4 6-GHz channels and mapping.....	12
3.4.1 6-GHz, 20-MHz bandwidth	12
3.4.2 40-MHz and 80-MHz bandwidths	13
3.5 Considerations of regulatory requirements on channel mapping and power	13
4 Regulatory testing methods and requirements.....	14
4.1 Commonly scoped Tier-1 regulatory requirements and differences	14
4.1.1 United States of America (USA) – FCC	14
4.1.1.1 FCC radiated bandedge	14
4.1.1.2 Power spectral density (PSD)	15
4.1.1.3 Outdoor vs. indoor power limits.....	15
4.1.1.4 Radiated harmonics.....	15
4.1.1.5 Antenna type	16
4.1.1.6 Contention-based protocol	16
4.1.1.7 Duty cycle	16
4.1.1.8 Channel 12 and Channel 13 requirements.....	16
4.1.2 Canada – ISED	17
4.1.2.1 Weather band.....	17
4.1.2.2 5150–5250 MHz frequency spectrum power limits.....	17
4.1.2.3 Outdoor vs. indoor power limits.....	17
4.1.2.4 Receiver emissions.....	17
4.1.3 Taiwan – NCC.....	17
4.1.3.1 Receiver emissions.....	17
4.1.3.2 Outdoor vs. indoor operation.....	17
4.1.4 Europe – ETSI	18
4.1.4.1 Power spectral density (PSD)	18
4.1.4.2 Outdoor vs. indoor operation.....	18
4.1.4.3 Adaptivity	18
4.1.4.4 Radiated immunity (RI).....	18
4.1.4.5 Electrostatic discharge (ESD)	18
4.1.4.6 Channel restrictions.....	18
4.1.5 China – SRRC	19

Table of contents

4.1.5.1	Frequency range	19
4.1.5.2	Outdoor vs. indoor operation	19
4.1.5.3	Radiated harmonics	19
4.1.5.4	Radiated bandedge	19
4.1.5.5	Channel restrictions	19
4.1.6	Japan – MIC	19
4.1.6.1	Power limits	19
4.1.6.2	Radiated harmonics	19
4.1.6.3	Radiated bandedge	19
4.1.6.4	Outdoor vs. indoor operation	20
4.1.6.5	Channel 14	20
4.1.6.6	Channel restrictions	20
4.1.7	South Korea – KCC	20
4.1.7.1	Power limits	20
4.1.7.2	PSD vs. temperature/humidity	20
4.1.7.3	Radiated bandedge	20
4.1.7.4	Radiated harmonics	20
4.1.7.5	Channel restrictions	20
4.2	Regulatory information, test methods, and configuration	20
4.2.1	FCC radiated bandedges (restricted) testing	21
4.2.2	FCC radiated bandedges (non-restricted) testing	22
4.2.3	FCC 5.8 GHz emissions mask testing	22
4.2.3.1	6 GHz in-band emissions	22
4.2.4	Contention-based protocol	23
4.2.4.1	Contention-based protocol test setup	24
4.2.5	Dual-client test, demonstration of proper power adjustment based on an associated AP	25
4.2.6	Channel/bandwidth/data rate testing	25
4.2.6.1	Channels testing and channel availability by country (Tier-1)	25
4.2.6.2	Bandwidth testing and availability	28
4.2.6.3	Data rate testing	28
4.2.6.4	WL command example for data rate testing for 2.4 GHz	28
4.2.6.5	WL command example for data rate testing for 11ax 2.4 GHz	29
4.2.6.6	WL command example for 2.4 GHz 11ax RU testing	30
4.2.6.7	WL command example for data rate testing for 5 GHz	31
4.2.6.8	WL command example for data rate testing for 11ax 5 GHz	32
4.2.6.9	WL command example for 5 GHz 11ax RU data rate testing	33
4.2.6.10	WL command example for continuous transmission for data rate testing for 2.4 GHz/5GHz/6GHz 11ax RU data rate testing (ANT0/ANT1/MIMO)	33
4.2.6.11	WL command example for data rate testing for 6 GHz	35
4.2.7	Adaptivity testing	36
4.2.7.1	Adaptivity test setup	36
4.2.7.2	WL command example for adaptivity for 2.4 GHz	36
4.2.7.3	WL command example for adaptivity for 5 GHz	38
4.2.7.4	WL command example for ETSI adaptivity for 6 GHz	38
4.2.7.5	WL command example for FCC Contention Base Protocol 6 GHz	39
4.2.7.6	Test requirement	41
4.2.7.7	Test information required for adaptivity testing	42
4.2.8	Dynamic frequency selection (DFS)	43
4.2.9	Radiated immunity testing	46
4.2.9.1	Test setup	46

Table of contents

4.2.9.2	Test requirements.....	47
4.2.10	Europe – ETSI SDR 5.8 GHz.....	47
4.2.11	Typical cycle time of certification for Tier-1 countries	47
4.3	Power characterization/optimization for Tier-1 countries.....	48
4.3.1	FCC/ISED/Taiwan power characterization.....	49
4.3.2	Setting initial regulatory power targets prior to testing.....	49
4.3.3	Recommended duty cycle by IEEE standard	49
4.3.4	Test items to determine power target compliance CLM input.....	49
4.3.4.1	Efficient channel/band testing and sequence	50
4.3.4.2	FCC WL commands example for power characterization for 2.4 GHz 2-Tx.....	51
4.3.4.3	FCC WL commands example for power characterization for 5 GHz 2-Tx.....	52
4.3.5	Europe – ETSI	53
4.3.6	Setting initial regulatory power targets prior to testing.....	53
4.3.7	Duty cycle	53
4.3.8	Test items to determine power target compliance CLM input.....	53
4.3.8.1	Efficient channel/band testing and sequence	54
4.3.8.2	ETSI WL command example for power characterization for 2.4 GHz 2-Tx	55
4.3.8.3	ETSI WL commands example for power characterization for 5 GHz 2-Tx.....	56
4.3.9	Japan/South Korea	56
4.3.10	Setting initial regulatory power targets prior to testing.....	56
4.3.11	Duty cycle	57
4.3.12	Test items to determine power target compliance CLM input.....	57
4.3.12.1	Efficient channel/band testing and sequence	57
4.3.12.2	Japan/Korea WL command example for power characterization for 2.4-GHz 2-Tx	58
4.3.12.3	Japan/Korea WL commands example for power characterization for 5-GHz 2-Tx	59
4.3.13	China – SRRC	60
4.3.14	Setting initial regulatory power targets prior to testing.....	60
4.3.15	Duty cycle	60
4.3.16	Test items to determine power target compliance CLM input.....	61
4.3.16.1	Efficient channel/band testing and sequence	61
4.3.16.2	China WL command example for power characterization for 2.4-GHz 2-Tx	62
4.3.16.3	China WL commands example for power characterization for 5-GHz 2-Tx	63
4.4	Power characterization and regulatory testing overall flow	64
5	Specific CLM terms and parameters	65
6	SKU planning guidance	67
6.1	Different methods of SKU management	67
6.2	Common problems with SKU planning	68
6.3	Minimum country set for world-wide support	69
6.4	SKU planning summary	69
7	How to work with Infineon CLM templates	70
7.1	Template contents	70
7.1.1	Project information and antenna gain worksheets	70
7.1.1.1	Project information worksheet	70
7.1.1.2	Antenna gain worksheet.....	71
7.1.2	802.11 Wi-Fi power target input worksheets	71
7.1.2.1	Required inputs.....	71
7.1.2.2	802.11 input worksheet descriptions.....	72
7.2	Step-by-step template walkthrough	73
7.2.1	Enabling macros on your template	73

Table of contents

7.2.2	Project information worksheet	74
7.2.3	Input worksheets	77
7.2.3.1	802.11b & g worksheet.....	77
7.2.3.2	802.11a worksheet	79
7.2.3.3	802.11n 2.4 GHz, 20 MHz worksheet.....	79
7.2.3.4	802.11n, ac 5 GHz, 20 MHz worksheet	80
7.2.3.5	802.11n 2.4 GHz, 40 MHz worksheet.....	81
7.2.3.6	802.11n, ac 5 GHz, 40 MHz worksheet	82
7.2.3.7	80 MHz worksheet.....	83
8	How to integrate CLM releases into your design.....	84
8.1	Integrating CLM blobs in AIROC™ platforms	84
8.2	Integrating CLM blobs with FMAC	85
8.3	Interaction of NVRAM and CLM parameters	85
9	Infineon regulatory WL command references	86
9.1	Regulatory WL command sequence used for testing	86
10	Overview of regulatory CLM process	87
	Revision history.....	88
	Disclaimer.....	89

Introduction

1 Introduction

This section provides a bit of background to the Infineon regulatory process for Wi-Fi devices and introduces the reader to common Infineon terms that will be referred to throughout this document.

The regulatory process at Infineon centers around what is called a CLM (Country Locale Matrix). CLM is an Infineon term and refers to a set of interlocking databases that represent a regulatory configuration for a given Country Code (CCode). Infineon implements Per Product CLM databases (for a specific customer project) as well as legacy master CLM databases (containing 10s of thousands of entries of information). The preferred interface method with Infineon is to use Per Product CLMs.

The Infineon CLM is ultimately integrated together with the Infineon Wi-Fi driver to provide the user with an operating mode (channels availability, output power, DFS, etc.) that customers will know is compliant to their target regulatory requirements.

This application note will cover the following critical topics to enhance your knowledge and information so that the Wi-Fi regulatory testing and approval process with Infineon products goes smoothly:

- **Wi-Fi basics:** This section provides the reader with an overview of Wi-Fi technology, specifically focusing on details that are pertinent to regulatory certification and communication with Infineon on regulatory compliance.
- **Recommended testing methods for regulatory compliance:** This section provides details that show you how to test your Infineon Wi-Fi design and confirm output power targets.
- **Specific CLM terms and parameters:** This section provides more terms and details that you will hear while going through the Infineon CLM process.
- **SKU planning:** This section provides guidance and lists typical challenges that are faced when releasing a Wi-Fi product to production.
- **How to work with Infineon CLM templates:** This section provides a detailed overview of the Infineon CLM templates used to communicate your requested regulatory configuration.
- **How to integrate CLM data into your design:** This section steps you through different types of releases that Infineon may make for a CLM Country Code request. It will also step through the details on how to integrate a CLM Blob into your design.
- **Infineon firmware/software regulatory APIs:** Describes the various APIs and features available with respect to regulatory compliance for AIROC™, Full Medium Access Control (FMAC), and Dongle Host Driver (DHD) releases.

At the end of this application note, you should understand the information that Infineon requires to complete a CLM country code request, templates that are required to be complete to start the process, methods, and tips available to test your design for regulatory compliance, and how to integrate the resulting Infineon CLM output into your product.

2 More information

2.1 Infineon Wi-Fi + Bluetooth® combo solutions

Infineon's AIROC™ Wi-Fi + Bluetooth® combos integrate IEEE 802.11a/b/g/n/ac/ax WLAN and Bluetooth® in a single-chip solution to enable small-form-factor IoT designs. To learn more, visit our [Wi-Fi + Bluetooth®](#) website.

2.2 Infineon Wi-Fi solutions

Infineon's AIROC™ Wi-Fi portfolio includes IEEE 802.11a/b/g/n WLAN System-on-Chip (SoC) solutions with 1x1 SISO and 2x2 MIMO up to 867 Mbps PHY data rates. To learn more, visit our [Wi-Fi](#) products website.

2.3 Infineon Bluetooth® and Bluetooth® module solutions

Whether you're in the market for a chip-down Bluetooth® solution or a ready-to-use Bluetooth® module, Infineon has a Bluetooth® portfolio second-to-none. Visit the [Bluetooth® LE and Bluetooth®](#) website for more details on the silicon solutions available from Infineon, or visit the [Bluetooth® Module](#) website to learn more about Infineon module offerings.

2.4 Infineon AIROC™ Bluetooth® community

Whether you're a customer, partner, or a developer interested in the latest Infineon innovations, the [Infineon Bluetooth® community](#) offers you a place to learn, share and engage with both Infineon experts and other embedded engineers around the world.

2.5 Technical support

If you have any questions, our technical support team is happy to assist you. You can create a support request by visiting [Technical Support](#).

Wi-Fi basics

3 Wi-Fi basics

This section introduces basic concepts related to Wi-Fi operation related to regulatory power setting and channel selection. Infineon Wi-Fi solutions support a number of frequency bands and data rates, and it is important to understand some of the basic terminology and interdependencies that exist when planning your regulatory strategy and configuration.

3.1 Wi-Fi bands, channels, and data rates

[Table 1](#) summarizes commonly known Wi-Fi terms and the linkage to technical terms that are used when configuring regulatory power limits, channels, and data rates.

Table 1 Common Wi-Fi terms and operating principals

Wi-Fi standard	Band	Data rate	Streams	Bandwidth		Modulation	Supported channels
	Frequency band	Maximum data rate	2 TX capable (MIMO)?	40-MHz capable?	80-MHz capable?		
802.11a	5.0 GHz	54 Mbps	No	No	No	OFDM	36-64, 100-144, 149-165
802.11b	2.4 GHz	11 Mbps	Yes	No	No	DSSS	1-13
802.11g	2.4 GHz	54 Mbps	No	No	No	OFDM	1-13
802.11n	2.4/5.0 GHz	600 Mbps	Yes	Yes	No	MCS, OFDM, HT	2.4G: 1-13 5G: 36-64, 100-144, 149-165
802.11ac	5.0 GHz	866 Mbps (2x2 MCS9 80 MHz)	Yes	Yes	Yes	MCS, OFDM, VHT	36-64, 100-144, 149-165
802.11ax	5.0 GHz	1200 Mbps (2x2 MCS11 80 MHz)	Yes	Yes	Yes	HE, MCS, OFDM, HE	36-64, 100-144, 149-165
Wi-Fi 6E	6.0 GHz	1200 Mbps (2x2 MCS11 80 MHz)	Yes	Yes	Yes	HE, MCS, OFDM, HE	1-233

Wi-Fi basics

Table 1 is important if you are looking to maximize your Wi-Fi performance while still maintaining regulatory compliance within targeted countries. Infineon solutions allow for each Wi-Fi standard shown in Table 1 to have power targets specified for each channel within a given Frequency Band, Bandwidth, and Modulation combination. The example below shows the flexibility available for a relatively simple Wi-Fi configuration:

- Frequency band: 802.11b
- Modulation: DSSS
- Channels supported: 20 MHz

With this Wi-Fi example configuration, Infineon Wi-Fi solutions can configure power targets for channels 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, and 13. A simple example like this makes the process seem like a breeze; however, taking the example above a step further and considering 802.11n, this allows for many more configuration options for each frequency band, bandwidth, channel, and Tx chains; creating an extremely large array of data and regulatory targets that can be difficult to conceptualize. The Infineon CLM process is meant to streamline and simplify the challenges associated with Wi-Fi regulatory compliance.

The coming sections provide more detail on the 2.4-GHz and 5-GHz spectrum for Wi-Fi regulatory compliance.

3.2 2.4-GHz channels and mapping

The 2.4-GHz spectrum includes channels 1–13 (Channel 14 is not discussed in this manual because it is not supported by Infineon). The 2.4-GHz band extends from 2.402 GHz to 2.483 GHz, with each channel occupying 22 MHz (for 802.11b) or 20 MHz (for 802.11g/n) of bandwidth. Each channel is separated by 5 MHz from neighboring channels.

The 2.4-GHz spectrum can be operated in both the 20-MHz and 40-MHz bandwidths for Infineon products. A graphical representation of the 2.4-GHz, 20-MHz channel mapping is shown in Figure 1.

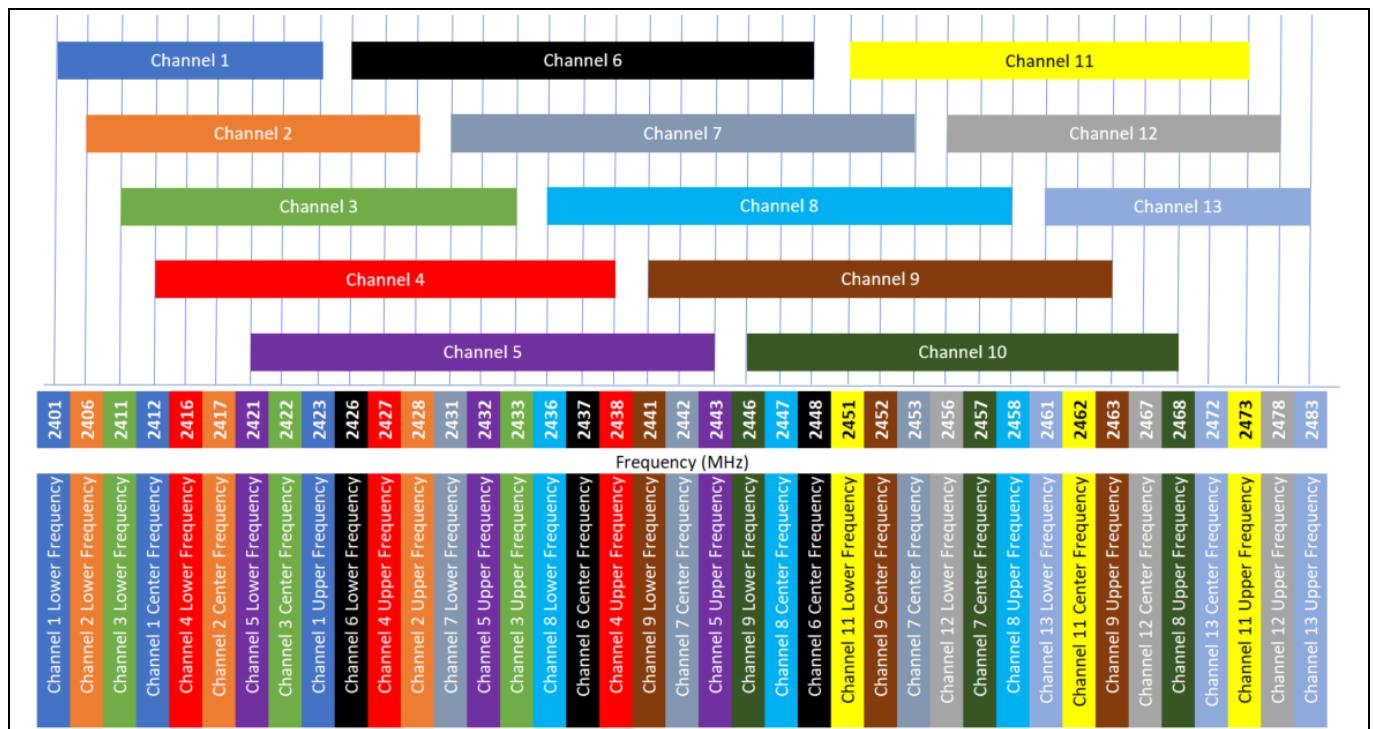


Figure 1 2.4-GHz channel mapping

Wi-Fi basics

Note: The special case about channels 1, 6, and 11 (or 2, 7, and 12 in countries that allow Channel 12/13): By reviewing the channel mapping representation in [Figure 1](#), you can see that these channel sets (1/6/11 and 2/7/12) do not overlap, allowing for less interference, thereby maximizing the performance.

Regardless of the data rate supported by 802.11b/g/n devices (for example 802.11b can support 1 Mbps, 2 Mbps, 5.5 Mbps, or 11 Mbps), the channel width and frequency range remain the same for a given bandwidth. The increase in data throughput is capable by using different modulation/encoding schemes for the same Wi-Fi standard (for example, 802.11b 1 Mbps and 2 Mbps utilize a Barker code chipping method, whereas 5.5 Mbps and 11 Mbps utilize CCK chipping method).

3.2.1 40-MHz bandwidth in the 2.4-GHz spectrum

Many of the Infineon Wi-Fi solutions support 40-MHz operation and can be configured to operate 2.4-GHz channels with 40-MHz bandwidth for 802.11n. However, it is recommended that these channels not be used due to inefficiencies and lack of non-overlapping channel sets. The channel mapping shown in [Figure 1](#) changes considerably when moving to 40-MHz operation in the 2.4-GHz spectrum.

To create a 40-MHz channel bandwidth, Wi-Fi uses a process called “channel bonding”, where two 20-MHz channels are bonded together to create a single 40-MHz channel. The channels bonded together are not neighboring – for example, Channel 3 in 40 MHz is a combination of Channel 1 and Channel 5 in 20 MHz.

The implementation of 2.4-GHz, 40-MHz channels affects channel availability for 40-MHz bandwidths:

- The upper and lower supported channels of the spectrum cannot be used. For North America, this means that channels 1, 2, 10, and 11 are not available (as two 20-MHz bandwidths are required for 40-MHz operation), leaving only channels 3-9 available. For countries that allow the use of channels 12 and 13 in 20-MHz bandwidth, this effectively allows for channels 3-11 to be used at the most.
- The loss of channels, and increased bandwidth of each channel lead to a decrease in the number of non-overlapping channels available. In North America, this translates to zero combinations that are non-overlapping (in other words, only a fixed-channel system can operate free of interference assuming that you restrict the use of that channel in your setup). For countries that allow channels 12 and 13 in the 20-MHz bandwidth, this provides a single set of non-overlapping channels (Channel 3 and 11).

Although a 40-MHz bandwidth does increase the throughput and speed of data transfer, the deficiencies that exist in crowded 2.4-GHz areas and the lack of non-overlapping channels generally lead to a lower performance than simply using 20-MHz channels.

It is important to note that if your solution utilizes the higher-bandwidth channels (40 MHz), the lower-bandwidth channels (20-MHz), specifically the channels that are required to be bonded together to support higher bandwidth, are required to be enabled with power targets.

Wi-Fi basics

3.3 5-GHz channels and mapping

Note: 5-GHz Wi-Fi solutions are not necessarily less immune to interference because there are fewer devices operating at 5 GHz. The 5-GHz spectrum also improves on the 2.4-GHz implementation by removing the channel overlap that exists in the 2.4-GHz spectrum.

5-GHz channels operate in one of the following bandwidths: 20 MHz (802.11a/n), 40 MHz (802.11n/ac, also known as HT - High Throughput), or 80 MHz (802.11ac, also known as VHT - Very High Throughput). The number of channels available in the 5-GHz, 20-MHz bandwidth is 25 (dependent on specific country regulations). Each channel in the 20-MHz bandwidth occupies 20 MHz and the center frequency of each channel is spaced at least 20 MHz from the neighboring channels, leading to no overlapping channels.

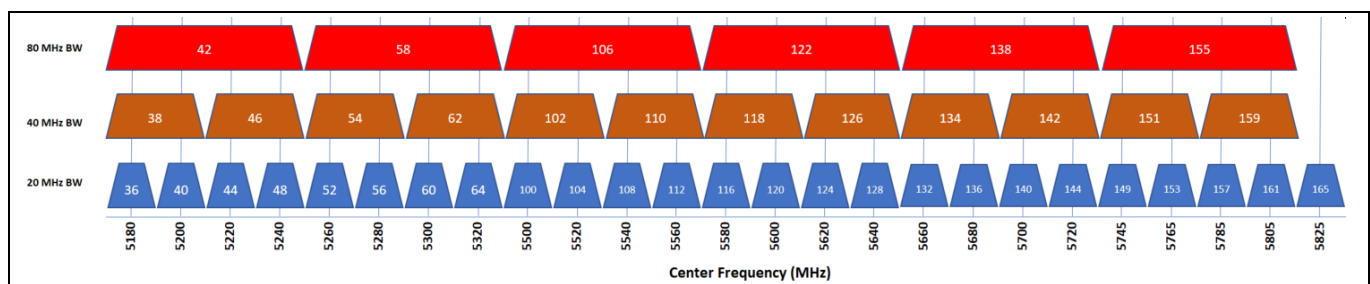


Figure 2 5-GHz channel mapping for 20-MHz, 40-MHz, and 80-MHz bands

3.3.1 5-GHz, 20-MHz bandwidth

The 20-MHz bandwidth for 5-GHz channels consist of 25 non-overlapping channels:

- 36, 40, 44, and 48 (U-NII-1)
- 52, 56, 60, and 64 (U-NII-2A)
- 100, 104, 108, 112, 116, 120, 124, 128, 132, 136, 140, and 144 (U-NII-2C)
- 149, 153, 157, 161, and 165 (U-NII-3)

Each channel is 20-MHz wide and is separated 20 MHz from the neighboring channels (center-to-center).

3.3.2 40-MHz and 80-MHz bandwidths

Many of the Infineon solutions support 40-MHz and 80-MHz operation in the 5-GHz spectrum. Just as in the 2.4-GHz spectrum, wider bandwidths are accomplished by bonding two smaller channels (for example, two 20-MHz channels combined to create a 40-MHz channel). However, unlike the 2.4-GHz spectrum, the bonded channels are neighboring channels in the 5-GHz spectrum. The channels bonded can be seen in [Figure 2](#). For example, Channel 38 in 40 MHz is the result of bonding channels 36 and 40 in 20 MHz. Likewise, Channel 42 in 80 MHz is the result of bonding channels 36, 40, 44, and 48 in 20 MHz (or channels 38 and 46 in 40 MHz).

It is important to note that if your solution utilizes the higher-bandwidth channels (40 MHz/80 MHz), the lower bandwidth channels (20 MHz and/or 40 MHz), specifically the channels that are required to be bonded together to support higher bandwidth, are required to be enabled with power targets.

Wi-Fi basics

3.4 6-GHz channels and mapping

6-GHz channels operate in one of the following bandwidths: 20 MHz, 40 MHz, or 80 MHz. The number of available channels in the 6-GHz, 20-MHz bandwidth is 59 (dependent on specific country regulations). Each channel in the 20-MHz bandwidth occupies 20-MHz and the center frequency of each channel is spaced at least 20-MHz from neighboring channels.

The 6-GHz spectrum channel mapping for the 20-MHz, 40-MHz, and 80-MHz bands is as follows:

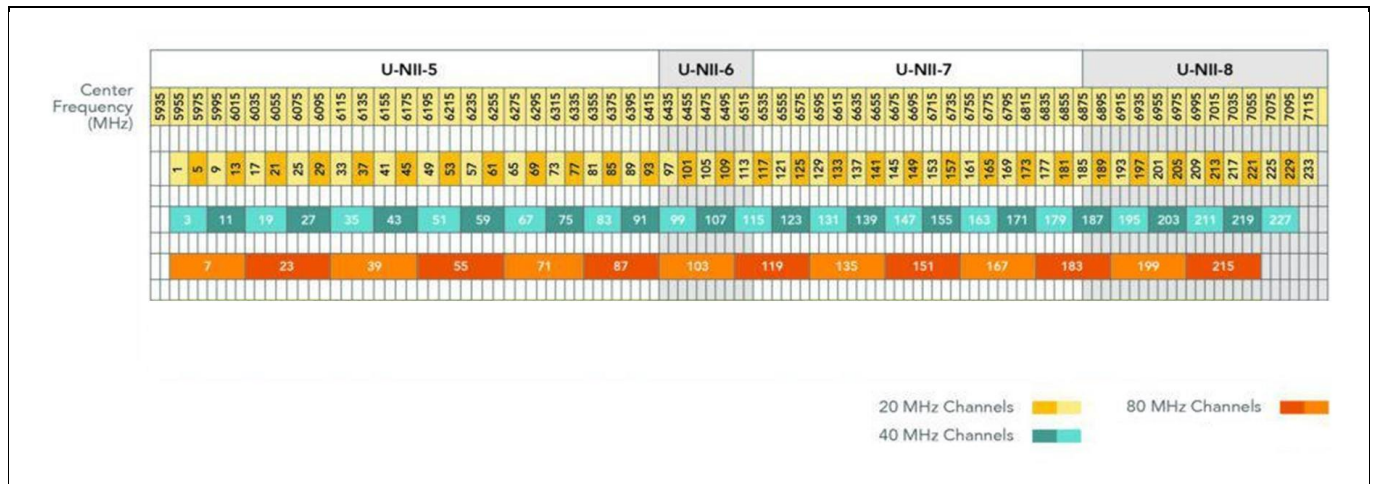


Figure 3 6-GHz channel mapping (source: <https://www.litepoint.com/blog/wi-fi-6e-standard-and-channels/>)

3.4.1 6-GHz, 20-MHz bandwidth

The 20-MHz bandwidth for 6-GHz channels consists of 59 non-overlapping channels:

- 1, 5, 9, 13, 17, 21, 25, 29, 33, 37, 41, 45, 49, 53, 57, 61, 65, 69, 73, 77, 81, 85, 89, 93 (U-NII-5)
- 97, 101, 105, 109, 113 (U-NII-6)
- 117, 121, 125, 129, 133, 137, 141, 145, 149, 153, 157, 161, 165, 169, 173, 177, 181 (U-NII-7)
- 185, 189, 193, 197, 201, 205, 209, 213, 217, 221, 225, 229, 233 (U-NII-8)

Each channel is 20-MHz wide and is separated 20 MHz from the neighboring channels (center-to-center).

3.4.2 40-MHz and 80-MHz bandwidths

Many of the Infineon solutions support 40-MHz and 80-MHz operation in the 6-GHz spectrum. Just as in the 2.4-GHz spectrum, wider bandwidths are accomplished by bonding two smaller channels (for example, two 20-MHz channels combined to create a 40-MHz channel). However, unlike the 2.4-GHz spectrum, the bonded channels are neighboring channels in the 6-GHz spectrum (see [Figure 3](#)). For example, Channel 3 in 40 MHz is the result of bonding channels 1 and 5 in 20 MHz. Likewise, Channel 7 in 80 MHz is the result of bonding channels 1, 5, 9, and 13 in 20 MHz (or channels 3 and 11 in 40 MHz).

It is important to note that if your solution utilizes the higher-bandwidth channels (40 MHz/80 MHz), the lower bandwidth channels (20 MHz and/or 40 MHz), specifically the channels that are required to be bonded together to support higher bandwidth, are required to be enabled with power targets.

3.5 Considerations of regulatory requirements on channel mapping and power

Each country (or Region representing a group of countries) specifies the allowable channels, maximum output power, and measurement method (conducted or radiated) required to certify a wireless product for importation and sale in a specific country.

For the 2.4-GHz spectrum, the maximum output power and measurement method is generally consistent for all 2.4-GHz channels allowed for each respective country. Some countries, such as the US, have specific restrictions and requirements related to band-edge and harmonics that must be taken to account for certification. However, with the exception of a handful of countries, the channel availability in the 2.4-GHz spectrum is fairly consistent.

For the 5-GHz spectrum, each country will specify the same information (maximum power, measurement method, and allowable channels). In the case of 5-GHz, both the maximum power and measurement method for a channel or group of channels may differ country-to-country, so it is important to understand the requirements that you are certifying against.

In either case, Infineon Wi-Fi solutions offer the ability to customize output power, measurement method, and channel availability for a specific country and frequency band, allowing you to optimize Wi-Fi performance while still remaining compliant to regulatory requirements.

Regulatory testing methods and requirements

4 Regulatory testing methods and requirements

This section provides the following information:

- Description and high-level differences between the most commonly scoped Tier-1 countries (US, Canada, China, Japan, and South Korea)
- Testing methods and process for optimizing the output power for Infineon Wi-Fi solutions
- Generally accepted testing methods and minimal channel/band/data rate test combinations required to complete regulatory certifications

4.1 Commonly scoped Tier-1 regulatory requirements and differences

This section provides information on the high-level requirements for the most commonly scoped countries. The regulatory agencies that oversee the requirements for the countries in question are as follows:

1. United States of America – Federal Communications Commission (FCC)
2. Canada – Innovation, Science and Economic Development (ISED)
3. Taiwan – National Communications Commission (NCC)
4. Europe – European Telecommunications Standards Institute (ETSI)
5. Japan – Ministry of Internal Affairs and Communications (MIC)

The following subsections outline the most common test items and standards where customers typically encounter certification delays because of lacking the understanding as to the importance of these tests during regulatory approval.

4.1.1 United States of America (USA) – FCC

The most common challenges experienced when certifying against FCC regulations are described in the following sub-sections.

4.1.1.1 FCC radiated bandedge

[Section 15.205](#) requires that any emission that falls in the restricted bands frequency list must meet the limits of [Section 15.209](#). Section 15.209 states that the device must comply to an average limit of 54 dBuV/m @ 3 meters distance. The peak limit of 74 dBuV/m @ 3 meters is also mandatory for emission compliance per [15.33\(b\)](#).

The high-level definition of bandedge requirements is that when you operate on a restricted band frequency/channel, the requirement is that the emissions level on these channels must be below the peak and average limits as stated above. The reason for this requirement is that the restricted band frequencies/channels are license-protected in the US, and unlicensed operation within this frequency must not interfere with the licensed usage of these channels.

The list of the restricted band channels can be found in [Section 15.205](#), and are summarized as follows:

2.4 GHz: Channels 1, 2, 10, 11, 12, and 13 are considered restricted band channels.

1. This applies to both 20-MHz and 40-MHz operating bands
2. 802.11b power outputs are much higher when compared to 802.11g on restricted channel list because of the narrow width of each channel (802.11b does not support 40-MHz operation)

5 GHz: 20-MHz restricted band channels include: Channel 36, 64, and 100.

Regulatory testing methods and requirements

Note: The 40-MHz and 80-MHz parent channels (as shown in [Figure 2](#)) are also considered restricted band channels (for example, Channel 38 in 40-MHz operation would be included because Channel 36 in 20-MHz operation is in a restricted band frequency).

The inner channels for example, 2.4-GHz Channel 6 are not impacted by bandedges so the power is set to maximum that the radio chip supports. The same is true for the inner channels in the 5-GHz band, which will be transmitting at the maximum power but could potentially be limited by PSD.

4.1.1.2 Power spectral density (PSD)

This test is mandatory to ensure that the energy created via Wi-Fi operation is being spread out evenly in the frequency spectrum. PSD is measured as conducted on a single spectrum point of the fundamental with specific spectrum analyzer settings (standard settings for spectrum analyzers).

Note: Section [15.247](#) (2.4-GHz operation) of the FCC requirements states that PSD is measured in a conducted method and does not require the antenna gain to be added; therefore, the PSD limit is not an Equivalent Isotropically Radiated Power (EIRP) limit. Conversely, Section [15.407](#) (5 GHz) states that PSD is measured in a conducted method, but the antenna gain must be added to the measured values; therefore, the PSD limit is an EIRP limit. If you are operating a dual-band Wi-Fi solution, you must complete testing and certification for both Section [15.247](#) and [15.407](#) for PSD.

In general, if you are using a narrow modulation bandwidth (for example, Direct Sequence Spread Spectrum (DSSS), this will result in a higher measured level of PSD for the operating channel. Conversely, if you have a wider modulation bandwidth (for example, Orthogonal Frequency Division Multiplexing (OFDM), this will result in a lower measured level of PSD for the operating channel.

The inner channels, for example, the 2.4-GHz Channel 6, are not impacted by bandedges, so the power is set to the maximum that the radio chip supports, but could be limited by PSD. The same is true for the inner channels in the 5-GHz band, which will be transmitting at the maximum power but could potentially be limited by PSD.

4.1.1.3 Outdoor vs. indoor power limits

FCC in the past has prohibited Wi-Fi devices for indoor operation only, but recently FCC now allows outdoor use for both 2.4-GHz and all 5-GHz bands.

4.1.1.4 Radiated harmonics

This testing is intended to measure unwanted emissions created by the operating channel. There are two key regulatory terms used to describe the limits and emission type during Radiated Harmonics testing:

1. “Non-restricted” emissions/limits
2. “Restricted” emissions/limits

The limit for non-restricted unwanted emissions is higher compared to the limit for the restricted unwanted emissions. See restricted unwanted emissions limits in FCC Section [15.209](#) and for non-restricted unwanted emissions limits, see [15.247\(2\)\(d\)](#) (2.4G) and [15.407\(5\)\(b\)\(1-4\)](#) (5G).

Unwanted emission for the 6-GHz band as stated in **987594 EMC Measurement(G)** will be measured with an RMS detector in restricted bands for measurements below 1 GHz and above 1 GHz.

Regulatory testing methods and requirements

Outdoor standard power access points and fixed client devices operating in U-NII-5 and U-NII-7 must not exceed the maximum EIRP allowed at elevation angles higher than 30 degrees from horizon. See [987594 EMC Measurement\(H\)\(1\)](#).

4.1.1.5 Antenna type

When selling/using different types of antennas with a radio, the following information must be known:

1. The FCC requires that each type of antenna is tested. Note that this is antenna types (for example, Dipole, Yagi, Dish, etc.), not antenna gains.
2. Only the highest gain antenna within an antenna type is required to be tested per the FCC. However, if you have multiple antenna gain options in your product, the FCC recommends that the lowest and highest antenna gain, and corresponding radio output power is also tested to ensure that the system complies to Radiated Bandedges and Harmonics. If you have multiple antenna gains or types, Infineon recommends that the worst-case antenna (in terms of Harmonics) be tested to reduce cost and test time.
3. Provide how the antenna beams are controlled to be kept below 30 degrees elevation angle. The explanation should include device installation instructions, mechanical control, electro-mechanical control or software algorithms, if the beams are electrically controlled by software as stated in [987594 EMC Measurement\(H\)\(2\)](#).

4.1.1.6 Contention-based protocol

Indoor access points, subordinate devices, and client devices operating in U-NII-5 through U-NII-8 bands (referred to as “unlicensed devices”) are required to use technologies that include a contention-based protocol to avoid co-channel interference with incumbent devices sharing the band. Unlicensed devices are required to detect co-channel radio frequency energy (energy detect) and avoid simultaneous transmission.

To ensure that incumbent operations are reliably detected in the band, low-power indoor devices must detect the RF energy throughout in their intended operating channel. See [987594 EMC Measurement\(I\)](#).

4.1.1.7 Duty cycle

During regulatory testing, the FCC requires that the radio be set to a specific duty cycle value which must be greater than 98%. If a 98% duty cycle is not possible because of radio chip limitations, the FCC allows the radio to be set to the maximum duty cycle that the radio can support and follow alternate test methods outlined in the DTS or UNII FCC test procedures. Note that not all bandwidth/channels/data rates supported by Infineon Wi-Fi solutions can support 98% duty cycle, so this information is valuable in the event that a lab indicates that they cannot test due to a duty cycle limitation (which is incorrect).

4.1.1.8 Channel 12 and Channel 13 requirements

FCC allows operation of channels 12 and 13, but there are two requirements that the radio must comply to use these channels:

1. Channels 12 and 13 must meet the Radiated Bandedge requirements. The drawback on this is that the power output will be much lower compared to channels 10 and 11. The lower power results could break the radio where it is recommended to disable channels 12 and 13. In other words, the performance of the radio is greatly affected, unless the lower powers have been verified to meet the EVM test.
2. The device will need a geo-location mechanism implemented to ensure that the radio will comply when operating in the United States. The geo-location mechanism will ensure that the correct regional code (country code) is being used by the device when operating in the United States. If a geo-location mechanism cannot be implemented, then channels 12 and 13 must be disabled.

Regulatory testing methods and requirements

A Permit but Ask (PBA) must be submitted to the FCC before the Telecommunication Certification Body (TCB) can approve the radio device. If such PBA is approved by the FCC, the TCB will grant the device so as to receive the FCC certificate. PBA can take from weeks to a month for approval, so include this into the product approval deadlines.

4.1.2 Canada – ISED

Most test requirements for ISED (RSS-247 is the ISED standard for Wi-Fi 2.4/5 GHz radios) are similar to FCC regulatory requirements with the following exceptions:

4.1.2.1 Weather band

The Weather band (TDWR) must be disabled (channels 120-128) for access points (APs)/routers and for STA/client devices. It should not be possible for the TDWR channels to be active when in Canada.

If a specific country code for Canada is developed for your product, or if Canada is a country grouped in a region you have defined, the TDWR channels should be disabled.

4.1.2.2 5150–5250 MHz frequency spectrum power limits

RSS-247 Section 6.2.1.1 power limits for both access points/routers and STA/client devices operating in the 5150-5250-MHz frequency range (5-GHz channels 36 – 48) is 200 mW (23 dBm) EIRP. FCC Section 15.407 (1)(iv) allows a power limit of 250 mW (24 dBm) EIRP. ISED does not yet allow high power limits of 1-watt (30 dBm) for access points/routers as the FCC does.

4.1.2.3 Outdoor vs. indoor power limits

Currently, ISED does not allow outdoor operation in the 5150-5250-MHz frequency range (5-GHz channels 36 – 48) except for automotive applications. All other frequencies can be used in outdoor operation. For 5250-5350-MHz frequencies to be used outdoors, additional test requirements need to be performed. See [RSS-247 Section 6.2.2.3](#) for details on the additional test requirements. 2.4-GHz frequencies, 5740-5725-MHz, and 5725-5850 MHz frequencies can operate in both indoor and outdoor scenarios.

4.1.2.4 Receiver emissions

ISED requires that all receivers are to be tested for radiated emissions. Whereas the FCC does not require receiver emissions testing if the receiver operates above 960 MHz.

4.1.3 Taiwan – NCC

Taiwan is very similar to FCC with the only exception noted as follows.

4.1.3.1 Receiver emissions

NCC requires that all receivers are to be tested for radiated emissions, whereas the FCC does not require receiver emissions test if the receiver operates above 960 MHz.

4.1.3.2 Outdoor vs. indoor operation

In the past, NCC has restricted Wi-Fi devices to indoor operation only, but recently NCC now allows outdoor use for all 2.4-GHz and 5-GHz bands.

Regulatory testing methods and requirements

4.1.4 Europe – ETSI

Europe does not follow FCC regulatory requirements. The regulatory test requirements are quite different between the FCC and ETSI. The most common challenges experienced when certifying against ETSI regulations are described below.

4.1.4.1 Power spectral density (PSD)

This test is mandatory to ensure that the energy created via Wi-Fi operation is being spread out evenly in the frequency spectrum. PSD is measured in a conducted manner on a single spectrum point using specific spectrum analyzer settings (standard settings for spectrum analyzers). The PSD limit for Europe is 10 dBm/MHz EIRP.

Note: Due to changes made to the ETSI 300 328 V.2.1.1, the PSD test measurement method has limited the output power of 11b (2.4-GHz DSSS) to be 18 dBm when using an antenna gain of 1 dBi or less. Output power for 11g (2.4-GHz OFDM) is limited to a power output of 19 dBm based on a 1 dBi antenna.

4.1.4.2 Outdoor vs. indoor operation

ETSI only allows indoor use for operating bands 5150-5250 and 5250-5350 MHz. Operating bands 5470-5725 and 5725-5850 MHz are allowed for both outdoor and indoor use. 2.4-GHz frequencies can operate in both indoor and outdoor scenarios.

4.1.4.3 Adaptivity

This test is to ensure that the spectrum, utilized by a Wi-Fi device, is being shared equally between users (Wi-Fi products). When the radio device detects an interference energy level above a preset trigger point (-70 dBm/MHz for 2.4 GHz; -75 dBm/MHz for 5 GHz), the radio must stop transmitting. The radio must not transmit on the specific channel until the interference energy level has disappeared.

4.1.4.4 Radiated immunity (RI)

This test ensures that the radio device can perform adequately when operating in a noisy environment. A radiated field level of 3 V/m is injected at a 3-meter distance from the device under test (DUT). This frequency is swept from 80 to 6000 MHz with a 1% step and the signal is modulated with an 80% AM (Amplitude Modulation).

4.1.4.5 Electrostatic discharge (ESD)

This test consists of exposing directly or indirectly an electrostatic discharge at a specific voltage level on the radio device. This test verifies that the radio device does not shutdown, reboot, lose connection, or suffer damage.

4.1.4.6 Channel restrictions

All member countries of ETSI regulations have the same channel requirements, with one exception in the United Kingdom. All ETSI member nations do not allow operation on Channel 144 of the 5-GHz spectrum, with the lone exception of the UK, which does allow operation on Channel 144 in the 5-GHz spectrum.

Regulatory testing methods and requirements

4.1.5 China – SRRC

China follows the European ETSI standards with the following exceptions.

4.1.5.1 Frequency range

Frequency range testing is to ensure that the operating frequency energy is entirely inside the operating band of 2400 to 2483.5-MHz for 2.4-GHz devices. This test is performed with conducted measurements and is only tested on Channel 1 and Channel 13.

The same test is performed for 5-GHz channels 149 and 165 with the same limit of -30 dBm/MHz.

Note: SRRC follows older EN300 328 V.1.6.1 and EN301 893 V.1.3.1 ETSI measurement standards.

4.1.5.2 Outdoor vs. indoor operation

SRRC only allows indoor use for operating bands 5150-5250 and 5250-5350 MHz. Operating bands in the 5725-5850 MHz frequency range are allowed for both outdoor and indoor use. 2.4-GHz channels can operate in both indoor and outdoor scenarios.

4.1.5.3 Radiated harmonics

Not required.

4.1.5.4 Radiated bandedge

Not required.

4.1.5.5 Channel restrictions

SRRC does not allow operation in channels 100 – 144 in the 5-GHz spectrum.

4.1.6 Japan – MIC

4.1.6.1 Power limits

There are no power limits stated for Japan; only PSD limits. PSD is performed in a conducted test method. In this test, labs will record the average output power of the radio with an average power meter and confirm the results are below the required limits.

4.1.6.2 Radiated harmonics

Not required.

4.1.6.3 Radiated bandedge

Not required.

Regulatory testing methods and requirements

4.1.6.4 Outdoor vs. indoor operation

MIC only allows indoor use for operating bands 5150-5250 and 5250-5350 MHz. Channels operating in the 5470-5725 MHz frequency range are allowed for outdoor use. 2.4-GHz channels can operate in both indoor and outdoor scenarios.

4.1.6.5 Channel 14

Japan is the only country that allows a Channel 14 in the 2.4-GHz spectrum.

Infineon does not support Channel 14 on our Wi-Fi devices.

4.1.6.6 Channel restrictions

MIC does not allow operation on channels 149 – 165 in the 5-GHz spectrum.

4.1.7 South Korea – KCC

4.1.7.1 Power limits

There are no power limits stated for KCC; only PSD limits. PSD is performed in a conducted test method. In this test, labs will record the average output power of the radio with an average power meter and confirm the results are below the required limits.

4.1.7.2 PSD vs. temperature/humidity

PSD must remain stable within a specified tolerance limit when the radio device is exposed to extreme temperature and humidity levels. PSD stability is performed in a conducted test method.

4.1.7.3 Radiated bandedge

Not required.

4.1.7.4 Radiated harmonics

Not required.

4.1.7.5 Channel restrictions

No restrictions all channels open similar to FCC.

4.2 Regulatory information, test methods, and configuration

For each Tier-1 country, allowable operating channels as well as the regulatory test requirements may differ. This section provides information about each test requirements that are frequently asked to Infineon:

1. FCC bandedge (Restricted)
2. FCC bandedge (Non-restricted)
3. FCC 5.8G emissions mask
4. Channels/bandwidth/data rate test modes
5. ETSI Adaptivity and FCC Contention Base Protocol
6. DFS (Dynamic Frequency Selection)

Regulatory testing methods and requirements

7. Radiated immunity
8. Europe - ETSI SDR 5.8G
9. Tier-1 test time

4.2.1 FCC radiated bandedges (restricted) testing

Radiated bandedges are required per the FCC Section 15.205 requirements as stated earlier. Measurements are performed in an anechoic chamber with a pick-up antenna (typically a horn antenna) placed three meters away from the device under test. The EMC lab should test radiated bandedges without the external amplifier to avoid overload. If the noise floor of the spectrum analyzer is too high, then it is acceptable to use an external amplifier with a 10 dB or 20 dB pad at the input connector of the external amplifier.

The EMC lab will measure the following channels for 2.4-GHz compliance:

For 802.11b, 802.11g, and 802.11n (20-MHz bandwidth): Channels **1, 2, 10, 11, 12, 13**

For 802.11n (40-MHz bandwidth): Channels **3, 4, 8, 9, 10, 11**

Note: In 802.11b mode, due to the narrow bandwidth, the energy present in the restricted bandedge can be very small; therefore, the power output will not need to be reduced. This is true for only channels 1-11. For channels 12 and 13, the power is expected to be lower.

Note: Discuss with the EMC lab to see whether they will agree to use 802.11n (20 MHz) to cover 802.11g operation. 802.11n (20 MHz) has a much wider bandwidth compared to 802.11g, which results in a worst-case mode for radiated bandedge testing.

Note: From experience, the power output from 802.11b should be higher compared to 802.11n (20 MHz). Additionally, 802.11n (20 MHz) should be higher compared to 802.11n (40 MHz) after completing radiated bandedge testing.

The EMC lab measures the following channels for 5-GHz operation:

For 802.11a and 802.11n (20 MHz): Channels **36, 64, 100, 140**

For 802.11n (40 MHz): Channels **38, 62, 102, 134**

For VHT80 (80 MHz): Channels **42, 58, 106**

The EMC lab measures the following channels for 6-GHz operation:

For 802.11, 802.11n, 802.11ac, 802.11ax (20 MHz): Channels **1 and 233**

For 802.11n, 802.11ac, 802.11ax (40 MHz): Channel **3 and 227**

For VHT80, 11ax (80 MHz): Channels **7 and 215**

Note: Discuss with the EMC lab to see whether they will agree to use 802.11n (20 MHz) to cover 802.11a operation. 802.11n (20 MHz) has a much wider bandwidth when compared to 802.11a, resulting in a worst-case mode for radiated bandedge testing. Also, 802.11n (20 MHz) or 802.11n (40 MHz) will be similar to VHT20 and VHT40 and can be used to cover VHT20 and VHT40 for all FCC test requirements.

Regulatory testing methods and requirements

Note: The use of a 100kHz Resolution BW can be used for channel 233 Bandedge compliance, but a 1MHz emission power summation shall be used to capture the emission level in 7125 to 7150MHz spectrum and compare to the -27dBm/MHz average limit.

Note: *From experience, the power output from 802.11n (20 MHz) should be higher when compared to 802.11n (40 MHz). Additionally, 802.11n (40 MHz) should be higher when compared to VHT80 after completing the radiated bandedge testing.*

4.2.2 FCC radiated bandedges (non-restricted) testing

Radiated bandedge non-restricted testing is also required. Measurements are performed in an anechoic chamber with a pick-up antenna (typically a horn antenna) placed three meters from the device under test. The EMC lab should test radiated bandedges without the external amplifier to avoid overload. If the noise floor of the spectrum analyzer is too high, it is acceptable to use an external amplifier with a 10- or 20-dB pad at the input connector of the external amplifier.

The EMC lab will measure the following channels for the 5 GHz band:

802.11a and 802.11n (20 MHz): Channel **140**

802.11n (40 MHz): Channel **134**

VHT80 (80 MHz): Channel **122**

The limit for the non-restricted bandedge for the 5 GHz band is -27 dBm/MHz and is a conducted limit. To convert the conducted limit to a field strength, use the following formula:

$$E(\text{dBuV/m}) = -27 \text{ dBm} + 95.2 \text{ dB}$$

$$E(\text{dBuV/m}) = 68.2$$

The test frequency that the EMC lab will test for compliance will be 5725 MHz. Any energy falling outside of 5725 MHz must be below 68.2 dBuV/m at 3 meters distance.

4.2.3 FCC 5.8 GHz emissions mask testing

FCC section 15.407(4)(i): All emissions shall be limited to a level of -27 dBm/MHz at 75 MHz or more above or below the bandedge increasing linearly to 10 dBm/MHz at 25 MHz above or below the band edge, and from 25 MHz above or below the band edge increasing linearly to a level of 15.6 dBm/MHz at 5 MHz above or below the band edge, and from 5 MHz above or below the band edge increasing linearly to a level of 27 dBm/MHz at the band edge.

This test can be performed using radiated (EIRP) or conducted measurement methods. Infineon suggests performing this test using radiated measurements as this test is performed with the actual antenna attached to the radio device.

4.2.3.1 6 GHz in-band emissions

987594 EMC Measurement(J): test procedure of 12.4.1 of ANSI C63.10-2013 will be followed to measure the 26-dB EBW. The channel bandwidth is defined as the 26 dB EBW for developing the emission mask. The emission power spectral density must be reduced below the peak power spectral density.

Regulatory testing methods and requirements

4.2.4 Contention-based protocol

Incumbent signal is assumed to be noise-like. A 10-MHz-wide additive white Gaussian noise (AWGN) signal is selected to simulate and represent the incumbent transmission. Incumbent and EUT signals may occupy different portions of the channel. Depending on the EUT transmission bandwidth and incumbent signal center frequency, the center frequency of the EUT signal may fall within the incumbent's occupied bandwidth, or outside of the incumbent's occupied bandwidth.

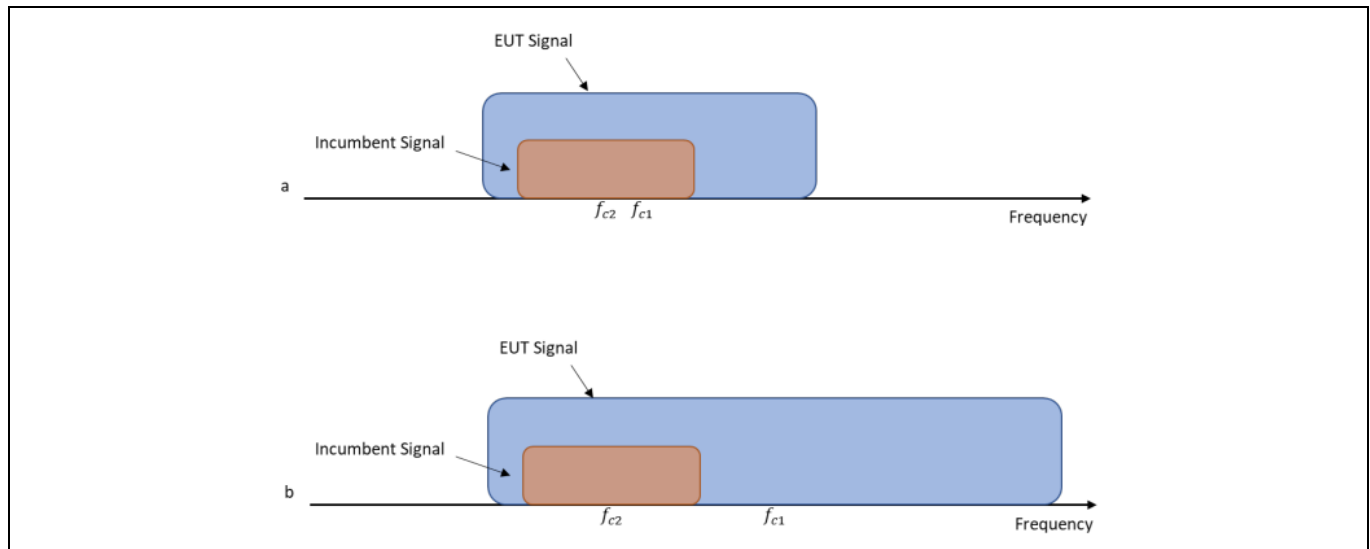


Figure 4 Scenarios where the center frequency of EUT transmission falls within incumbent's bandwidth(a), or outside of it(b)

The detection threshold test may be repeated multiple times with the incumbent signal tuned to different center frequencies within the EUT transmission bandwidth to verify that the EUT reliably detects the incumbent signal in the scenarios shown in [Figure 4](#). The following table determines how many times the detection threshold test must be performed.

Table 2 Threshold criteria detection

If	Number of tests	Placement of incumbent transmission
$BW_{EUT} \leq BW_{Inc}$	Once	Tune incumbent and EUT transmissions ($f_{c1} = f_{c2}$)
$BW_{Inc} < BW_{EUT} \leq 2BW_{Inc}$	Once	Incumbent transmission is contained within BW_{EUT}
$2BW_{Inc} < BW_{EUT} \leq 4BW_{Inc}$	Twice. Incumbent transmission is contained within BW_{EUT}	Incumbent transmission is located as closely as possible to the lower edge, and upper edge, respectively, of the EUT channel
$BW_{EUT} > 4BW_{Inc}$	Three times	Incumbent transmission is located as closely as possible to the lower edge of the EUT channel, in the middle of EUT channel, and as closely as possible to the upper edge of the EUT channel

Regulatory testing methods and requirements

4.2.4.1 Contention-based protocol test setup

Configure the EUT to transmit with a constant duty cycle for the EUT to detect co-channel energy. A signal generator that is capable of generating band-limited AWGN is required to simulate an incumbent signal. Based on the EUT antenna configuration, the AWGN signal can be provided to the EUT receiver via conducted or radiated method. In a conducted test setup, a band-limited AWGN signal is generated at low power level and injected into the EUT's antenna port. **Figure 5** shows the conducted setup. Refer to **987594 EMC Measurement(I)(c)**.

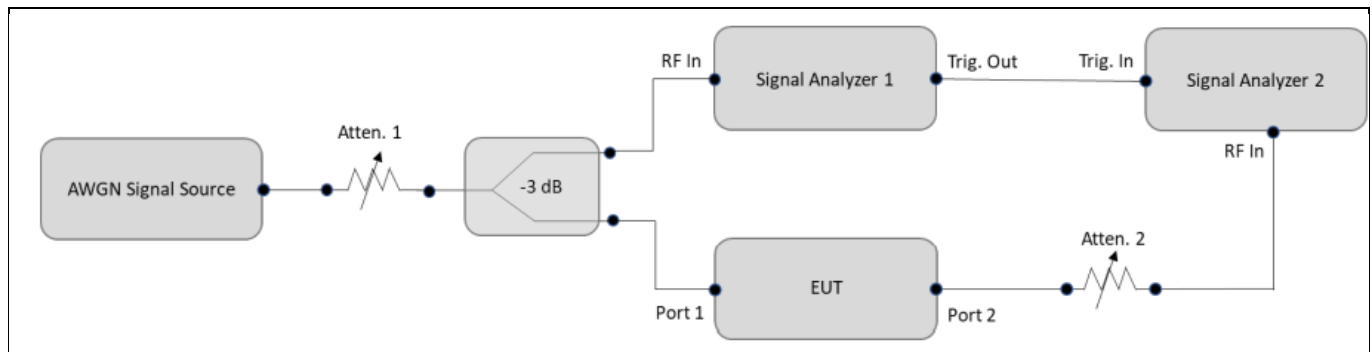


Figure 5 Contention-based protocol test setup, conducted method

In a radiated setup, the AWGN signal must be radiated in a controlled environment, with the AWGN signal illuminating the EUT antenna, and the AWGN signal power level accurately measured at the EUT antenna location. **Figure 6** shows the radiated test setup where the AWGN signal is generated and transmitted via antenna 1. Antenna 1 must be selected such that its 3-dB beamwidth can illuminate the EUT entirely. The EUT is replaced by antenna 2 which has a known gain, as shown in **Figure 6** to ensure that the AWGN signal level can be accurately measured at the EUT location. The radiated signal level is then measured using antenna 2. Antennas 1 and 2 are aligned and placed at a distance 'R' which is greater than the far-field distances of antennas 1 and 2. The AWGN signal power is measured by signal analyzer 1. The measured power P_{meas} is corrected by the gain of antenna 2, G_2 , and all cable losses and attenuations L . P_2 is then measured to determine the AWGN signal power level. P_2 is given by:

$$P_2 = P_{meas} + L - G_2 \quad (1)$$

The EUT is placed where antenna 2 was as shown in **Figure 7**.

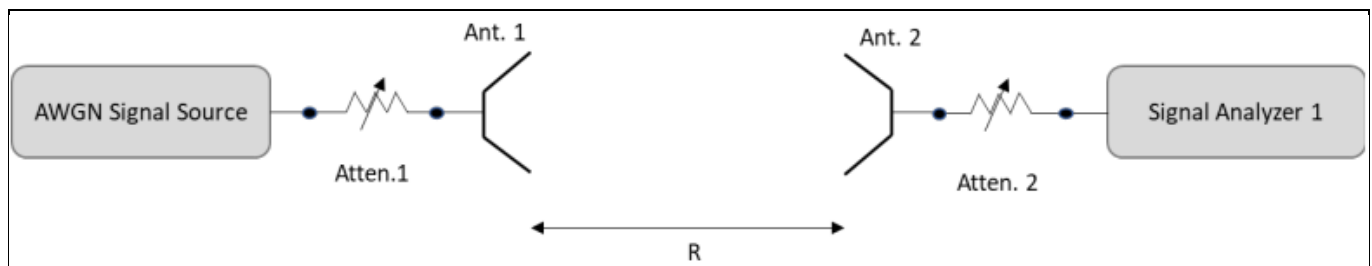


Figure 6 Contention-based protocol test setup, radiated method, power measurement

Regulatory testing methods and requirements



Figure 7 Contention-based protocol test setup, radiated method, detection threshold measurement

See [987594 EMC Measurement\(I\)\(d\)](#) for more information on contention-based protocol radiated setup.

4.2.5 Dual-client test, demonstration of proper power adjustment based on an associated AP

987594 EMC Measurement(K): A client device may connect to a standard power AP with a maximum power level of 30 dBm EIRP. A client may also connect to a low-power indoor AP, but the power level is limited to 24 dBm EIRP. If a client is flexible to connect to both APs, verification is needed to distinguish between the two configurations, and control the power levels accordingly.

4.2.6 Channel/bandwidth/data rate testing

To complete the channel, bandwidth, and data rate testing, your product should utilize Manufacturing Firmware and set the country code to “ALL”, which will open all channels, bandwidths, and data rates. At the same time, this also allows adjustment of power output in real time.

4.2.6.1 Channels testing and channel availability by country (Tier-1)

This section shows the channels that are allowed by country. Note that the requirements for allowable channels may differ from other channels that are permitted. The channels allowed per country are shown in the following tables:

Table 3 FCC/NCC (2.4 GHz)

Channels	11b	11g	11n20	11n40
1-11	3	3	3	3
12-13 ¹	3	3	3	3

Table 4 ETSI (2.4 GHz)

Channels	11b	11g	11n20	11n40
1-11	3	3	3	3
12-13	3	3	3	3

Table 5 FCC/NCC (5 GHz)

Channels	11a	11n20	11n40	VHT80
36-48	3	3	3	3

¹ Channel 12 and 13 in the US is permitted, but customers must adhere to the compliance requirements that are stated for these channels, which do differ from Channels 1-11.

Regulatory testing methods and requirements

Channels	11a	11n20	11n40	VHT80
52-64	3	3	3	3
100-140	3	3	3	3
144	3	3	3	3
149-165	3	3	3	3

Table 6 ETSI (5 GHz)

Channels	11a	11n20	11n40	VHT80
32-48	3	3	3	3
52-64	3	3	3	3
100-140	3	3	3	3
144 ¹	6	6	6	6
149-165	3	3	3	3

Table 7 ISSED (2.4 GHz)

Channels	11b	11g	11n20	11n40
1-11	3	3	3	3
12-13	3	3	3	3

Table 8 Japan MIC (2.4 GHz)

Channels	11b	11g	11n20	11n40
1-11	3	3	3	3
12-13	3	3	3	3

Table 9 ISSED (5 GHz)

Channels	11a	11n20	11n40	VHT80
36-48	3	3	3	3
52-64	3	3	3	3
100-116	3	3	3	3
120-128	6	6	6	6
132-140	3	3	3	3
144	3	3	3	3
149-165	3	3	3	3

Table 10 Japan MIC (5 GHz)

Channels	11a	11n20	11n40	VHT80
36-48	3	3	3	3
52-64	3	3	3	3

¹ The United Kingdom allows usage for Channel 144. All other ETSI member nations do not.

Regulatory testing methods and requirements

Channels	11a	11n20	11n40	VHT80
100-140	3	3	3	3
144	6	6	6	6
149-165	6	6	6	6

Table 11 China SRRC (2.4 GHz)

Channels	11b	11g	11n20	11n40
1-11	3	3	3	3
12-13	3	3	3	3

Table 12 South Korea KCC (2.4 GHz)

Channels	11b	11g	11n20	11n40
1-11	3	3	3	3
12-13	3	3	3	3

Table 13 China SRRC (5 GHz)

Channels	11a	11n20	11n40	VHT80
36-48	3	3	3	3
52-64	3	3	3	3
100-140	6	6	6	6
144	6	6	6	6
149-165	3	3	3	3

When testing various channels required for regulatory testing, Infineon recommends using country code “ALL”. To change the channels that the Infineon Wi-Fi device is operating on, you can use the following wl commands:

Mode	Command	Note
802.11b	wl chanspec 1	Sets up the radio to channel 1 for 11b mode
802.11g	wl chanspec 1	Sets up the radio to channel 1 for 11g mode
802.11n (20 MHz)	wl chanspec 1	Sets up the radio to channel 1 for 11n20 mode
802.11n (40 MHz)	wl chanspec -c 3 -b 2 -w 40 -s 1 or wl chanspec 36/40	Sets up the radio to channel 3 for 11n40 mode. -c is channel, -b is operating band (set to 2 for 2.4 GHz or 5 for 5 GHz), -w is bandwidth, -s is lower side for 40 MHz only (where 1 is enable, 0 is disable).
802.11ac (80 MHz)	wl chanspec 36/80	Sets up the radio to channel 42 for 80 MHz mode. Must use a 20 MHz channel over 80. Example to change to channel 58 using wl commands is wl chanspec 60/80. To

Regulatory testing methods and requirements

Mode	Command	Note
		set to channel 106, the wl command is <code>wl chanspec 100/80</code> .

4.2.6.2 Bandwidth testing and availability

All countries stated above (Tier-1 countries) allow operation in 20-MHz, 40-MHz, and 80-MHz bandwidths.

4.2.6.3 Data rate testing

All regulatory tests are to be performed with the lowest data available for a given IEEE standard. The data rates shown below are the data rates that should be used for this testing. The rates shown at in the form of the `wl` command parameters that you will use during testing.

Mode	Command	Note
802.11b	<code>wl 2g_rate -r 1</code>	Sets radio to 1-Mbps data rate for 11b mode
802.11g	<code>wl 2g_rate -r 6</code>	Sets radio to 6-Mbps data rate for 11g mode
802.11n (20 MHz)	<code>wl 2g_rate -h 0 -b 20</code> <code>wl 5g_rate -h 0 -b 20</code>	Sets radio to 6.5-Mbps data rate for 20-MHz mode
802.11n (40 MHz)	<code>wl 2g_rate -h 0 -b 40</code> <code>wl 5g_rate -h 0 -b 40</code>	Sets radio to 13.5-Mbps data rate for 40-MHz mode
802.11ac (80 MHz)	<code>wl 5g_rate -v 0x1 -b 80</code>	Sets radio to 32.5-Mbps data rate with 1 spatial stream for 80-MHz mode

4.2.6.4 WL command example for data rate testing for 2.4 GHz

Command	Notes
<code>wl pkteng_stop tx</code>	Stops the radio from transmitting
<code>wl out</code>	Powers down the radio
<code>wl down</code>	Sets the radio to standby
<code>wl frameburst 1</code>	Allows to increase data output during continuous transmission
<code>wl ampdu 1</code>	Set to 1 to activate frame aggregation
<code>wl country ALL</code>	Sets the specific regional code (country code)
<code>wl rateset 54b</code>	Required to avoid the presence of active beacons during regulatory testing
<code>wl bi 65000</code>	Increases the beacon sequence transmit so that it does not interfere during test
<code>wl phy_watchdog 0</code>	Disables calibration tone
<code>wl mpc 0</code>	Disables the minimum power consumption
<code>wl txchain 3</code>	For SISO, use 1 or 2; for MIMO, use 3. This command activates 1tx or 2tx operation.
<code>wl mimo_bw_cap 1</code>	Enables 40-MHz and 80-MHz bandwidth operation

Regulatory testing methods and requirements

Command	Notes
wl band b	Sets which operating band to active (b = 2.4 GHz only, a = 5 GHz only, Auto)
wl txbf 0	Set to 0 to disable TXBF mode.
wl chanspec -c 1 -b 2 -w 20 -s 0	Change -c 1 to -c 6 or -c 7 to change to channel 6 or 7. -c is channel, -b is operating band (set to 2 for 2.4 GHz or 5 for 5 GHz), -w is bandwidth, -s is lower portion for 40 MHz (1 = enable or 0 = disable).
wl 2g_rate -r 6	Always use the lowest data rate for radio regulatory testing. Command to set up data rate.
wl up	Turns on radio
wl disassoc	Disconnects radio from any access point STA may be connected
wl phy_forcecal 1	Calibrates the radio before starting transmission to avoid a center frequency tone
wl txpwr1 -o 19	Use this command to change the power in real time. Can also use the quarter step command wl txpwr1 -o -q XX, where XX is any value between 0 to 100.
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	Sets radio to continuous transmission

4.2.6.5 WL command example for data rate testing for 11ax 2.4 GHz

Command	Notes
wl mpc 0	
wl phy_watchdog 0	
wl country ALL	
wl band b	
wl chanspec 1/20	
wl down	
wl mimo_txbw -1	
wl up	
wl phy_forcecal 1	
wl scansuppress 1	
Wl ldpc_cap1	
wl txchain[x]	Core 0: x=1; Core 1: x=2;
wl txcore -s 1 -c[x]	Core 0: x=1; Core 1: x=2;
sl 2g_rate -e [0-11] -s 1 -b 20	-e = MCS0 – MCS11, -b 20 MHz

Regulatory testing methods and requirements

Command	Notes
For MIMO operation wl txbf 0 wl txchain 3 wl txcore -s 2 -c 3 wl 2g_rate -e [0-11] -s 2 -b 20 wl phy_txpwrctrl 1	s = 2 for two streams, c = 3 to enable both core 0 and core 1 -e MCS0 – MCS11, s = 2 for 2x2, -b 20 MHz
wl wl txpwr1 -1	
wl txpwr1 -o -d [x]	X = user-defined Tx power in dBm
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	
wl pkteng_stop tx	

4.2.6.6 WL command example for 2.4 GHz 11ax RU testing

Command	Notes
wl pkteng_stop tx	
wl down	
wl txchain <chain>	chain = 1/2/3
wl rxchain <chain>	
wl country ALL	
wl mpc 0	
wl PM 0	
wl phy_txpwrctrl 1	
wl band b	
wl chanspec <channel>/<bandwidth>	
wl 2g_rate -e <rate> -b <bandwidth> -s <stream>	-e = MCS0-MCS11, -b = 80-MHz BW, -s = 1/2
wl up	
wl phy_forcecal 1	
wl txpwr1 -o -q <power>	
wl load_trig_info -b <bw> -u <RU> -e <rate> -s 1 -len 1017.0.0 -g 1 -ls 1 -l 1 -p 2 -dcm 0 -lfm 0 -st 0	-b = bandwidth, -u = RU index, -e <rate>, -s = number of streams
wl pkteng_start 00:11:22:33:44:55 tgrtx 200 1017.0 0	

Regulatory testing methods and requirements

4.2.6.7 WL command example for data rate testing for 5 GHz

Any commands that are not commented below are contained in the 2.4-GHz example in Section 4.2.6.4.

Command	Notes
wl pkteng_stop tx	
wl out	
wl down	
wl frameburst 0	
wl ampdu 1	
wl country ALL	
wl rateset 54b	
wl bi 65000	
wl phy_watchdog 0	
wl apsta 0	Disables the SoftAP follows STA mode
wl ap 0	Disables SoftAP mode
wl mpc 0	Turns on radio
wl txchain 3	For SISO, use 1 or 2; for MIMO, use 3.
wl mimo_bw_cap 1	
wl band a	
wl txbf 0	
wl chanspec -c 36 -b 5 -w 20 -s 0	Change -c 36 to -c 64 or -c 100 to change to channel 64 or 100.
wl 5g_rate -h 0 -b 20	Always use the lowest data rate for radio regulatory testing.
wl up	
wl disassoc	
wl phy_forcecal 1	
wl txpwr1 -o 19	Use this command to change power in real time. Can also use the quarter step command wl txpwr1 -o -q XX, where XX is any value between 0 to 100.
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	

Regulatory testing methods and requirements

4.2.6.8 WL command example for data rate testing for 11ax 5 GHz

Command	Notes
wl down	
wl apsta 0	
wl ap 0	Disables SoftAP mode
wl mpc 0	
wl phy_watchdog 0	
wl country ALL	
wl band a	
wl chanspec 36/80	
wl mimo_txbw -1	
wl up	
wl phy_forcecal 1	
wl scansuppress 1	
wl ldpc_cap 1	
[For SISO] wl txchain [x] wl txcore -s 1 -c [x] wl 5g_rate -e [0-11] -b 80	Core 0: X = 1; Core 1: X = 2 Core 0: X = 1; Core 1: X = 2 -e = MCS0 – MCS11, -b = 80-MHz BW
[For MIMO] wl txbf 0 wl txchain 3 wl txcore -s 2 -c 3 wl 5g_rate -e [0-11] -s 2 -b 80	-s = 2 for two streams, -c = 3 to enable both Core 0 and Core 1 -e = MCS0 – MCS11, -s = 2 for 2x2, -b = 80-MHz BW
wl phy_txpwrctrl 1	
wl txpwr1 -1	Default target power
wl txpwr1 -o -d [X]	X = user defined Tx power, in dBm
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	
wl pkteng_stop tx	

Regulatory testing methods and requirements

4.2.6.9 WL command example for 5 GHz 11ax RU data rate testing

Command	Notes
wl pkteng_stop tx	
wl down	
wl txchain <chain>	chain = 1/2/3
wl rxchain <chain>	
wl country ALL	
wl apsta 0	Disables SoftAP follows STA mode
wl ap 0	Disable SoftAP mode
wl mpc 0	
wl PM 0	
wl phy_txpwrctrl 1	
wl band a	
wl chanspec <channel>/<bandwidth>	
wl 5g_rate -e <rate> -b <bandwidth> -s <stream>	-e = MCS0-MCS11, -b = 80-MHz BW, -s = 1/2
wl up	
wl phy_forcecal 1	
wl txpwr1 -o -q <power>	
wl load_trig_info -b <bw> -u <RU> -e <rate> -s 1 -len 1017.0.0 -g 1 -ls 1 -l 1 -p 2 -dcm 0 -lfm 0 -st 0	-b = bandwidth, -u = RU index, -e <rate>, -s = number of streams
wl pkteng_start 00:11:22:33:44:55 tgrtx 200 1017.0 0	

4.2.6.10 WL command example for continuous transmission for data rate testing for 2.4 GHz/5GHz/6GHz 11ax RU data rate testing (ANT0/ANT1/MIMO)

Command	Notes
if [\$chain -eq 3];then stream=2 idx="\$q_power \$q_power" ltfs=2 else stream=1 ltfs=1 if [\$chain -eq 2];then idx="127 \$q_power" else idx="\$q_power 127" fi	band=\$1 # a for 5G b for 2G s for 6G channel=\$2 # center frequency in channel number chain=\$3 # antenna setting 1 for A0 2 for A1 3 for both bw=\$4 # bandwidth 20 40 80 rate=\$5 # data 11ax HE rate index RU=\$6 # Resource unit index q_power=\$7 # tx index setting ex. 0 max power 127 min power bytes=\$8 # payload bytes
wl pkteng_stop tx	
wl reset_cnts	

Regulatory testing methods and requirements

Command	Notes
wl down	
wl band <band>	
wl country US/0	
wl txchain <chain>	chain = 1/2/3
wl rxchain <chain>	
wl mpc 0	
wl PM 0	
wl ampdu 0	
wl twt 0	
wl down	
wl phy_txpwrctrl 0	
wl band \$band	
wl chanspec \$channel/\$bw	
if [[\$band == "a"]] then wl 5g_rate -e \$rate -b \$bw -s \$stream elif [[\$band == "b"]] then wl 2g_rate -e \$rate -b \$bw -s \$stream else wl 6g_rate -e \$rate -b \$bw -s \$stream fi	-e = MCS0-MCS11, -b = 80-MHz BW, -s = 1/2
wl up wl sublocking 1 wl phy_watchdog 0 wl wd_disable 1 wl phy_Deaf 1 wl txbf 0 wl txbf_imp 0 sleep 1	
wl phy_forcecal 1 wl reset_cnts	
sleep 1	
wl phy_txpwrindex \$idx wl load_trig_info -b \$bw -u \$RU -e \$rate -s \$stream -len \$bytes -g 1 -ls \$ltfs -l 1 -p 2 -dcm 0 -lfm 0 -st 0 sleep 1 wl pkteng_start 00:11:22:33:44:55 tgrtx 20 \$bytes 0	
txfrm=`./wl counters   grep he_txbppdu   awk '{print \$2}'` while : do sleep 0.1; txfrm1=`./wl counters   grep he_txbppdu   awk '{print \$2}'` delta=\$((txfrm1 - txfrm))	-b = bandwidth, -u = RU index, -e <rate>, -s = number of streams

Regulatory testing methods and requirements

Command	Notes
<pre> if [[\$delta == 0]];then ./wl pkteng_stop tx ./wl load_trig_info -b \$bw -u \$RU -e \$rate -s \$stream -len \$bytes - g 1 -ls \$ltfs -l 1 -p 2 -dcm 0 -lfm 0 -st 0 sleep 0.1 ./wl pkteng_start 00:11:22:33:44:55 tgrtx 20 \$bytes 0 fi txfrm=\$txfrm1; echo "Delta counters = \$delta" done </pre>	

4.2.6.11 WL command example for data rate testing for 6 GHz

Command	Notes
wl pkteng_stop tx	
wl out	
wl down	
wl frameburst 1	
wl ampdu 1	
wl country ALL	
wl rateset 54b	
wl bi 65000	
wl phy_watchdog 0	
wl mpc 0	
wl txchain 3	For SISO use 1 or 2 and MIMO use 3
wl mimo_bw_cap 1	
wl txbf 0	
wl 6g_rate -e MxS	-e M[xS], --he=M[xS]; M = HE rate M [0-11], S = number of spatial streams
wl pe_ext 0	
wl wd_disable 1	Watchdog disable
wl spatial_policy 1	
wl stbc_rx 0	
wl stbc_tx 0	

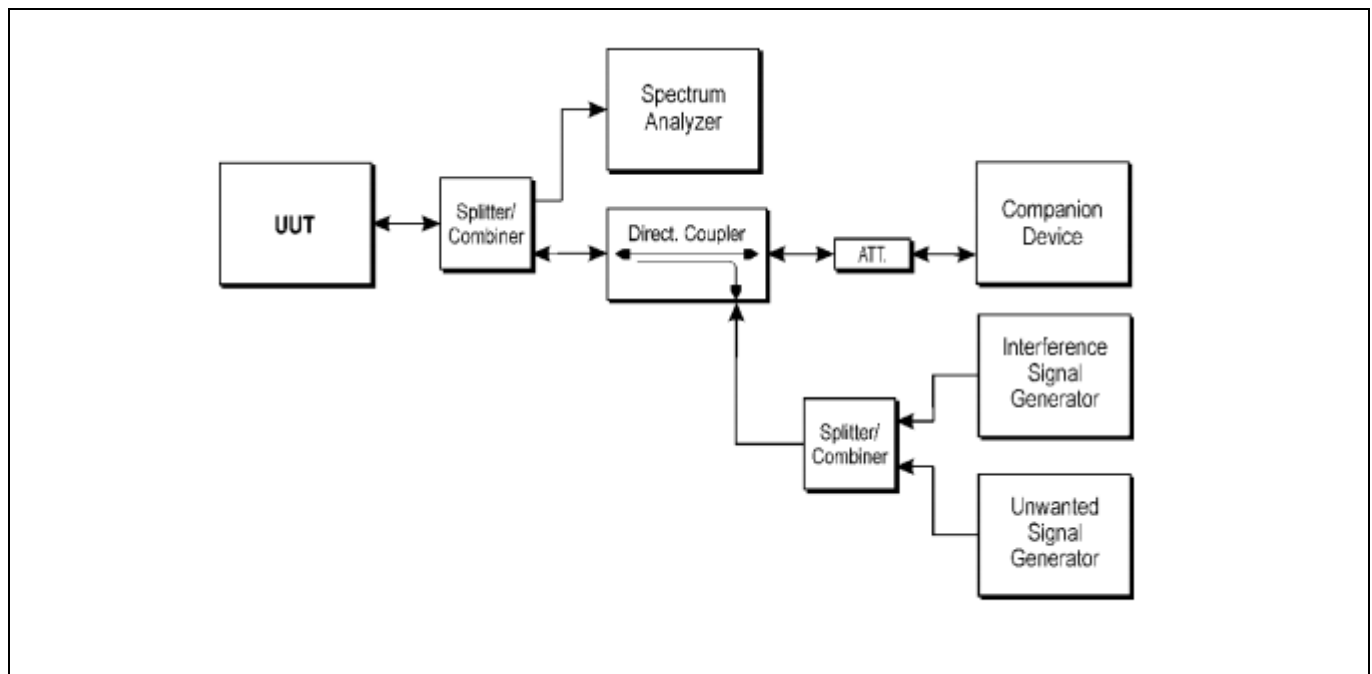
Regulatory testing methods and requirements

Command	Notes
wl ldpc_cap 1	
wl up	
wl phy_forcecal 1	
wl phy_txpwrctrl 1	
wl txpwr1 -o -q [x]	Wl txpwr1 -1 (default target power); [x] = user defined Tx power in qdBm
wl dump physcal	
wl phy_activecal	
wl phy_test_tssi 0	
wl pkteng_start 00:00:00:c0:ff:ee tx 250 52 0	

4.2.7 Adaptivity testing

This section provides information on the Adaptivity test setup, applicable Infineon wl commands to execute the test, and the compliance requirement for this test. This testing is applicable for ETSI member nations. Countries that follow the general ETSI regulatory requirements may or may not require this testing.

4.2.7.1 Adaptivity test setup



4.2.7.2 WL command example for adaptivity for 2.4 GHz

Command	Notes
wl out	

Regulatory testing methods and requirements

Command	Notes
wl down	
wl frameburst 0	
wl ampdu 1	
wl country FR/5	Set to a European country code FR, GB, the number 5 is a regrev which can range from 0 to 100
wl bi 65000	
wl phy_watchdog 0	
wl mpc 0	
wl txchain 1	Test SISO by using 1. If using MIMO 3, then connect both RF ports.
wl mimo_bw_cap 1	
wl 2g_rate -r 1	ETSI standard does not state which rate to use; therefore, set up per agreement with EMC test lab to produce 30% traffic.
wl up	
wl phy_forcecal 1	
wl join <SSID>	To connect to Access Point, need to know the SSID name. Replace <SSID> with the actual name of the Access Point.
wl status	Shows if STA/Client device connected to the Access Point.
wl phy_ed_thresh -70	This command is to adjust the Energy Detect (Adaptivity) sensitivity level. Range is from -20 to -75, but never below -75.
iperf -c <ip address of server> -u -b -l -i 1 -t 10000	NOTE 1

Note: `iperf -c 192.168.1.100 -u -b 5m -l 60000 -i 1 -t 10000`
 where
 -l = data length max of 65550
 -t = Timing
 -i = interval
 -u = UDP tests
 -b = Bandwidth settings

Note: The iperf example above uses the actual parameters that Infineon uses to create the 30% duty cycle traffic that is required for adaptivity testing at the EMC lab. The EMC lab will measure the 30% to confirm that the iperf settings used are accurate. Change only the -b or -l parameters if necessary to achieve 30% duty cycle traffic.

Regulatory testing methods and requirements

4.2.7.3 WL command example for adaptivity for 5 GHz

Command	Notes
wl out	
wl down	
wl frameburst 0	
wl ampdu 1	Set to a European country code FR, GB, the number 5 is a regrev which can range from 0 to 100
wl country FR/5	
wl bi 65000	
wl phy_watchdog 0	
wl mpc 0	
wl txchain 1	Test SISO by using 1. If using MIMO 3, then connect both RF ports.
wl mimo_bw_cap 1	
wl 5g_rate -r 1	ETSI standard does not state which rate to use; therefore, set up per agreement with EMC test lab to produce 30% traffic.
wl up	
wl phy_forcecal 1	
wl join <SSID>	To connect to Access Point, need to know the SSID name. Replace <SSID> with the actual name of the Access Point.
wl status	Shows if STA/Client device connected to the Access Point
wl phy_ed_thresh -73	This command is to adjust the Energy Detect (Adaptivity) sensitivity level. Range is from -20 to -75, but never below -75.
iperf -c <ip address of server> -u -b -l -i 1 -t 10000	NOTE 1

4.2.7.4 WL command example for ETSI adaptivity for 6 GHz

Command	Notes
wl out	
wl down	
wl frameburst 0	
wl ampdu 1	
wl country FR/5	Set to a European country code FR, GB, the number 5 is a regrev which can range from 0 to 100.

Regulatory testing methods and requirements

Command	Notes
wl bi 65000	
wl phy_watchdog 0	
wl mpc 0	
wl band s	S = 6 GHz
wl chanspec 1/20	
wl txchain 1	Test SISO by using 1. If using MIMO 3, then connect both RF ports.
wl rxchain 1	Test SISO by using 1. If using MIMO 3, then connect both RF ports.
wl mimo_bw_cap 1	
wl 6g_rate -r 6 -b 20	ETSI standard does not state which rate to use; therefore, set up per agreement with EMC test lab to produce 30% traffic.
wl up	
wl phy_forcecal 1	
wl join <SSID>	To connect to Access Point, need to know the SSID name. Replace <SSID> with the actual name of the Access Point.
wl status	Shows if STA/Client device connected to the Access Point
wl phy_ed_thresh -73	This command is to adjust the Energy Detect (Adaptivity) sensitivity level. Range is from -20 to -75, but never below -75.
iperf -c <ip address of server> -u -b 100m -i 1 -t 10000	

4.2.7.5 WL command example for FCC Contention Base Protocol 6 GHz

Command	Notes
wl down	
wl frameburst 1	
wl ampdu 1	
wl country US/0	
wl bi 65000	
wl phy_watchdog 0	
wl mpc 0	
wl band s	S = 6 GHz
wl chanspec 1/20	
wl txchain 1	Test SISO by using 1. If using MIMO 3, then connect both RF ports.
wl rxchain 1	Test SISO by using 1. If using MIMO 3, then connect both RF ports.

Regulatory testing methods and requirements

Command	Notes
wl mimo_bw_cap 1	
wl 6g_rate -r 6 -b 20	ETSI standard does not state which rate to use; therefore, set up per agreement with EMC test lab to produce 30% traffic.
wl up	
wl phy_forcecal 1	
wl join <SSID>	To connect to Access Point, need to know the SSID name. Replace <SSID> with the actual name of the Access Point.
wl status	Shows if STA/Client device connected to the Access Point
iperf -c <ip address of server> -u -i 1 -b 100m -t 10000	

The recommended data rates to be used for testing are as follows by IEEE standard:

IEEE standard	Suggested data rate for testing (Mbps)	Comments
802.11b	5.5	Suggested to use 5.5-Mbps data rate. Ensure that your iperf wl command setting is such that the Duty Cycle Traffic during testing is $\geq 30\%$.
802.11g	24	Suggested to use 5.5-Mbps data rate. Ensure that your iperf wl command setting is such that the Duty Cycle Traffic during testing is $\geq 30\%$.
802.11a	6	Use the lowest data rate available.
802.11n	6.5	Use the lowest data rate available.
802.11ac	32.5 (MCS0 SS1)	Use wl 5g_rate -v 0x1 -b 80.

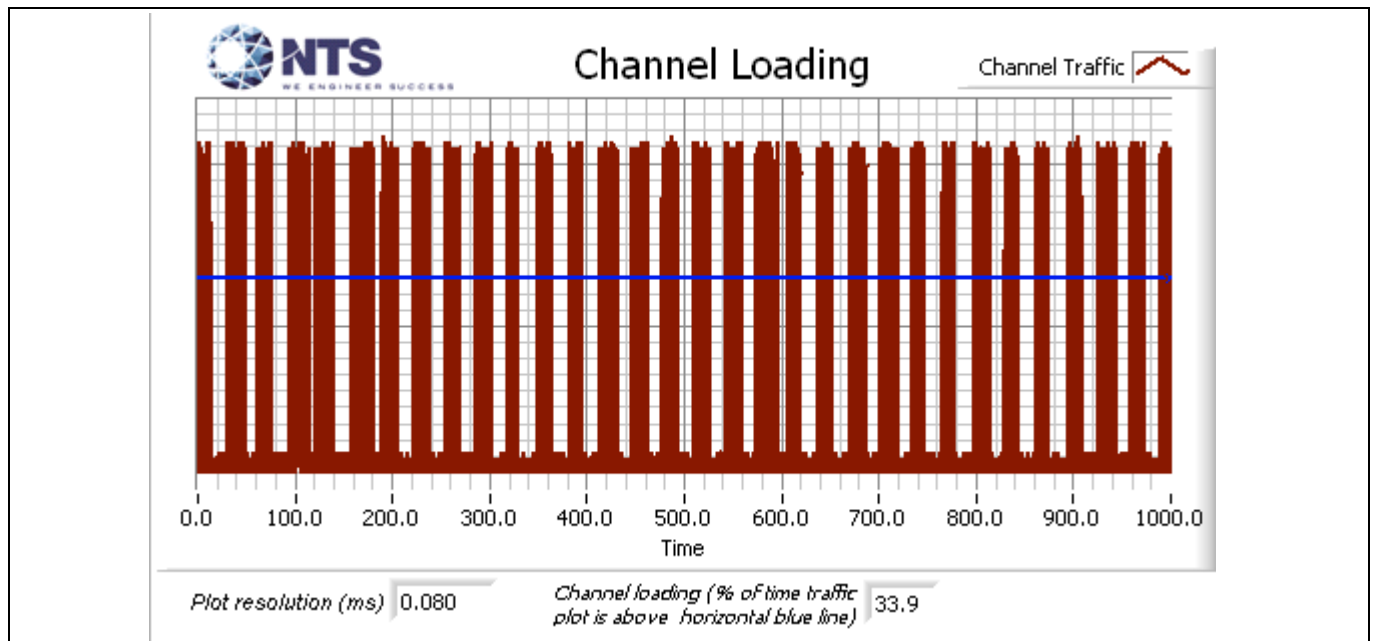
Regulatory testing methods and requirements

4.2.7.6 Test requirement

For 2.4-GHz radios, the applicable test levels are as follows:

1. AWGN interference signal generator = -70 dBm/MHz
2. Rx Blocker signal generator = -35 dBm
3. Duty cycle traffic between Unit Under Test and Support Link Partner = > 30%

See [300 328 V2.2.2](#), sections [5.4.6.2](#) and [5.4.6.2.1.4](#) for additional information. A plot from a spectrum analyzer showing a 30% duty cycle traffic is shown as follows:



For 5-GHz radios, the applicable test levels are as follows:

- AWGN interference signal generator = -75 dBm/MHz
- Traffic between Unit Under Test and Support Link Partner = Configure the traffic source so that it fills the unit under test (UTT) buffers to a level causing the UUT to always have transmissions queued (buffer-ready-for-transmission condition) towards the companion device. User Datagram Protocol (UDP) may be used to avoid any adverse effects on the measurements results.

See [301 893 V2.1.1](#), Sections [5.4.9](#) and [5.4.9.3.2](#) for additional information.

Regulatory testing methods and requirements

4.2.7.7 Test information required for adaptivity testing

To complete adaptivity testing, EMC labs will generally request the information provided in this section, which is specific to the supplier Wi-Fi solution. The following information should be provided for Infineon Wi-Fi solutions.

EN300 328 V2.2.2 (2.4 GHz) required information for adaptivity

Generally, EMC labs will only require the following information for EN300 328 V2.2.2:

1. The EMC lab will want to confirm if the device is load-based equipment. Infineon Wi-Fi radios are all load-based equipment; therefore, the section that applies to Infineon Wi-Fi radios for adaptivity testing is Section 4.3.2.6.3.2.3 of EN300 328 V2.2.2.
2. The COT for Infineon Wi-Fi solutions is 13 ms. COT = channel occupancy time.

EN301 893 V2.1.1 (5 GHz) required information for adaptivity

If your product uses the 5-GHz spectrum, the EMC labs will require the information contained in this section. The following questions are taken from Annex G in EN301 893 V2.1.1, which are the questions that customers are required to answer for adaptivity testing. Each question is followed by the information pertinent to Infineon Wi-Fi solutions.

Adaptivity (Channel access mechanism)

Infineon Wi-Fi solutions are load-based equipment.

For load-based equipment that supports multi-channel operation

Infineon Wi-Fi load-based equipment supports Option 2 as described in Clause 4.2.7.3.2.3 of EN301 893. The lab will request the following additional information for Infineon Wi-Fi solutions:

- The (maximum) number of channels used for multi-channel operation: **4**
- These channels are adjacent channels: **Yes**
- In case of non-adjacent channels, whether or not these channels are in different sub-bands: **No**

Operating mode

Customers are required to define the end-product operating type. There are three different types listed in the ETSI specifications:

1. Does the load-based equipment operate as a supervising device?
Infineon Information: “Supervising” means access point/router device. This is dependent on your end-product implementation.
2. Does the load-based equipment operate as a supervised device?
Infineon Information: “Supervised” means STA/client device. This is dependent on your end-product implementation.
3. Does the load-based equipment operate as a supervising and as a supervised device?
Supervising and Supervised indicates that the device can operate as either a STA/client or AP/router.

Note: Infineon cannot pre-determine the type of solution that you have developed, so you must select the definition that applies to your product from this list.

Regulatory testing methods and requirements

Priority classes

Priority classes implemented by the load-based equipment can be one of the following. The specific priority class for Infineon Wi-Fi devices is also provided:

1. When operating as a supervising device:
 - a) Infineon Wi-Fi device answer: Priority Class 2
2. When operating as a supervised device:
 - a) Infineon Wi-Fi device answer: Priority Class 2

Additional adaptivity information for Infineon Wi-Fi solutions

The following information will likely also be asked by the EMC lab for adaptivity testing:

1. Does the load-based equipment operate as an initiating device?
Initiating means access point/router device.
2. Does the load-based equipment operate as a responding device?
Responding means STA/client device.
3. Does the load-based equipment operate as an initiating device and as a responding device?

Note: Infineon cannot pre-determine the type of solution that you have developed, so you must select the definition that applies to your product from the above list.

Energy detection threshold and protocol

With respect to the energy detection threshold for the load-based equipment under test, the EMC lab will need to know if the load-based equipment has implemented either Option 1 of Clause 4.2.7.3.2.5 of ETSI EN 301 893 V2.1.1 or Option 2 of Clause 4.2.7.3.2.5 of ETSI EN 301 893 V2.1.1. The answer to this question for Infineon Wi-Fi solutions is “Option 1”.

In addition, you will likely be asked to specify the protocol that has been implemented for the Infineon Wi-Fi solutions. The answer to this question is “IEEE 802.11”.

4.2.8 Dynamic frequency selection (DFS)

DFS requirements are categorized by the type of device that is being tested. Wi-Fi devices can be categorized in one of five areas:

1. Master Device with Radar Signal Detection
2. Master Device with no Radar Signal Detection
3. Slave Device with SoftAP functionality and no Radar Signal Detection
4. Slave Device with SoftAP functionality with Radar Signal Detection
5. Slave Device with no SoftAP functionality and no Radar Signal Detection

The specific testing that is carried out for DFS compliance are as follows:

1. In-service monitoring – Only applies to device categories 1 and 4 (devices with Radar Signal Detection)
2. Non-occupancy period (NOP) – Applies to all categories of devices
3. Bandwidth detection – Only applies to device categories 1 and 4 (devices with Radar Signal Detection)
4. Channel availability check (CAC) – Only applies to device categories 1 and 4 (devices with Radar Signal Detection)
5. Channel move and close – Applies to all categories of devices

Regulatory testing methods and requirements

In-service monitoring

This test applies to devices that are capable of detecting radar signals. In-Service Monitoring is a test requirement to prove that the DFS mechanisms can detect radar signals. Each regulatory standard states how many test trials are to be used as well as the pass/fail detection percentage criteria.

Non-occupancy period (NOP)

This test applies to any device that plans to operate using DFS channels. The Non-Occupancy Period (NOP) test is where the radio detects radar signals and moves to another channel. When radar signals are detected on a specific target channel, the device should move to another channel and not occupy the original target channel for a period of 30 minutes. The test house will monitor the original target channel after radar signal detection for 30 minutes to confirm compliance. This is to assure that the radio will not come back to the original channel it moved from, because the original channel should be blacklisted or disallowed for 30 minutes.

Bandwidth detection

This test applies to devices that are capable of detecting radar signals. The Bandwidth Detection test is to confirm that the radio can detect radar signals across the operating channel bandwidth. To comply with this test, a DFS channel is selected and tested to confirm that it is utilizing 99% of the channel bandwidth (for example for 802.11a, this test confirms that 99% of the 18 MHz channel bandwidth is utilized). The test house will typically run a total of 9 or 10 trials in 1 MHz increments across the channel bandwidth.

Channel availability check (CAC)

This test applies to devices that are capable of detecting radar signals. The Channel Availability Check (CAC) test is where the radio scans a target DFS channel (that it would like to utilize) for radar signals for a period of 60 seconds. After the 60 second scan, if no radar signals are detected, the radio can transmit on the target DFS channel. During the 60-second scan, if radar signals are detected on a target DFS channel, the radio must move to another DFS channel and re-start the 60-second CAC. To prove that the CAC mechanism works, the radio will be set to start the CAC and the test house will inject a radar test signal 6 seconds after CAC has started. The passing criteria is determined by monitoring if the radio moves to another channel. A failing case is where the radio starts transmitting after the 60 seconds has elapsed on the DFS channel that the radar signal was injected on. This test is then repeated with the radar test signal injection occurring 54 seconds after the CAC has started.

Channel move and close

This test applies to any device that plans to operate using DFS channels. The Channel Move and Close testing will confirm that the radio moves and closes specific DFS channels when radar signals are detected, or when instructed to move and/or close a channel from a Master device. When a radar signal is detected over the air on an operating DFS channel, the radio will close the current operating DFS channel and move to another channel. There are specific limits to the close and move requirements, all of which are timing related. The timing requirements for close and move operations are specifically called out in each of the regulatory standards for FCC/Europe/Japan/China/South Korea/Brazil and others.

Radar detecting and non-radar detecting devices

As mentioned at the start of this section, devices capable of detecting radar signals must comply and be tested to each of the five test cases above. Radar detecting devices will operate and comply to the requirements without the control or command from a Master device.

Non-radar detecting devices will still be tested for the test cases mentioned above; however, compliance to these test cases will be carried out through control/command given by a Master device in the test setup. Non-

Regulatory testing methods and requirements

Radar detecting devices do not determine themselves if a DFS channel is available to communicate on; they rely on the Master device to move and close specific DFS channels as well as when to not occupy (NOP) a specific blacklisted DFS channel.

Recommended DFS testing method

Infineon highly recommends testing DFS via radiated testing for FCC approval. The FCC will request a sample to validate this test and the FCC test setup is radiated. For EU, Japan, China, South Korea, and Brazil, the DFS tests can be performed via conducted testing.

Per FCC and European requirements, the lowest antenna gain must be used for both Radiated and Conducted DFS testing. Infineon does not recommend the use of negative antenna gains in a Wi-Fi design. The Infineon DFS mechanism uses sensitivity threshold values that are preset for an antenna gain value that will work correctly for any antenna gain greater than 0 dBi. Additionally, FCC and European (ETSI) standards specifically state that the DFS test level is in reference to a 0 dBi antenna gain value. As noted, a negative antenna gain will reduce the DFS radar detection sensitivity level. Adjusting the DFS sensitivity threshold levels for a negative antenna gain is not recommended as this will cause the radio to be prone to higher false radar detections.

Infineon pre-adjusts the DFS threshold sensitivity levels for an Infineon device to pass each country's DFS test requirements. No adjustments of the DFS threshold levels is allowed. Infineon can share certain technical documents that describe how Infineon DFS mechanism works and the wl commands to be used during DFS testing. Any adjustments to the DFS threshold sensitivity levels must be requested directly to Infineon via a Salesforce case. The case should state the chip model, FW version, lowest antenna gain, the specific test that is failing (if any), conditions of failure, and an analyzer plot or test result of the failure.

Command	Notes
wl down	This command can also be used to reset the blacklisted DFS channels, but it is also used to set the radio to standby mode.
wl ap 1	Enables SoftAP mode
wl mpc 0	Disables the minimum power consumption
wl country US/0	Use US/0 for FCC, BE/4 for EU, JP/972 for Japan.
wl band a	Set the radio to only be operated in the 5-GHz band.
wl mimo_bw_cap 1	To enable 40 and 80 MHz
wl dfs_preism 1	To enable and disable the CAC (60 sec). 0 = disable, 1 = enable
wl spect 1	To enable 802.11H for DFS testing
wl txchain 3	Test SISO by using 1. If using MIMO 3, then connect both RF ports.
wl chanspec 100/20	Controls the operating channel of radio. For 40, use wl chanspec 100/40 and for 80 use wl chanspec 100/80.
wl up	Turns on the radio
wl radar 1	Enables radar detection
wl ssid DFS_AP	Sets the SSID to identify the radio as DFS_AP
wl status	Shows the Access Point (SoftAP) information of channel.

Regulatory testing methods and requirements

Command	Notes
<code>wl dfs_ism_monitor 0</code>	To enable and disable the channel from moving when radar signal is detected. 0 = enable, 1 = disable.
<code>wl chan_info</code>	To show radar sensitive, passive, and temporary out of service for 30 min after radar signal detected on any of the DFS channels

4.2.9 Radiated immunity testing

4.2.9.1 Test setup

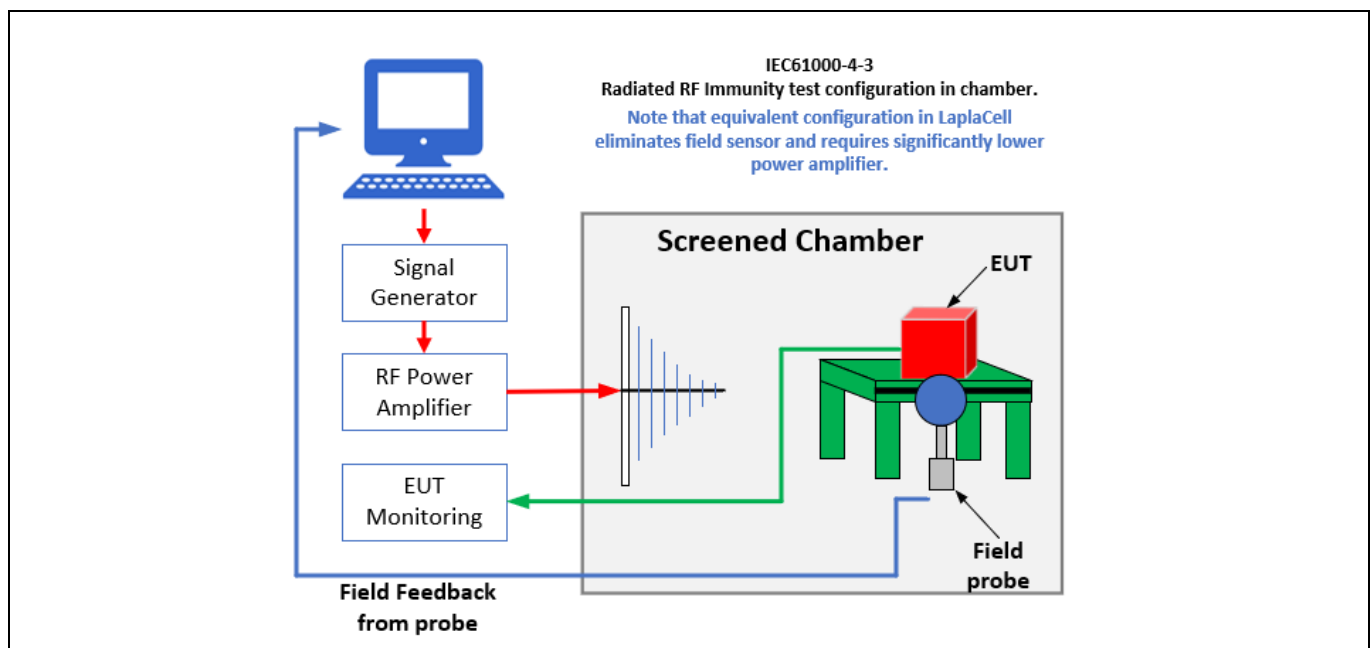


Figure 8 Test setup

The Equipment Under Test (EUT) is set up on a table as demonstrated in Figure 8. To monitor the performance of the EUT, the following must be done:

1. Use an Access Point or a STA device (Link Partner) to create a link.
2. Connect the UUT, which can be either the Access Point or the STA device, to the Link Partner.
3. Use ping to create traffic and use this as a monitoring method to ensure that the link is stable during the Radiated Immunity test.
4. If any failures are reported, do the following:
 - a) Repeat the failing frequency. If repeatable, it fails; if not repeatable, it passes.
 - b) Ensure that the support laptop is not being affect by the RI. This can be confirmed by placing the support laptop at the bottom of the table.
 - c) Ensure that the EMC lab is including the exclusion bands per [301 489-17 Section 4.3](#).
 - d) Ask the EMC lab to measure the harmonic content of the signal generator and the output of the power amplifier.
 - e) Add any ferrites on the support connection cable as this can also cause the link to drop.

Regulatory testing methods and requirements

4.2.9.2 Test requirements

IEC61000-4-3 immunity testing requires that the EUT operates satisfactorily when subject to a strong radiated electromagnetic field (such as what might be created by a cell phone or other intentional radiators, and from RF noise that might be present, inadvertently or otherwise, by industrial processes).

This requires a frequency scan at a certain fixed level of field strength (specified by the standard). The 'scan' will comprise a series of 'steps' in frequency. Each step is specified as a percentage of current frequency value. This percentage is variable from 0.2% to 5%.

At each step, the frequency is held, the level adjusted to the required field strength (V/m) as measured by a field sensor, and a prescribed modulation mode is initiated and then the conditions held for a 'dwell' time. The EUT should be monitored to detect any faulty operation during the test. Typical values for field strength are 3 V/m per EN301 489-1 and EN301 489-17.

Other higher stress levels are often specified for certain types of products (for example, medical and automotive end products). These fields are so intense that this test must be carried out in a screened chamber or test cell. Test cells such as the GTEM or LaplaCell are highly efficient but restricted to small products. Screened chambers can be large but can be relatively expensive and require powerful amplifiers to create the required stress levels.

The following list of equipment is required for this Radiated Immunity testing:

1. Signal Generator
2. Power Amplifier
3. Antenna
4. Field Probe
5. Control software and PC
6. EUT monitoring system
7. Test cell or chamber

4.2.10 Europe – ETSI SDR 5.8 GHz

Europe allows the operation in the 5725-5825 MHz frequency range, but does require testing the device to the Short Range Device ETSI standard ([300 440 V2.1.1](#)).

The power limit for the 5725-5825 MHz frequency range is 14 dBm EIRP.

The ETSI committee is working on introducing the 5725-5825 MHz frequency range into the ETSI EN301 893 standard.

4.2.11 Typical cycle time of certification for Tier-1 countries

Regulatory agency	Operation	TX chains	Typical time (weeks)			
			Typical test time	Typical test report time	Typical approval time	Total (Typical)
FCC/ISED/NCC	2.4 GHz Only	1	1	1	1	3
FCC/ISED/NCC	2.4 GHz Only	2	2	1	1	4
FCC/ISED/NCC	5 GHz Only	1	2	1	1	4
FCC/ISED/NCC	5 GHz Only	2	4	1	1	6

Regulatory testing methods and requirements

Regulatory agency	Operation	TX chains	Typical time (weeks)			
			Typical test time	Typical test report time	Typical approval time	Total (Typical)
FCC/ISED/NCC	2.4 + 5 GHz	1	3	1	1	5
FCC/ISED/NCC	2.4 + 5 GHz	2	6	1.5	1	8.5
ETSI	2.4 GHz Only	1	1.5	1	- ¹	2.5
ETSI	2.4 GHz Only	2	3	1	-	4
ETSI	5 GHz Only	1	2.5	1	-	3.5
ETSI	5 GHz Only	2	5	1	-	6
ETSI	2.4 + 5 GHz	1	3.5	1	-	4.5
ETSI	2.4 + 5 GHz	2	7	1	-	8
ETSI	Immunity Testing	-	1	1	-	2
MIC	2.4 GHz Only	1	0.5	1	2	3.5
MIC	2.4 GHz Only	2	1	1	2	4
MIC	5 GHz Only	1	1	1	2	4
MIC	5 GHz Only	2	2	1	2	5
MIC	2.4 + 5 GHz	1	1.5	1	2	4.5
MIC	2.4 + 5 GHz	2	3	1.5	2	6.5

For China and South Korea (in-country testing required), the test time will vary. Infineon suggests that you use a third-party vendor to minimize the test time, cost, and approval times.

4.3 Power characterization/optimization for Tier-1 countries

One of the benefits to Infineon Wi-Fi solutions is the ability to customize the output power targets by Channel/Data Rate/Band. This allows for optimal performance in your Wi-Fi design. To determine optimal power targets for each Channel/Data Rate/Band, it is required to conduct testing and power characterization for your solution to ensure that the target powers are compliant with regulatory requirements.

This section will walk you through the methods and process for power characterization with respect to the Tier-1 countries referenced in this document.

The process for determining compliant power targets by channel/band/data rate is to do the following:

1. Start with an initial power target setting.
2. Work with the EMC lab to decrease or increase these power targets to the point where they are compliant with regulatory requirements.
3. Record the final power targets and include these in the Infineon CLM templates provided and submit them to Infineon.

¹ Approval time is immediate for ETSI once the test report is completed and received. The ETSI standard is self-declaratory.

Regulatory testing methods and requirements

4.3.1 FCC/ISED/Taiwan power characterization

4.3.2 Setting initial regulatory power targets prior to testing

This process is completed to provide the EMC lab with actual operating powers for your device. To determine the wl command parameters (power levels) that the lab should use to set specific channel/data rate/band powers, follow this initial process.

1. Install/Program the Infineon Manufacturing FW.
2. Use an average gated power meter.
3. Use wl commands to set the radio to transmit continuously.
4. Set the output power to the desired level using the `wl txpwr1` command.
5. Record the average power level and the power settings and provide to the EMC lab.

Note: The wl command examples to complete the process above are included at the end of this section.

4.3.3 Recommended duty cycle by IEEE standard

The duty cycles that should be used by the EMC lab as follows:

1. 802.11b = >98% duty cycle (**Note:** Using 1 Mbps data rate)
2. 802.11g = ~98% duty cycle (**Note:** Using 6 Mbps data rate)
3. 802.11n (20 MHz) = 95% duty cycle (**Note:** Using MCS0 data rate)
4. 802.11n (40 MHz) = 90% duty cycle (**Note:** Using MCS0 data rate)
5. VHT80 (80 MHz) = 75% duty cycle (**Note:** Using VHT0 SS1 data rate)

The above rates shall be used in conjunction with the following packet engine parameters:

```
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0
```

4.3.4 Test items to determine power target compliance CLM input

The recommended test items and sequence to capture the target powers that will be provided in the CLM power templates is listed as follows. The following sequence is developed to ensure that test items that are impacted by others are completed first.

1. Radiated bandedges
2. Radiated harmonic
3. PSD
4. Power Output
5. Conducted harmonic emissions

Note: For Radiated bandedges in solutions that include multiple Tx chain configurations, there are two options:

1. Individually maximized power for 1-Tx and then individually maximized power for 2-Tx
2. Maximized power for 2-Tx only and use the 2-Tx powers for 1-Tx

Regulatory testing methods and requirements

If using Option 1, for Radiated Harmonics testing, all channels/bandwidth for 2-Tx should be set to 1-Tx maximum power. This will automatically cover 1-Tx modes because testing 2-Tx modes at maximum power is the worst-case scenario.

If using Option 2, only test all channels/bandwidth to 2-Tx maximum power.

4.3.4.1 Efficient channel/band testing and sequence

To maximize the efficiency when testing channel and band combinations, use the following sequences when verifying power targets.

For 2.4 GHz, it is recommended to test 1-Tx configuration first, then test 2-Tx configurations. The recommended channels and order of testing is listed as follows to confirm passing power targets.

1. 802.11b channels 1 and 11 (or test 1, 11, 12, 13 if channels 12 and 13 are enabled)
2. 802.11n (20 MHz): Use this mode to cover 802.11g mode. Test channels 1, 2, 10, and 11 (or test 1, 2, 10, 11, 12, and 13 if channels 12 and 13 are enabled)
3. 802.11n (40 MHz): Channels 3, 4, 8, and 9 (or test 3, 4, 8, 9, 10, 11 if channels 10 and 11 are enabled)

The inner channels (that is, 2.4-GHz Channel 6) are not impacted by bandedges; therefore, the power is set to the maximum output power that the radio chip supports. The same is true for the inner channels in the 5-GHz band, which will be transmitting at maximum power but could potentially be limited by PSD.

For 5 GHz, it is recommended to test 1-Tx configuration first, then test 2-Tx configuration. The recommended channels and order of testing is listed as follows to confirm passing power targets.

1. 802.11n (20 MHz): Use this mode to cover 802.11a and VHT20. Test channels 36, 64, 100, 140, 149, and 165
2. 802.11n (40 MHz): Use this mode to cover VHT40 as well. Test channels 38, 62, 102, 134, 151, and 159
3. VHT80 (80 MHz): Test channels 42, 58, 106, 122, and 155

For 2.4-GHz designs that support 2-Tx operation and use 2-Tx power characterization to cover 1-Tx power targets, the recommended channels and order of testing is listed as follows to confirm passing power targets.

1. 802.11b: Test channels 1 and 11 (or test 1, 11, 12, 13 if Channel 12 and 13 are enabled)
2. 802.11n (20 MHz): Use this mode to cover 802.11g mode as well. Test channels 1, 2, 10, and 11 (or test 1, 2, 10, 11, 12, 13 if channels 12 and 13 are enabled)
3. 802.11n (40 MHz): Test channels 3, 4, 8, and 9 (or test 3, 4, 8, 9, 10, 11 if channels 10 and 11 are enabled)

The inner channels (that is, 2.4-GHz Channel 6) are not impacted by bandedges; therefore, the power is set to maximum that the radio chip supports. The same is true for the inner channels in the 5-GHz band, which will be transmitting at the maximum power but could potentially be limited by PSD.

For 5-GHz designs that support 2-Tx operation and will use 2-Tx power characterization to cover 1-Tx power targets, the recommended channels and order of testing is listed as follows to confirm passing power targets.

1. 802.11n (20 MHz): Use this mode to cover 802.11a and VHT20 as well. Test channels 36, 64, 100, 140, 149, and 165.
2. 802.11n (40 MHz): Use this mode to cover VHT40 as well. Test channels 38, 62, 102, 134, 151, and 159.
3. VHT80 (80 MHz): Test channels 42, 58, 106, 122, and 155.

Regulatory testing methods and requirements

4.3.4.2 FCC WL commands example for power characterization for 2.4 GHz 2-Tx

Command	Notes
wl pkteng_stop tx	Stops the radio from transmitting
wl out	Powers down the radio
wl down	Sets the radio to standby
wl frameburst 1	Allows to increase data output during continuous transmission
wl ampdu 1	Set to 1 to activate frame aggregation
wl country ALL	Sets the specific regional code (country code)
wl rateset 54b	Required to avoid the presence of active beacons during regulatory testing
wl bi 65000	Increases the beacon sequence transmit so that it does not interfere during test
wl phy_watchdog 0	Disables calibration tone
wl mpc 0	Disables the minimum power consumption
wl txchain 3	For SISO, use 1 or 2; for MIMO, use 3. This command activates 1tx or 2tx operation.
wl mimo_bw_cap 1	Enables 40-MHz and 80-MHz bandwidth operation
wl band b	Sets which operating band to active (b = 2.4 GHz only, a = 5 GHz only, Auto)
wl txbf 0	Set to 0 to disable TXBF mode.
wl chanspec -c 1 -b 2 -w 20 -s 0	Change -c 1 to -c 6 or -c 7 to change to channel 6 or 7. -c is channel, -b is operating band (set to 2 for 2.4 GHz or 5 for 5 GHz), -w is Bandwidth, -s is lower portion for 40 MHz (1 = enable or 0 = disable).
wl 2g_rate -r 6	Always use the lowest data rate for radio regulatory testing. Command to set up data rate.
wl up	Turns on radio
wl disassoc	Disconnects radio from any access point STA may be connected
wl phy_forcecal 1	Calibrate the radio before starting transmission to avoid a center frequency tone.
wl txpwr1 -o 19	Use this command to change the power in real time. Can also use the quarter step command wl txpwr1 -o -q XX, where XX is any value between 0 to 100.
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	Sets radio to continuous transmission

Regulatory testing methods and requirements

4.3.4.3 FCC WL commands example for power characterization for 5 GHz 2-Tx

Command	Notes
wl pkteng_stop tx	
wl out	
wl down	
wl frameburst 1	
wl ampdu 1	
wl country ALL	
wl rateset 54b	
wl bi 65000	
wl phy_watchdog 0	
wl mpc 0	
wl txchain 3	For MIMO use 3
wl mimo_bw_cap 1	
wl band a	
wl txbf 0	
wl chanspec -c 36 -b 5 -w 20 -s 0	Change -c 36 to -c 64 or -c 100 to change to channel 64 or 100.
wl 5g_rate -h 0 -b 20	Always use the lowest data rate for radio regulatory testing.
wl up	
wl disassoc	
wl phy_forcecal 1	
wl txpwr1 -o 19	Use this command to change power in real time. Can also use the quarter step command wl txpwr1 -o -q XX, where XX is any value between 0 to 100.
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	

Regulatory testing methods and requirements

4.3.5 Europe – ETSI

4.3.6 Setting initial regulatory power targets prior to testing

This process is completed to provide the EMC lab with actual operating powers for your device. To determine the wl command parameters (power levels) that the lab should use to set specific channel/data rate/band powers, the customers should follow this initial process.

1. Install/Program the Infineon Manufacturing FW
2. Use an average gated power meter
3. Use wl commands to set the radio to transmit continuously
4. Set the output power to the desired level using the wl txpwr1 command
5. Recall that the ETSI power limits are in EIRP. Measure the conducted power and make sure that both conducted + antenna gain (dBi) are below the maximum allowed EN300 328 or EN301 893 power limits
6. Record the average power level and the power settings and provide to the EMC lab

Note: The wl command examples to complete the process above are included at the end of this section.

4.3.7 Duty cycle

The recommended duty cycle that should be used by IEEE standard is listed as follows:

1. 802.11b = >98% duty cycle (Note: Using 1 Mbps data rate)
2. 802.11g = ~98% duty cycle (Note: Using 6 Mbps data rate)
3. 802.11n (20 MHz) = 95% duty cycle (Note: Using MCS0 data rate)
4. 802.11n (40 MHz) = 90% duty cycle (Note: Using MCS0 data rate)
5. VHT80 (80 MHz) = 75% duty cycle (Note: Using VHT0 SS1 data rate)

The above rates shall be used in conjunction with the following packet engine parameters:

```
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0
```

4.3.8 Test items to determine power target compliance CLM input

The recommended test items and sequence to capture the target powers that will be provided in the CLM power templates are listed as follows:

1. PSD
2. Power vs. temperature
3. Power output
4. Conducted harmonic emissions
5. Radiated harmonic emissions (50-Ω load)

Note: There are two options available for testing when considering multiple Tx chain configurations:

1. Individually maximized power for 1-Tx and then individually maximized power for 2-Tx.
2. Maximized power for 2-Tx only and use the 2-Tx powers for 1-Tx.

Regulatory testing methods and requirements

If using Option 1 for Radiated Harmonic testing, all channels/bandwidth for 2-Tx should be set to 1-Tx maximum powers. This will automatically cover 1-Tx modes because testing 2-Tx modes at maximum power is the worst-case scenario.

If using Option 2, then test all channels/bandwidth at the 2-Tx maximum powers.

4.3.8.1 Efficient channel/band testing and sequence

To maximize efficiency when testing channel and band combinations, use the following sequences when verifying power targets.

For 2.4 GHz, it is recommended to test 1-Tx configuration first, then test 2-Tx configurations. The recommended channels and order of testing are listed as follows to confirm passing power targets.

1. 802.11b: Test channels 1, 7, and 13
2. 802.11n (20 MHz): Use this mode to cover 802.11g mode. Test channels 1, 7, and 13
3. 802.11n (40 MHz): Test channels 3, 7, and 11

For 5 GHz, it is recommended to test 1-Tx configuration first, then test 2-Tx configurations. The recommended channels and order of testing are listed as follows to confirm passing power targets.

1. 802.11n (20 MHz): Use this mode to cover 802.11a and VHT20 as well. Test channels 36, 64, 100, 140, 149, 153, and 165.
2. 802.11n (40 MHz): Use this mode to cover VHT40 as well. Test channels 38, 62, 102, 134, 151, and 159.
3. VHT80 (80 MHz): Test channels 42, 58, 106, 122, and 155.

For 2.4-GHz designs that support 2-Tx operation and will use 2-Tx power characterization to cover 1-Tx power targets, the recommended channels and the order of testing are listed as follows to confirm passing power targets.

1. 802.11b: Test channels 1, 7, and 13.
2. 802.11n (20 MHz): Use this mode to cover 802.11g mode. Test channels 1, 7, and 13.
3. 802.11n (40 MHz): Test channels 3, 7, and 11.

For 5-GHz designs that support 2-Tx operation and will use 2-Tx power characterization to cover 1-Tx power targets, the recommended channels and the order of testing are listed as follows to confirm passing power targets.

1. 802.11n (20 MHz): Use this mode to cover 802.11a and VHT20 modes. Test channels 36, 64, 100, 140, 149, and 165.
2. 802.11n (40 MHz): Use this mode to cover VHT40 mode. Test channels 38, 62, 102, 134, 151, and 159.
3. VHT80 (80 MHz): Test channels 42, 58, 106, 122, and 155.

Regulatory testing methods and requirements

4.3.8.2 ETSI WL command example for power characterization for 2.4 GHz 2-Tx

Command	Notes
<code>wl pkteng_stop tx</code>	Stops the radio from transmitting
<code>wl out</code>	Powers down the radio
<code>wl down</code>	Sets the radio to standby
<code>wl frameburst 1</code>	This allows to increase data output during continuous transmission.
<code>wl ampdu 1</code>	Set to 1 to activate frame aggregation
<code>wl country ALL</code>	Sets the specific regional code (country code)
<code>wl rateset 54b</code>	Required to avoid the presence of active beacons during regulatory testing.
<code>wl bi 65000</code>	Increases the beacon sequence transmit so that it does not interfere during test
<code>wl phy_watchdog 0</code>	Disables calibration tone
<code>wl mpc 0</code>	Disables the minimum power consumption
<code>wl txchain 3</code>	For SISO, use 1 or 2; for MIMO, use 3. This command activates 1tx or 2tx operation.
<code>wl mimo_bw_cap 1</code>	Enables 40 MHz and 80 MHz bandwidth operation
<code>wl band b</code>	Sets which operating band to active (b = 2.4 GHz only, a = 5 GHz only, Auto)
<code>wl txbf 0</code>	Set to 0 to disable TXBF mode
<code>wl chanspec -c 1 -b 2 -w 20 -s 0</code>	Change -c 1 to -c 6 or -c 7 to change to channel 6 or 7. -c is channel, -b is operating band (set to 2 for 2.4 GHz or 5 for 5 GHz), -w is Bandwidth, -s is lower portion for 40 MHz (1 = enable or 0 = disable).
<code>wl 2g_rate -r 6</code>	Always use the lowest data rate for radio regulatory testing. Command to set up data rate.
<code>wl up</code>	Turns on radio
<code>wl disassoc</code>	Disconnects radio from any access point STA may be connected
<code>wl phy_forcecal 1</code>	Calibrate the radio before starting transmission to avoid a center frequency tone.
<code>wl txpwr1 -o 19</code>	Use this command to change the power in real time. Can also use the quarter step command <code>wl txpwr1 -o -q xx</code> , where XX is any value between 0 to 100.
<code>wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0</code>	Sets radio to continuous transmission

Regulatory testing methods and requirements

4.3.8.3 ETSI WL commands example for power characterization for 5 GHz 2-Tx

Command	Notes
wl pkteng_stop tx	
wl out	
wl down	
wl frameburst 1	
wl ampdu 1	
wl country ALL	
wl rateset 54b	
wl bi 65000	
wl phy_watchdog 0	
wl mpc 0	
wl txchain 3	For MIMO use 3
wl mimo_bw_cap 1	
wl band a	
wl txbf 0	
wl chanspec -c 36 -b 5 -w 20 -s 0	Change -c 36 to -c 64 or -c 100 to change to channel 64 or 100.
wl 5g_rate -h 0 -b 20	Always use lowest data rate for radio regulatory testing.
wl up	
wl disassoc	
wl phy_forcecal 1	
wl txpwr1 -o 19	Use this command to change power in real time. Can also use the quarter step command <code>wl txpwr1 -o -q XX</code> , where XX is any value between 0 to 100.
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	

4.3.9 Japan/South Korea

4.3.10 Setting initial regulatory power targets prior to testing

This process is completed to provide the EMC lab with actual operating powers for your device. To determine the wl command parameters that the lab should use to set specific channel/data rate/band powers, you should follow this initial process.

1. Install/Program the Infineon Manufacturing FW.
2. Use an average gated power meter.
3. Use wl commands to set the radio to transmit continuously.
4. Set the output power to the desired level using the `wl txpwr1` command.
5. Recall that the Japan/South Korea are PSD limit in EIRP. There are no power limits. Measure the PSD values in a spectrum analyzer.
6. Measure the RF output power with an average power meter.

Regulatory testing methods and requirements

7. Record the average power level and the power settings and provide to the EMC lab.

Note: The wl command examples to complete the process above are included at the end of this section.

4.3.11 Duty cycle

The recommended duty cycle that should be used by IEEE standard is listed as follows:

1. 802.11b = >98% duty cycle (**Note:** Using 1 Mbps data rate)
2. 802.11g = ~98% duty cycle (**Note:** Using 6 Mbps data rate)
3. 802.11n (20 MHz) = 95% duty cycle (**Note:** Using MCS0 data rate)
4. 802.11n (40 MHz) = 90% duty cycle (**Note:** Using MCS0 data rate)
5. VHT80 (80 MHz) = 75% duty cycle (**Note:** Using VHT0 SS1 data rate)

The above rates shall be used in conjunction with the following packet engine parameters:

```
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0
```

4.3.12 Test items to determine power target compliance CLM input

The recommended test items and sequence to capture the target powers that will be provided in the CLM power templates are as follows:

1. PSD
2. Power Vs Temperature
3. Conducted harmonic emissions

Note: There are two options available for testing when considering multiple Tx chain configurations:

1. Individually maximized power for 1-Tx followed by individually maximized power for 2-Tx
2. Maximized power for 2-Tx only and use the 2-Tx powers for 1-Tx

If using Option 1 for conducted harmonic testing, all channels/bandwidth for 2-Tx should be set to 1-Tx maximum power. This will automatically cover 1-Tx modes because testing 2-Tx modes at maximum power is the worst-case scenario.

If using Option 2, then test all channels/bandwidth at the 2-Tx maximum powers.

4.3.12.1 Efficient channel/band testing and sequence

To maximize efficiency when testing channel and band combinations, it is recommended to follow the sequences below when verifying power targets.

For 2.4 GHz, it is recommended to test 1-Tx configuration first, then test 2-Tx configurations. The recommended channels and the order of testing are listed as follows to confirm passing power targets.

1. 802.11b: Test channels 1, 7, and 13.
2. 802.11n (20 MHz): Use this mode to cover 802.11g mode as well. Test channels 1, 7, and 13.
3. 802.11n (40 MHz): Test channels 3, 7, and 11.

For 5 GHz, it is recommended to test 1-Tx configuration first, then test 2-Tx configurations. The recommended channels and the order of testing are listed as follows to confirm passing power targets.

Regulatory testing methods and requirements

- 802.11n (20 MHz): Use this mode to cover 802.11a and VHT20. For Japan, test channels 36, 40, 48, 52, 6064, 100, 120, and 140; for Korea, test channels 36, 40, 48, 52, 60, 64, 100, 120, 140, 149, 153, and 165.
- 802.11n (40 MHz): Use this mode to cover VHT40 mode. For Japan, test channels 38, 46, 54, 62, 102, 118, and 134; for South Korea, test channels 38, 46, 54, 62, 102, 118, 134, 151, and 159.
- VHT80 (80 MHz): For Japan, test channels 42, 58, 106, and 122; for South Korea, test channels 42, 58, 106, 122, and 155.

For 2.4-GHz designs that support 2-Tx operation and will use 2-Tx power characterization to cover 1-Tx power targets, the recommended channels and the order of testing are listed as follows to confirm passing power targets.

- 802.11b: Test channels 1, 7, and 13.
- 802.11n (20 MHz): Use this mode to cover 802.11g mode. Test channels 1, 7, and 13.
- 802.11n (40 MHz): Test channels 3, 7, and 11.

For 5-GHz designs that support 2-Tx operation and will use 2-Tx power characterization to cover 1-Tx power targets, the recommended channels and the order of testing are listed as follows to confirm passing power targets.

- 802.11n (20 MHz): Use this mode to cover 802.11a and VHT20 modes. For Japan, test channels 36, 40, 48, 52, 60, 64, 100, 120, and 140; for Korea, test channels 36, 40, 48, 52, 6064, 100, 120, 140, 149, 153, and 165.
- 802.11n (40 MHz): Use this mode to cover VHT40 mode. Test channels 38, 46, 54, 62, 102, 118, 134, 151, and 159.
- VHT80 (80 MHz): Test channels 42, 58, 106, 122, and 155.

4.3.12.2 Japan/Korea WL command example for power characterization for 2.4-GHz 2-Tx

Command	Notes
<code>wl pkteng_stop tx</code>	Stops the radio from transmitting
<code>wl out</code>	Powers down the radio
<code>wl down</code>	Sets the radio to standby
<code>wl frameburst 1</code>	This allows to increase data output during continuous transmission.
<code>wl ampdu 1</code>	Set to 1 to activate frame aggregation.
<code>wl country ALL</code>	Sets the specific regional code (country code)
<code>wl rateset 54b</code>	Required to avoid the presence of active beacons during regulatory testing.
<code>wl bi 65000</code>	Increases the beacon sequence transmit so that it does not interfere during test
<code>wl phy_watchdog 0</code>	Disables calibration tone
<code>wl mpc 0</code>	Disables the minimum power consumption
<code>wl txchain 3</code>	For SISO, use 1 or 2; for MIMO, use 3. This command activates 1tx or 2tx operation.
<code>wl mimo_bw_cap 1</code>	Enables 40-MHz and 80-MHz bandwidth operation
<code>wl band b</code>	Sets which operating band to active (b = 2.4 GHz only, a = 5 GHz only, Auto)

Regulatory testing methods and requirements

Command	Notes
wl txbf 0	Set to 0 to disable TXBF mode.
wl chanspec -c 1 -b 2 -w 20 -s 0	Change -c 1 to -c 6 or -c 7 to change to channel 6 or 7. -c is channel, -b is operating band (set to 2 for 2.4 GHz or 5 for 5 GHz), -w is Bandwidth, -s is lower portion for 40 MHz (1 = enable or 0 = disable).
wl 2g_rate -r 6	Always use the lowest data rate for radio regulatory testing. Command to set up data rate.
wl up	Turns on radio
wl disassoc	Disconnects radio from any access point STA may be connected
wl phy_forcecal 1	Calibrate the radio before starting transmission to avoid a center frequency tone.
wl txpwr1 -o 19	Use this command to change the power in real time. Can also use the quarter step command wl txpwr1 -o -q XX, where XX is any value between 0 to 100.
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	Sets radio to continuous transmission

4.3.12.3 Japan/Korea WL commands example for power characterization for 5-GHz 2-Tx

Command	Notes
wl pkteng_stop tx	
wl out	
wl down	
wl frameburst 1	
wl ampdu 1	
wl country ALL	
wl rateset 54b	
wl bi 65000	
wl phy_watchdog 0	
wl mpc 0	
wl txchain 3	For MIMO use 3
wl mimo_bw_cap 1	
wl band a	
wl txbf 0	
wl chanspec -c 36 -b 5 -w 20 -s 0	Change -c 36 to -c 64 or -c 100 to change to channel 64 or 100.
wl 5g_rate -h 0 -b 20	Always use lowest data rate for radio regulatory testing.

Regulatory testing methods and requirements

Command	Notes
wl up	
wl disassoc	
wl phy_forcecal 1	
wl txpwr1 -o 19	Use this command to change power in real time. Can also use the quarter step command <code>wl txpwr1 -o -q XX</code> , where XX is any value between 0 to 100.
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0	

4.3.13 China – SRRC

4.3.14 Setting initial regulatory power targets prior to testing

This process is completed to provide the EMC lab with actual operating powers for your device. To determine the `wl` command parameters that the lab should use to set specific channel/data rate/band powers, you should follow this initial process.

1. Install/Program the Infineon Manufacturing FW.
2. Use an average gated power meter.
3. Use `wl` commands to set the radio to transmit continuously.
4. Set the output power to the desired level using the `wl txpwr1` command.
5. Measure the RF output power with an average power meter.
6. Record the average power level and the power settings and provide to the EMC lab.

4.3.15 Duty cycle

The recommended duty cycle that should be used by IEEE standard is as follows:

1. 802.11b = >98% duty cycle (Note: Using 1 Mbps data rate)
2. 802.11g = ~98% duty cycle (Note: Using 6 Mbps data rate)
3. 802.11n (20 MHz) = 95% duty cycle (Note: Using MCS0 data rate)
4. 802.11n (40 MHz) = 90% duty cycle (Note: Using MCS0 data rate)
5. VHT80 (80 MHz) = 75% duty cycle (Note: Using VHT0 SS1 data rate)

The above rates shall be used in conjunction with the following packet engine parameters:

```
wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0
```

Regulatory testing methods and requirements

4.3.16 Test items to determine power target compliance CLM input

The recommended test items and sequence to capture the target powers that will be provided in the CLM power templates are as follows:

1. PSD
2. Frequency range
3. Power limit
4. Conducted harmonic emissions

Note: There are two options available for testing when considering multiple Tx chain configurations:

1. *Individually maximized power for 1-Tx and then individually maximized power for 2-Tx*
2. *Maximized power for 2-Tx only and use the 2-Tx powers for 1-Tx*

If using Option 1 for Conducted Harmonic testing, all channels/bandwidth for 2-Tx should be set to 1-Tx maximum powers. This will automatically cover 1-Tx modes because testing 2-Tx modes at maximum power is the worst-case scenario.

If using Option 2, then test all channels/bandwidth at the 2-Tx maximum powers.

4.3.16.1 Efficient channel/band testing and sequence

To maximize efficiency when testing channel and band combinations, use the following sequences when verifying power targets.

For 2.4 GHz, it is recommended to test 1-Tx configuration first, then test 2-Tx configurations. The recommended channels and the order of testing are listed as follows to confirm passing power targets.

1. 802.11b: Test channels 1, 7, and 13.
2. 802.11n (20 MHz): Use this mode to cover 802.11g mode. Test channels 1, 7, and 13.
3. 802.11n (40 MHz): Test channels 3, 7, and 11.

For 5 GHz, it is recommended to test 1-Tx configuration first, then test 2-Tx configurations. The recommended channels and the order of testing are listed as follows to confirm passing power targets.

1. 802.11n (20 MHz): Use this mode to cover 802.11a and VHT20 modes. Test channels 36, 40, 48, 52, 60, 64, 149, 153, and 165.
2. 802.11n (40 MHz): Use this mode to cover VHT40 mode. Test channels 38, 46, 54, 62, 151, and 159.
3. VHT80 (80 MHz): Test channels 42, 58, and 155.

For 2.4-GHz designs that support 2-Tx operation and will use 2-Tx power characterization to cover 1-Tx power targets, the recommended channels and the order of testing are listed as follows to confirm passing power targets.

1. 802.11b: Test channels 1, 7, and 13.
2. 802.11n (20 MHz): Use this mode to cover 802.11g mode. Test channels 1, 7, and 13.
3. 802.11n (40 MHz): Test channels 3, 7, and 11.

For 5 GHz designs that support 2-Tx operation and will use 2-Tx power characterization to cover 1-Tx power targets, the recommended channels and the order of testing are listed as follows to confirm passing power targets.

Regulatory testing methods and requirements

1. 802.11n (20 MHz): Use this mode to cover 802.11a and VHT20 modes. Test channels 36, 40, 48, 52, 60, 64, 149, 153, and 165.
2. 802.11n (40 MHz): Use this mode to cover VHT40 mode. Test channels 38, 46, 54, 62, 151, and 159.
3. VHT80 (80 MHz): Test channels 42, 58, and 155.

4.3.16.2 China WL command example for power characterization for 2.4-GHz 2-Tx

Command	Notes
wl pkteng_stop tx	Stops the radio from transmitting
wl out	Powers down the radio
wl down	Sets the radio to standby
wl frameburst 1	This allows to increase data output during continuous transmission.
wl ampdu 1	Set to 1 to activate frame aggregation
wl country ALL	Sets the specific regional code (country code)
wl rateset 54b	Required to avoid the presence of active beacons during regulatory testing
wl bi 65000	Increases the beacon sequence transmit so that it does not interfere during test
wl phy_watchdog 0	Disables calibration tone
wl mpc 0	Disables the minimum power consumption
wl txchain 3	For SISO, use 1 or 2; for MIMO, use 3. This command activates 1tx or 2tx operation.
wl mimo_bw_cap 1	Enables 40 MHz and 80 MHz bandwidth operation
wl band b	Sets which operating band to active (b = 2.4 GHz only, a = 5 GHz only, Auto)
wl txbf 0	Set to 0 to disable TXBF mode.
wl chanspec -c 1 -b 2 -w 20 -s 0	Change -c 1 to -c 6 or -c 7 to change to channel 6 or 7. -c is channel, -b is operating band (set to 2 for 2.4 GHz or 5 for 5 GHz), -w is Bandwidth, -s is lower portion for 40 MHz (1 = enable or 0 = disable).
wl 2g_rate -r 6	Always use the lowest data rate for radio regulatory testing. Command to set up data rate.
wl up	Turns on radio
wl disassoc	Disconnects radio from any access point STA may be connected
wl phy_forcecal 1	Calibrate the radio before starting transmission to avoid a center frequency tone

Regulatory testing methods and requirements

Command	Notes
<code>wl txpwr1 -o 19</code>	Use this command to change the power in real time. Can also use the quarter step command <code>wl txpwr1 -o -q XX</code> , where XX is any value between 0 to 100.
<code>wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0</code>	Sets radio to continuous transmission

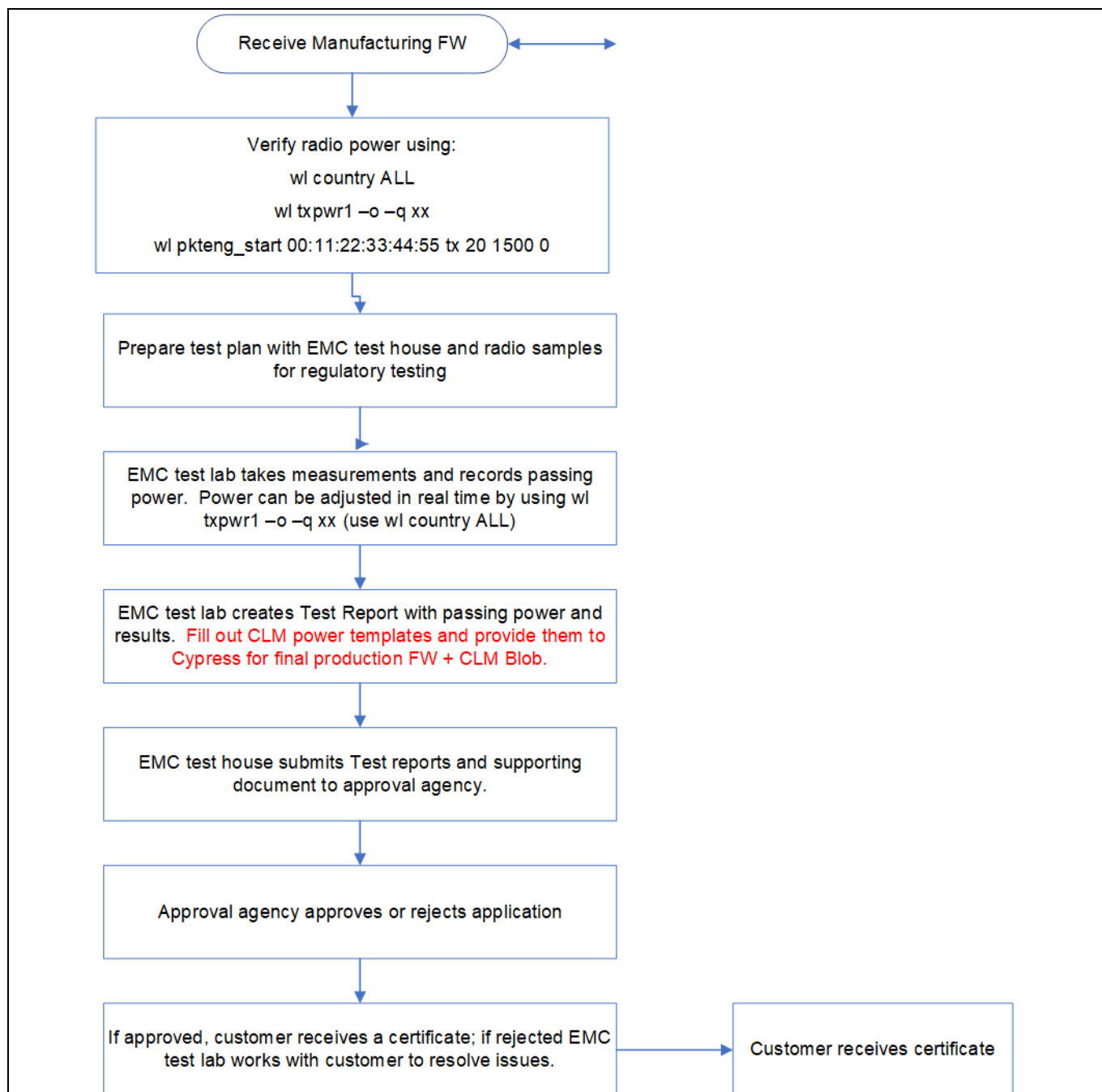
4.3.16.3 China WL commands example for power characterization for 5-GHz 2-Tx

Command	Notes
<code>wl pkteng_stop tx</code>	
<code>wl out</code>	
<code>wl down</code>	
<code>wl frameburst 1</code>	
<code>wl ampdu 1</code>	
<code>wl country ALL</code>	
<code>wl rateset 54b</code>	
<code>wl bi 65000</code>	
<code>wl phy_watchdog 0</code>	
<code>wl mpc 0</code>	
<code>wl txchain 3</code>	For MIMO use 3
<code>wl mimo_bw_cap 1</code>	
<code>wl band a</code>	
<code>wl txbf 0</code>	
<code>wl chanspec -c 36 -b 5 -w 20 -s 0</code>	Change -c 36 to -c 64 or -c 100 to change to channel 64 or 100
<code>wl 5g_rate -h 0 -b 20</code>	Always use the lowest data rate for radio regulatory testing.
<code>wl up</code>	
<code>wl disassoc</code>	
<code>wl phy_forcecal 1</code>	
<code>wl txpwr1 -o 19</code>	Use this command to change power in real time. Can also use the quarter step command <code>wl txpwr1 -o -q XX</code> , where XX is any value between 0 to 100.
<code>wl pkteng_start 00:11:22:33:44:55 tx 20 1500 0</code>	

Regulatory testing methods and requirements

4.4 Power characterization and regulatory testing overall flow

The overall flow for the details provided in sections 4.1, 4.2, and 4.3 is shown as follows:



Specific CLM terms and parameters

5 Specific CLM terms and parameters

This section defines terms and parameters associated with the Infineon CLM process. Terms outlined in this section may be used during the Country Code request process, so it is important to understand their meaning.

- **CLM:** An Infineon acronym that stands for Country Locale Matrix. This term describes the database that represents the regulatory configuration for a project or a group of projects. This database can be represented in a Per Product or Master Database form.
- **Per product CLM:** A database representing the regulatory configuration for a single product and customer combination. This is the default and preferred implementation when completing the Country Code request process. All Country Codes within a Per Product CLM have a Regulatory Revision (Regrev) of “/0”.
- **Master CLM:** A database of legacy Country Code. The Master CLM database contains thousands of previous country codes generated for previous and current Infineon products. The Master CLM database is the old method of completing Country Code requests and is not used unless the chip set in question cannot support Per Product CLM releases.
- **ISO country code:** IEEE-defined and maintained two letter code (for example, “US”) that represents a physical country. An ISO country code is the default two letter code used for new requests that represent only a single country.
- **Country code:** Two-letter code that may or may not represent an actual country. For example, a Country Code requests that include multiple countries (for example, combining all European countries into a single code) in a single code could elect to use the ISO country code from one of the included countries (for example, “FR”) or it could elect to use a non-ISO code (for example, “XZ”). Either option is acceptable.
- **Regrev:** Regulatory Revision. A Regrev is the numeric identifier included after the country code and a forward slash. For example, “US/0” includes an ISO country code of “US”, a separator of “/” and a Regrev of “0”. In Per Product CLM releases, every Regrev is equal to “0”, regardless of how many countries are included in the Per Product CLM release.
- **Region:** The combination of a Country Code and a Regrev. “US/0” is an example of a region. Similarly, “XZ/0” is also a region. A region can define a single target country or a group of countries.
- **Pseudo-Country:** A Region that represents a collection of actual countries. Typically, a Pseudo-Country is used when multiple countries have similar or the same regulatory requirements. For example, European countries that follow the ETSI standard.
- **Locale:** An internal Infineon term that represents the regulatory configuration for a specific band and bandwidth (for example, 2.4-GHz legacy standards 802.11b and g have a locale; 5-GHz legacy 802.11a have a different locale; 802.11n standards have a locale for 2.4 GHz and one for 5 GHz; 802.11ac shares the same locale as 802.11n for 5 GHz).

Locales are used in both Per Product or Master CLM databases, which allow for country code to support different configurations for the different Wi-Fi standards. Locale names and definitions are handled by Infineon and are not required to have customer input.

- **DFS:** Dynamic Frequency Selection. The term DFS will be associated with 5-GHz channels 52-64 and channels 100-144 (actual channels requiring DFS may vary by country but would be contained in this superset).

There are two DFS implementations generally adopted world-wide, one defined by the FCC, and the other defined by CE/ETSI. If a customer has requested to enable DFS channels in 5 GHz, Infineon will apply the appropriate DFS operating standard based on the country or countries requested.

The concept of DFS is in place to decrease interference of a Wireless system with government or military communications in the 5-GHz spectrum.

Specific CLM terms and parameters

- TPC:** Transmit power control. TPC is a feature of 802.11h. TPC is used by APs supporting this feature to command STAs to decrease their output power level by an amount specified by the AP.
 This function is intended to further reduce potential interference with government or military communications. If TPC is supported, it also applies generally to channels 52-64 and channels 100-144 in the 5 GHz spectrum. Each country may vary in the required channels or if TPC is required at all.
- Adaptivity:** A Wi-Fi mechanism to reduce communication on channels that a STA or AP is already using. This mechanism is used to promote increased bandwidths and fewer collisions between wireless devices. The Wi-Fi standard (IEEE) supports adaptivity by default, which is the concept of not communicating over another device already using a channel. However, the term ‘adaptivity’ when used in the context of Regulatory Compliance testing is associated with the ETSI testing standard.
 The concept of adaptivity with respect to ETSI is the same as IEEE, but testing parameters and requirements are tightened to promote more clean air transmission and receiving. If a Country Code request includes a country that has adopted the CE/ETSI requirements, Infineon will apply a flag in your CLM release that will enable ETSI adaptivity rules.
- Subchannels:** Describes the channels involved in channel bonding described in the [Wi-Fi basics](#) section of this document.
- For example, Channel 38 (5-GHz Band, 40-MHz Bandwidth) makes use of subchannels 36 and 40 in the 5-GHz band, 20-MHz bandwidth. Subchannels are defined in the IEEE Wi-Fi requirements if more information is needed.
- Restricted channel:** A channel that has been requested to only actively communicate after it has detected an AP that is communicating on this specific channel. Until the Wi-Fi enabled device detects this channel in use, it will not actively communicate on this channel. Restricted channels should only be used in pseudo-country configurations.
- 802.11d:** A standard that is part of the 802.11 specification created to support information sharing to Wi-Fi systems. 802.11d can be used to determine the location (at a country level) that a STA or AP is currently in. 802.11d information includes (but is not limited to): Country Code, Maximum Output Power supported by Channel Range, among other information. 802.11d operation (for APs) and listening (for STAs/APs) is configured at the hardware/firmware level.
 Infineon CLM comes into the picture by including the maximum output powers available by channel range. This information does not require customer input, as the values are set according to the required maximum power levels for each country.
- TX beamforming:** A method and feature of 802.11 that effectively allows APs to concentrate a transmitted signal toward the location where the STA is located. This capability allows for increased signal (and subsequent signal to noise ratio), which allows for greater range and/or higher throughput. TX Beamforming requires the use of at least two antennas (two TX Chains) to operate.
- TX chains:** Refers to the number of antennas/radiators present on a Wi-Fi solution. A Wi-Fi system that has two antennas could also be referred to as having 2 TX chains. ‘TX chains’ is a term used in the Infineon power target templates to denote operating modes and data rates where multiple antennas are being used.

Understanding these terms will allow you to progress through the CLM process. If you encounter questions or terms from Infineon regulatory or applications engineering that you don’t understand, have that term defined to ensure that you know what is being asked and that your answer is correct.

SKU planning guidance

6 SKU planning guidance

This section provides guidance to customers on the concept of Stock Keeping Units (SKU) planning related to Wi-Fi system designs and release. SKU planning is not a direct part of the Infineon Regulatory CLM process but is important to consider early on in your product development.

SKU planning is an essential aspect of your product's regulatory strategy. Within the process of SKU planning, you will determine how many individual SKUs are required to produce your product and ship it to the countries you have identified.

SKU planning is required for Wi-Fi solutions because of the many differences that exist country-to-country. Before deciding how many different SKUs you will be manufacturing, the first critical questions that must be answered are the following:

1. How many countries are you planning to ship your product to?
 - a) Do you know which specific countries these are?
2. Of the countries you have identified, do any of them have restrictions that make it difficult to produce only a single SKU (for example, US Ch. 12/13 regulations)
3. Of the countries you have identified, do you want to group them into pseudo-countries, or treat each independently?
4. Are you planning on using a form of geo-location to determine the country your product is in and set the country code based on this?
 - b) If yes, what method of geo-location will be used? 802.11d, GPS, cellular?

The answers to the questions above will lead you to a determination on how many different FW images (with differing CLM configurations) you need to maintain in your production line. Of the questions above, the most critical is question #2, which can limit the flexibility of your SKU strategy (more on this in the next section).

The process of SKU planning is completed to determine precisely how many of your target countries you will/can combine into a single SKU. Note that these countries can be combined in a pseudo-region or represented by an individual ISO country code in a single CLM file.

6.1 Different methods of SKU management

The downside of SKU planning is when you inevitably encounter a country that will not allow certain configurations that you want to have enabled in the other countries you ship to (for example, Channel 12/13 in US). In most cases like this, the only compliant method is to ship a specific SKU number to that country that is different than the rest. The goal with SKU planning for Wi-Fi products is to minimize the number of outliers you have in the target ship-to locations.

There are several methods that can be used to determine what CLM configuration your product should be operating under. The following are the common methods for managing wireless product SKUs:

- **Static SKU method:** Each target country is programmed with a CLM configuration that is specific to that country. The benefit of this approach is that it is simple, and for products with limited target ship-to countries, requires little planning. The downside to this approach is the need for a unique SKU and firmware image for every target ship-to country.
- **Dynamic SKU method:** In this method, a FW image will contain multiple target ship-to countries and their regulatory configuration (CLM) vis-a-vis an ISO country code. In this method, the product will determine its location (in terms of country) through mechanisms available in 802.11d or other means (GPS, cellular), and will set the operating country within its firmware accordingly. Note that Infineon does not support auto-country but does support the building blocks required for a customer to implement such a feature (that is,

SKU planning guidance

802.11d, multiple CLM configurations). In this method, the SKU would be required to have a default country code that would be used in the scenario where the product is in a country that has not been defined for this product.

- **Single-SKU method:** In this method, the product would have only a single country-code (or pseudo country code) that would apply for all target ship-to countries. This country code would be configured to have regulatory power limits and channel selection be equal to the union of requirements for all target countries.

Generally, a combination of methods from above are used to release a product to manufacturing. Note that a Static SKU method requires that the unit be programmed with the target firmware image once you know the country that it will be shipped to. In general, the Dynamic and Single SKU methods do not require knowing the final location prior to shipment.

6.2 Common problems with SKU planning

One might ask “Can’t I just query the surrounding APs to determine what country I am in and then set the country according to this?”. The answer is dependent on the countries that you are supporting and the CLM configuration that you have set.

This section details some of the major hurdles that you may face when defining your SKU strategy:

1. **US/FCC channel 12/13 restrictions:** This, more than any other, requirement leads to a multiple SKU approach. The requirements stated by the FCC for channels 12/13 is that if these channels are supported, they must follow the requirements below:
 - a) The product complies to the required output power, bandedge, and harmonics requirements (Note that this typical results in an output power than is near zero or less).
 - b) At no point and in no way can the product be operating at output power levels on channel 12/13. It means that any country codes that you have combined with this SKU would need to also have FCC-compliant power targets for these channels, which would be significantly lower than what you would normally have.
2. **DFS standards:** The DFS standard or operation should be considered when looking to complete a single-SKU method by combining multiple countries into a single pseudo-country code. In general, countries with different DFS standards (US vs. Europe) should not be combined together if DFS channels are supported.
3. **Weather band:** Knowing whether your target ship-to country allows for communication on the weather band (TDWR – channels 120-128 in 5 GHz) may be a deciding factor to confirm if you need an additional SKU for that country. For example, Canada does not allow transmission on weather band channels. However, the US does allow for communication. At a minimum, a separate distinct country code should be assigned for Canada that disables communication on these channels.
4. **Adaptivity:** Adaptivity testing is required for ETSI/CE (Europe). It is acceptable to group countries that do not require adaptivity testing together with those that do, but not the other way around. If you group a European country (for example, Italy) together with a set of countries that do not require adaptivity and you do not enable this feature, you cannot pass testing for Italy. Take note, however, that Infineon would not allow this to occur, and in this situation, all of the countries would support Adaptivity regardless of if it was requested.
5. **How low can you go?:** When considering a Single-SKU strategy, it is important to understand how much of a reduction your output or channels will take. For example, you may want to combine all Southeast Asian countries together to simplify your SKU planning; this seems logical. However, when you realize that Indonesia does not allow for any 40-MHz or 80-MHz operation, and only allows for you to operate on channels 149-161 in the 5-GHz spectrum, you may want to change your approach.

There are many pitfalls to regulatory SKU planning, but the trick is to begin the planning early in the development cycle. Infineon will be able to provide insights and details to help you reach your final decision.

SKU planning guidance

6.3 Minimum country set for world-wide support

To support “world-wide” shipments, you should target at least three SKUs/regions. The minimum set of required country codes/SKUs needed is provided below, assuming that you maximize the channel availability within each of the regions.

1. Support for one US related SKU (including Canada, US, and US islands). If support for channels 12/13 is not a feature you care about, this region could be combined with others. However, support for channels 12/13 in Europe is typically requested.
2. Support for Europe (CE/ETSI) countries using a single Country Code. European nations follow the same requirements and restrictions, and therefore naturally lends themselves to be combined and simplifying your SKU strategy.
3. Support for ROW (Rest of World): In this region, the union of all power targets and channel availability would be used.

As mentioned earlier in this section, determining exactly which countries you will be shipping to will help to decrease the amount of work required in your SKU planning. If you embark on attempting to support every country in the world, this will increase the amount of work required on your part to plan your product. In practice, there are very few instances where a product needs world-wide coverage. Most products have defined market targets, which include focus/target countries.

6.4 SKU planning summary

To summarize, SKU planning is an essential part of creating a design that can be efficiently released to your manufacturing. The key aspects for each target country that need to be considered to appropriately plan are as follows:

1. Are there any restrictions that my target countries have?
 - a) Are these restrictions strict, or can I combine with other countries via passive operation?
2. Are my country groupings alike in terms of DFS and adaptivity standards?
3. Are the resulting output powers and channel availability acceptable for my product when grouping specific countries together?
 - a) If not, then separate specific countries that overly restrict the configuration in your grouping.

At the core, SKU planning is a grouping and combinatorial problem that can be easily solved with the right amount of information and thought put behind it. Infineon is here to help with your questions, but the decisions are up to you for your product.

7 How to work with Infineon CLM templates

As mentioned previously in this document, to release a CLM blob for your specific project, Infineon needs you to provide target output powers for each of the bands, data rates, and channels that you would like to support. To ensure consistency among different customers, Infineon provides templates for communicating the power targets for your project. The templates are available to download as part of this application note; this section will navigate you through the templates and the inputs that are required to start the process.

Infineon provides six templates to be used depending on the countries that are requested to be supported. The templates and the countries that they represent are stated as follows:

- *FCC Canada Taiwan Regulatory Power Template.xlsm* – This template is to be used if the countries requested are either US, Canada, or Taiwan
- *CE Regulatory Power Template.xlsm* – This template is to be used if the countries requested follow the CE (ETSI) standards
- *China Regulatory Power Template.xlsm* – This template is to be used if China is a desired country to be supported
- *South Korea Regulatory Power Template.xlsm* – This template is to be used if South Korea is a desired country to be supported
- *Japan Regulatory Power Template.xlsm* – This template is to be used if Japan is a desired country to be supported
- *Other Countries Regulatory Power Template.xlsm* – This template is to be used if Other countries are requested that are not covered by the five templates above

The templates provided are based on the frequency of requests. The underlying task in each of the templates is to provide the target outputs powers by channel for your solution. The difference between each template is that Infineon has provided country-specific channel restrictions for the most frequently requested countries.

The “Other Countries” template is provided for any countries not covered by the country-specific templates. In the case of this template, there are no channel restrictions note. Infineon will review request and inform customers if they have requested channels that are disallowed in a specific country, but this is not communicated specifically in the template.

The templates provided are macro-supported Microsoft Excel files that will dynamically decrease or increase the number of required inputs based on the information provided by the customer. Infineon also provides a non-macro version for customers who do not have Microsoft Excel versions that support the macros used. If you have an Excel version of 2016 or newer, it is recommended to use the macro-enabled version so that only data pertinent to your project is requested and required.

7.1 Template contents

The following sections guide you through the different tabs in each template and provide details on what customer input is required.

7.1.1 Project information and antenna gain worksheets

7.1.1.1 Project information worksheet

The Project Information Worksheet consists of a series of questions. Each question in this worksheet is required. The answer will help to decrease the amount of additional questions and/or inputs that are required by the customer to complete a CLM request.

How to work with Infineon CLM templates

7.1.1.2 Antenna gain worksheet

The Antenna Gain worksheet is used to input the antenna gains for your system. This worksheet provides the composite gains for 2-, 3-, and 4-Tx implementations. This worksheet is also used in non-US/Canada/Taiwan requests to provide a calculated power target based on the regulatory requirements for each country.

7.1.2 802.11 Wi-Fi power target input worksheets

This section details the Power Target input worksheets. These worksheets are the location where your target not-to-exceed output powers are provided.

7.1.2.1 Required inputs

Within each of the Power Target Input worksheets, there are three columns that require your input:

1. The column titled “Target Power” is the input column where the not-to-exceed output power is provided. The units for this input column are in the nearest 0.25 dBm. These cells are free form but should have two significant digit numbers rounded to the nearest 0.25.

Note: If a specific channel is requested to be disabled, then an entry of “NS” should be made. “NS” stands for “Not Supported”.

2. The column titled “Measure” indicates how the measurement method that the target output power should be considered. The cells in this column are selected from a drop-down list with two choices: “Conducted” or “EIRP”.
 - If selecting “Conducted”, the effective output power on the antenna pin of the Infineon Wi-Fi chipset will be targeted at the value provided in the “Target Power” column.
 - If selecting “EIRP (Radiated)”, the effective output power on the antenna pin of the Infineon Wi-Fi chipset will be equal to that calculated based on the following equation:

$$\text{Output power} = [(\text{Target Power}) - (\text{Antenna Gain NVRAM Setting})]$$
 This equation states that the value provided in the Target Power column will be decremented by the value of the antenna gain programmed into NVRAM.
3. The column titled “Active/Restricted?” indicates whether you would like the channel to actively transmit and receive, or whether you would like it to operate on a restricted basis (that is, it will not transmit on this channel until it detects an Access Point using this channel). Note that the term “Restricted” is similar to “Passive” operation for DFS channels, but the term “Restricted” is used to ensure that there is no confusion between DFS channel operation (requirement) versus Restricted channel operation (which is customer preference). Notes on this column are listed as follows:
 - The default value for this field is “Active” for all channels except for DFS channels. Restricted operation is not recommended unless a pseudo-region is being requested, where specific channels are not allowed in specific countries in the pseudo-region but are allowed in others. Note that this method is not sufficient in some countries (for example, U.S. Channel 12 and 13 restrictions cannot be worked around by making these channels Restricted).
 - DFS channels do not provide an option to operate this actively. Infineon will not complete the production CLM requests to operate DFS channel as Active.
 - The “Active/Restricted?” input column is only present in the “802.11b & g” and “802.11a” worksheets:
 - The inputs in the 802.11b and g worksheet are inherited in all other 2.4-GHz worksheets.
 - The inputs in the 802.11b and g worksheet are inherited in all other 2.4-GHz worksheets.

How to work with Infineon CLM templates

- The inputs in the 802.11a worksheet are inherited in all other 5-GHz worksheets.
- Infineon cannot customize Active/Restricted operation for different bandwidths of a desired frequency band. The selection made on these two worksheets is inherited for each channel in all bandwidths and data rates.

7.1.2.2 802.11 input worksheet descriptions

Each Power Target input worksheet has a clear title; each of the Power Target input worksheets are described in the following sections.

“802.11b & g” worksheet

This worksheet consists of the inputs required for 802.11b and 802.11g standards. It includes 2.4-GHz channels for DSSS (802.11b) and OFDM (802.11g). The required inputs on this worksheet are as follows: “Target Power”, “Measurement”, and “Active/Restricted?” inputs.

Note: Active/Restricted selections in this worksheet will be inherited by all 2.4-GHz channels.

“802.11a” worksheet

This worksheet consists of the inputs required for the 802.11a standard. It includes 5-GHz channels for the OFDM modulation scheme (802.11a). The required inputs on this worksheet are as follows: “Target Power”, “Measurement”, and “Active/Restricted?” inputs.

Note: Active/Restricted selections in this worksheet will be inherited by all 5-GHz channels.

“802.11n 2.4 GHz, 20 MHz” worksheet

This worksheet consists of the inputs required for the 802.11n standard for 2.4-GHz channels utilizing the 20-MHz bandwidth. It also includes required inputs field if 802.11b MIMO operation is supported. This worksheet includes inputs for 1Tx through 4Tx implementations. Only the required inputs cells will be shown if using the macro-enabled version of the template and answering all questions on the “Project Information” worksheet. Required inputs on this worksheet are “Target Power”, and “Measurement”.

“802.11n, ac 5 GHz, 20 MHz” worksheet

This worksheet consists of the inputs required for the 802.11n/ac standard for 5-GHz channels utilizing the 20-MHz bandwidth. This worksheet includes inputs for 1Tx through 4Tx implementations. Only the required inputs cells will be shown if using the macro-enabled version of the template and answering all questions on the “Project Information” worksheet. The required inputs on this worksheet are “Target Power”, and “Measurement”.

“802.11n 2.4 GHz, 40 MHz” worksheet

This worksheet consists of the inputs required for the 802.11n standard for 2.4-GHz channels utilizing the 40-MHz bandwidth. This worksheet includes inputs for 1Tx through 4Tx implementations. Only the required inputs cells will be shown if using the macro-enabled version of the template and answering all questions on the “Project Information” worksheet. The required inputs on this worksheet are “Target Power”, and “Measurement”.

How to work with Infineon CLM templates

“802.11n, ac 5 GHz, 40 MHz” worksheet

This worksheet consists of the inputs required for the 802.11n/ac standards for 5-GHz channels utilizing the 40-MHz bandwidth. This worksheet includes inputs for 1Tx through 4Tx implementations. Only the required inputs cells will be shown if using the macro-enabled version of the template and answering all questions on the “Project Information” worksheet. The required inputs on this worksheet are “Target Power”, and “Measurement”.

“80MHz” worksheet

This worksheet consists of the inputs required for the 802.11ac standard for 5-GHz channels utilizing the 80-MHz bandwidth. This worksheet includes inputs for 1Tx through 4Tx implementations. Only the required inputs cells will be shown if using the macro-enabled version of the template and answering all questions on the “Project Information” worksheet. The required inputs on this worksheet are “Target Power”, and “Measurement”.

7.2 Step-by-step template walkthrough

This section shows screenshots of the “FCC Canada Taiwan Regulatory Power Template Rev 3.0.xlsm” to walk you through an example of a completed template.

7.2.1 Enabling macros on your template

Enabling macros in the Infineon template can be done by selecting “Enable Content” when the Excel workbook is first opened. The Infineon macro-enabled template is saved as a “.xlsm” extension. The flat file version of the template is saved as “.xls”.

Macros contained in Infineon templates are local to the spreadsheet and do not store or collect any information. They are provided to ease in identification of the required input cells to improve the efficiency of the regulatory process with Infineon.

Figure 9 shows a screen shot of the Security Warning that may appear when first opening the Infineon template. Click **Enable Content** to enable macros.

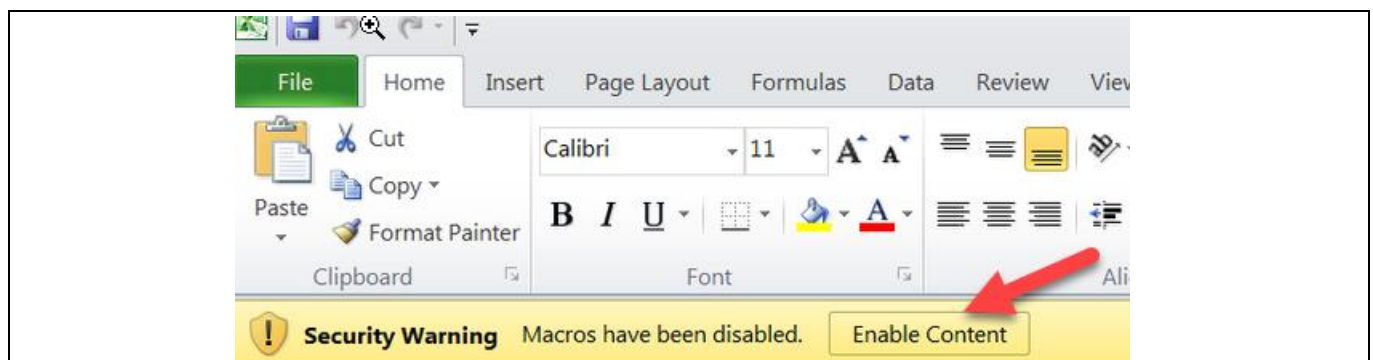


Figure 9 Enabling macros in the Infineon template

How to work with Infineon CLM templates

7.2.2 Project information worksheet

After enabling macros, the first step to complete the template is to complete the “Project Information” worksheet. Figure 10 shows the “Project Information” worksheet of the template.

#	Question	Response	Comments
1	End customer name		The end customer name is the actual end customer selling the final product, not the CM build the product. General products produced by Cypress partners may use the partner name for this field.
2	Project name		This is the name of the final end project. This is important as it is used for retrieving prior releases if future changes are required.
3	What Cypress chip set is this for?		Please indicate the "CYW" device used for this project.
4	Is this project AP or Client device?		AP devices may have slightly different requirements compared to client-only devices.
5	Does this project support 2.4GHz only, 5GHz only, or both?		Indicated which frequency bands your project supports. This will indicate what additional information is required to be completed on this page and subsequent tabs in this template.
6	Does this project support 1Tx, 2Tx, 3Tx, or 4Tx operation in 2.4GHz?		Confirm how many antennas (radiators) are used for the 2.4GHz spectrum in the design. This selection will determine additional information required to be completed in the subsequent tabs.
7	Does this project support 1Tx, 2Tx, 3Tx, or 4Tx operation for 5GHz?		Confirm how many antennas (radiators) are used for the 5GHz spectrum in the design. This selection will determine additional information required to be completed in the subsequent tabs.
8	Does this project support 20MHz, 40MHz, and/or 80MHz		
9	Does your product support and has been tested for 802.11b MIMO modes (2Tx or 3Tx)		
10	Does it support TXBF (TX Beamforming)?		NOTE: TxBF should be lower in power than CDD rates.
11	Does the product support TPC (Transmit Power Control)?		NOTE: To support TXBF, a minimum of 2TX operation is required.
12	Does the product supports VHTTXBF rates?		TPC = Transmit Power Control
13	Does the product supports VHT10-11 rates?		VHTTXBF = Very High Throughput TX Beamforming. If your device does not support a minimum of 2TX operation with 802.11ac, the answer to this question should be 'No'.
14	Does the product support weather band (TDWR)?		VHT10-11 rates are proprietary rates used on with Broadcom or Cypress WiFi devices. If you are not aware of these rates, answer no.
15	Do you need any special Tag for this code Request?		TDWR is typically used for channels 120-128 in 5GHz (Frequencies 5.60GHz - 5.65GHz)
16	Confirm that Channel 12 and Channel 13 in 2.4GHz are disabled for US/FCC?		Please indicate any tags that you would like to include in the release. Tags may be used to identify the release.
17	Is this request to update or modify an existing code? If Yes please provide your previously released ccode/regrev (e.g. "US/125") in the Comments Column.		Note: If Channel 12 and/or 13 is requested to be enabled, it is the responsibility of the manufacturer to implement the appropriate geo-location mechanism in their application code, and get approval for each SKU from the FCC. This additional work and FCC approvals can be avoided by disabling Channel 12 and 13 for SKUs shipping to the US.

Figure 10 “Project information” worksheet questions

The Project Information worksheet for the US/Canada/Taiwan template consists of 17 questions, each is described as follows.

Note: Answers to some questions will remove (black out) other questions in the Project Information worksheet.

- End customer name:** This is the name of the end customer completing the final assembled product. This information is used by Infineon to store individual Per Product CLM releases, and therefore is necessary to be completed and accurate. If the requestor is an Infineon module partner and the end product is a wireless module, then the Partner name may be used as the end customer.
- Project name:** This is the name of the final end product using an Infineon silicon device or partner module. This information is used by Infineon to store individual Per Product CLM releases, and therefore is necessary to be completed and accurate.

Note: The Customer Name and Project name must be restricted to a maximum of 20 characters.

- What Infineon chip set is this for?:** Fill in the CYW* part number that is being used in this project. Note that CLM releases are part number-specific, and this field is used to confirm that the final build of your CLM blob is targeted at the correct wireless chip set from Infineon.
- Is this project AP or client device?:** This question provides helpful information for review of the requested regulatory configuration. Access Points may have additional regulatory requirements compared to STAs (Station – Client device), depending on the country you are targeting. This field is a drop-down box, and allows for one of three options to be selected:
 - AP
 - Client
 - AP + Client

How to work with Infineon CLM templates

5. **Does this project support 2.4 GHz only, 5 GHz only, or both?:** This question is answered through a drop-down box. One of three answers are allowed:

- a) 2.4 GHz
- b) 5 GHz
- c) 2.4 GHz + 5 GHz

This question is specific to the 2.4 GHz band. The answer to this question will dynamically hide non-applicable input fields in the Infineon template. Answering this question correctly will limit the required inputs to only those that are needed.

6. **Does this project support 1Tx, 2Tx, 3Tx, or 4Tx operation in 2.4 GHz?:** The answer to this question provides the 2.4-GHz configuration/implementation of your wireless system. A drop-down box selection is provided that allows one of four options:

- a) 1Tx – One Antenna
- b) 2Tx – Two Antennas
- c) 3Tx – Three Antennas
- d) 4Tx – Four Antennas

This question is specific to the 2.4-GHz band. The answer to this question will dynamically hide non-applicable input fields in the Infineon template. Answering this question correctly will limit the required inputs to only those that are needed.

7. **Does this project support 1Tx, 2Tx, 3Tx, or 4Tx operation for 5 GHz?:** The answer to this question provides the 5-GHz configuration/implementation of your wireless system. A drop-down box selection is provided that allows one of four options:

- a) 1Tx – One Antenna
- b) 2Tx – Two Antennas
- c) 3Tx – Three Antennas
- d) 4Tx – Four Antennas

This question is specific to the 5-GHz band. The answer to this question will dynamically hide non-applicable input fields in the Infineon template. Answering this question correctly will limit the required inputs to only those that are needed.

8. **Does this project support 20 MHz, 40 MHz, and/or 80 MHz?:** This question is requesting you to provide the bandwidths that you plan to support in your product. A drop-down box is provided that allows one of three options:

- b) 20 MHz Only
- c) 20 MHz + 40 MHz
- d) 20 MHz + 40 MHz + 80 MHz

Note: That it is not a valid request to support upper bandwidths without the lower bandwidths supported (for example, only supporting 80 MHz or supporting 40/80 MHz without 20 MHz supported).

The answer to this question will dynamically hide non-applicable input fields in the Infineon template. Answering this question correctly will limit the required inputs to only those that are needed.

How to work with Infineon CLM templates

9. **Does your product support and has been tested for 802.11b MIMO modes (2Tx, 3Tx or 4Tx)?:** This question asks whether your product supports 802.11b MIMO configurations. To support this feature, you must have a hardware configuration of at least two antennas. Typically, this option is not requested, but is available if desired. The answer to this question will dynamically hide non-applicable input fields in the Infineon template. Answering this question correctly will limit the required inputs to only those that are needed.
10. **Does it support TXBF (TX beamforming)?:** To support TX Beamforming, your hardware implementation must support at a minimum two antennas (2Tx). If you have a SISO (1Tx) implementation, the answer to this question is "No". The answer to this question will dynamically hide non-applicable input fields in the Infineon template. Answering this question correctly will limit the required inputs to only those that are needed.
11. **Does the product support TPC (Transmit power control)?:** TPC stands for Transmit Power Control and is a method by which APs can set STAs (clients) to operate at a lower power than their current output power. TPC is applicable in certain countries and applies to the same channels that must adhere to DFS rules (channels 52-64 and channels 100-144).
12. **Does the product support VHTTXBF rates?:** VHTTXBF stands for Very High Throughput TX Beamforming. This question is asking whether your project supports TX Beamforming for 802.11ac (VHT). If you do not support 802.11ac rates and your hardware configuration is not a minimum of 2Tx, then the answer to this question is "No".
13. **Does the product support VHT10-11 rates?:** VHT10-11 rates are proprietary Infineon and Broadcom rates that are not part of the IEEE standard. If you are not familiar with these rates, the answer to this question should be "No". If you are aware of these data rates and have designed your system to enable them, then answer "Yes" to this question
14. **Does the product support weather band (TDWR)?:** The weather band includes 5-GHz channels 120, 124, and 128. Some countries restrict the use of these channels completely (for example, Canada). If your product does intend to support the TDWR, answer "Yes", and if not, then an answer of "No" will remove weather band channels in the required inputs of the template.
15. **Do you need any special tag for this CCode request?:** A Tag is an identifier that Infineon can mark on your specific country code request. This can be used for additional verification if subsequent modifications or changes to your project CLM configuration is needed. This cell is a free-form input. Common Tags used include Board Name, SKU number, etc. The project name inputted in Question 2 will already be applied to your request, so this is provided is additional differentiation is needed.
16. **Confirm that Channel 12 and Channel 13 in 2.4 GHz are disabled for US/FCC?:** This question is specific to the US/Canada/Taiwan template, and is explicitly asked to ensure that Channel 12 and 13 requirements and/or restrictions in the US are known. If you are not aware of the regulations related to these channels, you should answer this question as "Yes (Disabled)" to disable these channels in these countries. If you have already planned a geo-location method and plan to get approval from the FCC on said method, then you can answer "No (Enabled)". Before requesting these channels to be enabled, ensure that you understand the requirements needed for these channels. The answer to this question will dynamically hide non-applicable input fields in the Infineon template. Answering this question correctly will limit the required inputs to only those that are needed.
17. **Is this request to update or modify an existing code?:** If "Yes", provide your previously released ccode/regrev (for example, "US/125") in the Comments column. This question is confirming if this is a modification request to a previously released CLM configuration. If a Per Product release has been made, then your regulator revision (regrev) will only be "/0". If you received legacy CLM configurations in the past, you may have a non-0 regrev. In any case, if the request is to modify a previously release, specify the country code you would like modified in the "Comments" column.

How to work with Infineon CLM templates

Details are provided within the Infineon template on what the question is meant to confirm. Most of these questions above will be common in other templates (for example, Europe, Japan, etc.), but some templates may add or remove questions (for example, Question 14 above only applies to the US/Canada/Taiwan template). Descriptions for each question contained in a specific template are included in the “Comments” column of the Project Information worksheet.

7.2.3 Input worksheets

Infineon templates include seven input worksheets. The number of worksheets that you are required to provide inputs in and the number of inputs on each worksheet depends on your specific configuration. The Infineon template will adjust and remove input cells that are not applicable based on the answers provided in the “Project Information” worksheet.

While completing the Project Information worksheet, each individual Power Target Input tab will change the tab color. A tab color of Red indicates that the worksheet is not required for your configuration. A tab color of Green indicates that your configuration requires inputs to complete the Country Code request process.

For example, if you were to answer “20 MHz Only” to question 8 above, the tab colors will appear as shown below:

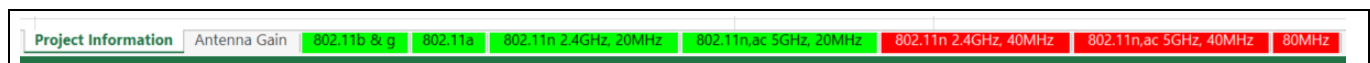


Figure 11 Required input worksheet color example

In this example, only 20-MHz bandwidth worksheets are required to be reviewed and provide inputs, and you needn’t worry about the 40-MHz and 80-MHz bandwidth configurations.

7.2.3.1 802.11b & g worksheet

The “802.11b & g” worksheet is required for 802.11b and 802.11g standards in the 2.4-GHz band. A screenshot of this worksheet is in [Figure 12](#). Depending on the answers provided in the [Project information worksheet](#), this worksheet may be smaller than as shown in the following figure.

802.11 Reference Standard	Channels	Band	Rate	Target Power	Measure	Active/Passive?	Power Limiting Factors
802.11b	1	2.4	DSSS			Active	Band edge restricted.
802.11b	2	2.4	DSSS			Active	Band edge restricted.
802.11b	3	2.4	DSSS			Active	Spurious emissions and harmonics restricted.
802.11b	4	2.4	DSSS			Active	Spurious emissions and harmonics restricted.
802.11b	5	2.4	DSSS			Active	Spurious emissions and harmonics restricted.
802.11b	6	2.4	DSSS			Active	Spurious emissions and harmonics restricted.
802.11b	7	2.4	DSSS			Active	Spurious emissions and harmonics restricted.
802.11b	8	2.4	DSSS			Active	Spurious emissions and harmonics restricted.
802.11b	9	2.4	DSSS			Active	Spurious emissions and harmonics restricted.
802.11b	10	2.4	DSSS			Active	Band edge restricted.
802.11b	11	2.4	DSSS			Active	Band edge restricted.
802.11b	12	2.4	DSSS			Active	Band edge restricted. Refer to Note 2.
802.11b	13	2.4	DSSS			Active	Band edge restricted. Refer to Note 2.
802.11g	1	2.4	OFDM			Active	Band edge restricted.
802.11g	2	2.4	OFDM			Active	Band edge restricted.
802.11g	3	2.4	OFDM			Active	Spurious emissions and harmonics restricted.
802.11g	4	2.4	OFDM			Active	Spurious emissions and harmonics restricted.
802.11g	5	2.4	OFDM			Active	Spurious emissions and harmonics restricted.
802.11g	6	2.4	OFDM			Active	Spurious emissions and harmonics restricted.
802.11g	7	2.4	OFDM			Active	Spurious emissions and harmonics restricted.
802.11g	8	2.4	OFDM			Active	Spurious emissions and harmonics restricted.
802.11g	9	2.4	OFDM			Active	Spurious emissions and harmonics restricted.
802.11g	10	2.4	OFDM			Active	Band edge restricted.
802.11g	11	2.4	OFDM			Active	Band edge restricted.
802.11g	12	2.4	OFDM			Active	Band edge restricted. Refer to Note 2.
802.11g	13	2.4	OFDM			Active	Band edge restricted. Refer to Note 2.

Figure 12 “802.11b & g” input worksheet

How to work with Infineon CLM templates

The columns shown in the red box in [Figure 12](#) are the only columns that required your inputs. All supported channels are shown in each of the worksheets, so there is no need to add or remove rows. Comments on the three input columns required are as follows:

1. **Target power:** As mentioned earlier, this is the value (to the nearest 0.25 dBm) that you would like the Infineon solution to operate at. This is a not-to-exceed value, meaning the Infineon Wi-Fi device will never exceed the value that you specify. This value is collected before or during regulatory testing and indicates the output power level where your product successfully passes regulatory requirements.
2. **Measure:** The “Measure” input cells is a drop-down selection with two options: “Conducted” or “EIRP”. No other values are valid for this column. The operation of Conducted or EIRP measurement methods as it pertains to the Infineon solution is detailed in [Section 7.1.2.1](#).

Note: Although it is possible to select different measurement methods for each channel, it is recommended that this field be consistent for a particular country code request (that is, all entries being Conducted; or all entries being EIRP).

3. **Active/Restricted?:** The operation of Active or Restricted channels is detailed in [Section 7.1.2.1](#). This column provides a drop-down selection with two options: “Active” or “Restricted”. As mentioned previously, if the country code requested has known regulatory requirements on channel availability, then only “Active” channel operation should be selected. Restricted operation is provided for pseudo-country codes, where a collection of countries is combined to support a larger region, each having different channel restrictions.

Note: The selection of Active/Restricted is inherited for all other 2.4-GHz channels in 20-MHz and 40-MHz bandwidths.

The remaining columns of the “802.11b & g” spreadsheet consist of the following information:

1. **802.11 reference standard:** This column indicates the 802.11 standard that applies to each row of the spreadsheet.
2. **Channels:** This column lists all available channels for the standard. If a channel is not desired to be enabled, simply input “NS” in the Target Power column.
3. **Band:** This column indicates the band (2.4 GHz or 5 GHz) that each row corresponds to.
4. **Rate:** This column indicates the modulation rate that is used for each row. DSSS = 802.11b; OFDM = 802.11g.
6. **Power limiting factors:** This column provides reference to notes on this worksheet. It also indicates whether a channel has specific band-edge, harmonics, or emissions restrictions that the customer should be aware of. These restrictions and tests will typically result in a lower output power required to pass compliance testing.

How to work with Infineon CLM templates

7.2.3.2 802.11a worksheet

This input worksheet is required for 802.11a operation. The worksheet contains all available 5-GHz, 20-MHz bandwidth channels. The 802.11a worksheets is the same in structure and format as the [802.11b & g worksheet](#). A screenshot of the 802.11a worksheet is shown in [Figure 13](#).

802.11 Reference Standard	Channels	Band	Rate	Target Power	Measure	Active/Passive?	Power Limiting Factors
802.11a	36	5	OFDM			Active	Bandedge restricted
802.11a	40	5	OFDM			Active	PPSD restricted
802.11a	44	5	OFDM			Active	PPSD restricted
802.11a	48	5	OFDM			Active	PPSD restricted
802.11a	52	5	OFDM			Passive (DFS)	PPSD restricted
802.11a	56	5	OFDM			Passive (DFS)	PPSD restricted
802.11a	60	5	OFDM			Passive (DFS)	PPSD restricted
802.11a	64	5	OFDM			Passive (DFS)	Bandedge restricted
802.11a	100	5	OFDM			Passive (DFS)	Bandedge restricted
802.11a	104	5	OFDM			Passive (DFS)	PPSD restricted
802.11a	108	5	OFDM			Passive (DFS)	PPSD restricted
802.11a	112	5	OFDM			Passive (DFS)	PPSD restricted
802.11a	116	5	OFDM			Passive (DFS)	PPSD restricted
802.11a	120	5	OFDM			Passive (DFS)	PPSD restricted DFS restrictions for Master devices CH120-128 in Canada. Refer to Note 2.
802.11a	124	5	OFDM			Passive (DFS)	PPSD restricted DFS restrictions for Master devices CH120-128 in Canada. Refer to Note 2.
802.11a	128	5	OFDM			Passive (DFS)	PPSD restricted DFS restrictions for Master devices CH120-128 in Canada. Refer to Note 2.
802.11a	132	5	OFDM			Passive (DFS)	PPSD and Power restricted
802.11a	136	5	OFDM			Passive (DFS)	PPSD and Power restricted
802.11a	140	5	OFDM			Passive (DFS)	PPSD and Power restricted
802.11a	144	5	OFDM			Passive (DFS)	Spurious emissions and harmonics restricted.
802.11a	149	5	OFDM			Active	Spurious emissions and harmonics restricted.
802.11a	153	5	OFDM			Active	Spurious emissions and harmonics restricted.
802.11a	157	5	OFDM			Active	Spurious emissions and harmonics restricted.
802.11a	161	5	OFDM			Active	Spurious emissions and harmonics restricted.
802.11a	165	5	OFDM			Active	Spurious emissions and harmonics restricted.

Figure 13 “802.11a” input worksheet

The required input columns are highlighted in the red box in [Figure 13](#). The only differences in the “802.11a” worksheet and the [802.11b & g worksheet](#) are summarized as follows:

- Channels:** Lists all channels available in the 5-GHz spectrum
- Rate:** OFDM in 5 GHz = 802.11a
- Active/Restricted?:** Note that only the non-DFS channels allow for the option to make them Active or Restricted. Channels 52-64 and channels 100-144 are operated as Restricted due to DFS restrictions required in these countries. This requirement is consistent world-wide.

All other columns have the same definition as in the [802.11b & g worksheet](#).

7.2.3.3 802.11n 2.4 GHz, 20 MHz worksheet

This input worksheet is required for 802.11n operation for 20-MHz bandwidth channels. The worksheet contains the following sections (each required depending on your inputs in the [Project information worksheet](#)):

- 1Tx operating mode
- 2Tx non-TXBF operating mode, including 802.11b MIMO
- 2Tx TXBF operating modes

Note: In the 2TX TXBF input section, channels 1, 11, 12, and 13 do not require inputs, as they will use the non-TXBF power targets.

4. 3Tx non-TXBF operating mode
5. 3Tx TXBF operating modes

Note: There are two input cells per channel to allow for output power control of TXBF and $N_{ss} = 2$. N_{ss} = Number of Spatial Streams. Details on the specific rates for each are shown in the Rate column.

How to work with Infineon CLM templates

6. 4Tx non-TXBF operating mode
7. 4Tx TXBF operating modes

Note: There are three input cells per channel to allow for output power control of TXBF $N_{ss} = 2$, and $N_{ss} = 3$.

The number of inputs required on this worksheet may decrease depending on the answers provided in the [Project information worksheet](#). The required input columns in the 802.11n 2.4-GHz, 20-MHz worksheet are shown below:

1. Target Power
2. Measure (Conducted or EIRP)

Note: The Active/Restricted column is not contained in this worksheet. This is because all 2.4-GHz channels will inherit the Active/Restricted configuration selected in the [802.11b & g worksheet](#). All other columns and definitions are consistent with the [802.11b & g worksheet](#).

7.2.3.4 802.11n, ac 5 GHz, 20 MHz worksheet

This input worksheet is required for 802.11n/ac operation for 5-GHz, 20-MHz bandwidth channels. The worksheet contains the following sections (each required depending on your project configuration):

1. 1Tx operating mode
2. 2Tx non-TXBF operating mode
3. 2Tx TXBF operating modes

Note: In the 2TX TXBF input section, channels 1, 11, 12, and 13 do not require inputs, as they will use the non-TXBF power targets.

4. 3Tx non-TXBF operating mode
5. 3Tx TXBF operating modes

Note: There are two input cells per channel to allow output power control for TXBF and $N_{ss} = 2$. N_{ss} = Number of Spatial Streams. Details on the specific rates for each are shown in the Rate column.

6. 4Tx non-TXBF operating mode
7. 4Tx TXBF operating modes

Note: There are three input cells per channel to allow output power control for TXBF $N_{ss} = 2$, and $N_{ss} = 3$.

The number of inputs required on this worksheet may decrease depending on the answers provided in the [Project information worksheet](#). The required input columns in the 802.11n, ac 5-GHz, 20-MHz worksheet are the following items:

1. Target Power
2. Measure (Conducted or EIRP)

Note: The Active/Restricted column is not contained in this worksheet. This is because all channels will inherit the Active/Restricted configuration selected in the [802.11a worksheet](#). All other columns and definitions are consistent with the [802.11a worksheet](#).

How to work with Infineon CLM templates

7.2.3.5 802.11n 2.4 GHz, 40 MHz worksheet

This input worksheet is required for 802.11n operation for 2.4-GHz, 40-MHz bandwidth channels. The worksheet contains the following sections (each required depending on your inputs in the [Project information worksheet](#)):

1. 1Tx operating mode
2. 2Tx non-TXBF operating mode
3. 2Tx TXBF operating modes

Note: In the 2TX TXBF input section, channels 1, 11, 12, and 13 do not require inputs, as they will use the non-TXBF power targets.

4. 3Tx non-TXBF operating mode
5. 3Tx TXBF operating modes

Note: There are two input cells per channel to allow output power control for TXBF and $N_{ss} = 2$. N_{ss} = Number of Spatial Streams. Details on the specific rates for each are shown in the Rate column.

7. 4Tx non-TXBF operating mode
8. 4Tx TXBF operating modes

Note: There are three input cells per channel to allow output power control for TXBF $N_{ss} = 2$, and $N_{ss} = 3$.

The number of inputs required on this worksheet may decrease depending on the answers provided in the Project Information worksheet.

The required input columns in the 802.11n, ac 2.4 GHz, 40 MHz worksheet are the following items:

1. Target Power
2. Measure (Conducted or EIRP)

Note: The Active/Restricted column is not contained in this worksheet. This is because all 2.4 GHz channels will inherit the Active/Restricted configuration selected in the [802.11b & g worksheet](#).

Note: “Notes/Comments” is one additional column contained in the 2.4-GHz, 40-MHz worksheet. The Notes/Comments column is not an input column but provides information and/or restrictions to the requestor depending on the 2.4-GHz, 20-MHz channel configuration in prior worksheets. If a restriction is determined in the 40-MHz worksheet, then it will highlight the restriction in red text, and will black out the Target Power and Measure column for that channel.

All other columns and definitions are consistent with the [802.11b & g worksheet](#).

How to work with Infineon CLM templates

7.2.3.6 802.11n, ac 5 GHz, 40 MHz worksheet

This input worksheet is required for 802.11n/ac operation for 5-GHz, 40-MHz bandwidth channels. The worksheet contains the following sections (each required depending on your inputs in the [Project information worksheet](#)):

1. 1Tx operating mode
2. 2Tx non-TXBF operating mode
3. 2Tx TXBF operating modes

Note: In the 2TX TXBF input section, channels 1, 11, 12, and 13 do not require inputs, as they will use the non-TXBF power targets.

4. 3Tx non-TXBF operating mode
5. 3Tx TXBF operating modes

Note: There are two input cells per channel to allow output power control for TXBF and Nss = 2. Nss = Number of Spatial Streams. Details on the specific rates for each are shown in the Rate column.

6. 4Tx non-TXBF operating mode
7. 4Tx TXBF operating modes

Note: There are three input cells per channel to allow output power control for TXBF Nss = 2, and Nss = 3.

The number of inputs required on this worksheet may decrease depending on the answers provided in the [Project information worksheet](#).

The required input columns in the 802.11n, ac 2.4-GHz, 40-MHz worksheet are the following items:

1. Target Power
2. Measure (Conducted or EIRP)

Note: The Active/Restricted column is not contained in this worksheet. This is because all channels will inherit the Active/Restricted configuration selected in the [802.11a worksheet](#).

Note: "Notes/Comments" is one additional column contained in the 5-GHz, 40-MHz worksheet. The Notes/Comments column is not an input column but provides information and/or restrictions to the requestor depending on the 5-GHz, 20-MHz channel configuration in prior worksheets. If a restriction is determined in the 40-MHz worksheet, then it will highlight the restriction in red text, and will black out the Target Power and Measure column.

All other columns and definitions are consistent with the [802.11a worksheet](#).

How to work with Infineon CLM templates

7.2.3.7 80 MHz worksheet

This input worksheet is required for 802.11ac operation for 5-GHz, 80-MHz bandwidth channels. The worksheet contains the following sections (each required depending on your inputs in the [Project information worksheet](#)):

1. 1Tx operating mode
2. 2Tx non-TXBF operating mode
3. 2Tx TXBF operating modes

Note: In the 2TX TXBF input section, channels 1, 11, 12, and 13 do not require inputs, as they will use the non-TXBF power targets.

4. 3Tx non-TXBF operating mode
5. 3Tx TXBF operating modes

Note: There are two input cells per channel to allow output power control for TXBF and $N_{ss} = 2$. N_{ss} = Number of Spatial Streams. Details on the specific rates for each are shown in the Rate column.

6. 4Tx non-TXBF operating mode
7. 4Tx TXBF operating modes

Note: There are three input cells per channel to allow output power control for TXBF $N_{ss} = 2$, and $N_{ss} = 3$.

The number of inputs required on this worksheet may decrease depending on the answers provided in the [Project information worksheet](#).

The required input columns in the 80-MHz worksheet are the following items:

1. Target Power
2. Measure (Conducted or EIRP)

Note: The Active/Restricted column is not contained in this worksheet. This is because all channels will inherit the Active/Restricted configuration selected in the [802.11a worksheet](#).

Note: "Notes/Comments" is one additional column contained in the 80-MHz worksheet. The Notes/Comments column is not an input column but provides information and/or restrictions to the requestor depending on the 5-GHz, 20-MHz or 40-MHz channel configurations in prior worksheets. If a restriction is determined in the 80-MHz worksheet, then it will highlight the restriction in red text, and will black out the Target Power and Measure column.

All other columns and definitions are consistent with the [802.11a worksheet](#).

How to integrate CLM releases into your design

8 How to integrate CLM releases into your design

The process that occurs after you have completed and submitted a Power Target template to Infineon is as follows:

1. Infineon regulatory engineering will review the template you have provided. If there are any questions or inconsistencies, they will be clarified during the review process.
2. Once clarifications have been confirmed, Infineon regulatory engineering will create one of the following types of releases:
 - a) **A master CLM XML release:** These releases are reserved for legacy devices that have not been transitioned to Per Product releases.
 - b) **A per product CLM XML release:** This release is how the majority of CLM releases are handled today. The CLM release will be stored on Infineon servers according to the “Customer Name” and “Project Name” information provided.
3. The *CLM.xml* release above will be taken by engineering, who in turn will create one of the following releases for the requestor:
 - a) **“Built-In master” release:** This is a firmware image release that contains the specific country codes created/modified for the requestor in the CLM Master database. This type of release can contain non-0 regrevs. This firmware release will be completed only if the part number being used does not support Per Product CLM releases.
 - b) **“Built-In per product” release:** This is a firmware image release that contains the specific country codes created/modified for the requestor as a Per Product database specific to your request. This type of release will only include regrevs with values equal to zero. This type of release will only be used if the part number in question does not support the CLM Blob download method.
 - c) **“Downloadable blob” release:** This is the preferred release method, which includes only a *.clm_blob* file that contains the regulatory information and configurations that were requested by the customer. This method of release allows for compilation and integration of the CLM configuration to be completed on the customer side, instead of relying on Infineon to build the firmware image. This method of release is only completed with Per Product CLM releases.

When comparing the different types of CLM release methods, downloadable blob releases are the preferred method for Infineon and customers, because they improve on the other method in release cycle time, flexibility of integration (customer-side compilation), and standardization (use of only “/0” regrevs).

Built-in release methods (either Master or Per Product) are complete binary images that are released to a requestor and can only be programmed into the final device. Source code is not released as part of this release method.

Downloadable blob releases do require the customer to process the *.clm_blob* file on their end to integrate the CLM configuration into their product. The integration of the *.clm_blob* release is a straight-forward process, whether it be in an AIROC™ embedded project, or in use with the Infineon FMAC driver releases.

8.1 Integrating CLM blobs in AIROC™ platforms

To integrate a *clm_blob* with the AIROC™-based platform, copy and replace the blob file that ends with *.clm_blob* into the firmware directory located at *43xxx_Wi-Fi/resources/firmware/<chip_specific_directory>*.

Example for CYW4343W:

clm_blob: 43xxx_Wi-Fi/resources/firmware/4343W/4343WA1.clm_blob

How to integrate CLM releases into your design

8.2 Integrating CLM blobs with FMAC

To integrate the `clm_blob` with the FMAC-based platform, copy the blob file that ends with `.clm_blob` into the wireless driver's directory at `/lib/firmware/brcm` along with firmware and the `nvrnm.txt` file. Make sure that the firmware, `nvrnm.txt`, and `clm_blob` adhere to the following pattern while naming:

`clm_blob:` `brcmfmac<chip_name>-<interface>.clm_blob`

`Firmware:` `brcmfmac<chip_name>-<interface>.bin`

`Nvrnm.txt:` `brcmfmac<chip_name>-<interface>.txt`

Example for CYW43430:

`Clm_blob:` `brcmfmac43430-sdio.clm_blob` `firmware:` `brcmfmac43430-sdio.bin` `Nvrnm.txt:` `brcmfmac43430-sdio.txt`

8.3 Interaction of NVRAM and CLM parameters

All Infineon boards are provided with a NVRAM file. This file contains parameters to control RF performance, maximum output power, power amplifier and various other variables to match the board RF architecture. See the [AN214807](#) application note for complete details of NVRAM parameters. As explained in Section 4, a CLM release also contains output power related parameters based on the regulatory considerations.

There is an overlap among a few parameters in the CLM and NVRAM configuration files. If a parameter is present in both the files (NVRAM and CLM), the minimum value for that parameter will be considered. For example, if the output power target for Channel 1 is set to 20 in NVRAM but is set to 18 in the CLM file for the product, the value of 18 will be used as the operating condition.

Note: *Infineon highly recommends that customers do not change the default power parameters in NVRAM because modifications to these values may affect regulatory measurements.*

9 Infineon regulatory WL command references

Based on the regions and respective regulatory authorities, the regulatory rates can be broadly classified as follows:

For USA, CANADA, TAIWAN, SOUTH KOREA, JAPAN, EUROPE, MEXICO, BRAZIL, ECUADOR AND REST OF WORLD, the regulatory rates and the corresponding wl command parameters for each are as follows:

11b = 1 Mbps = wl **2g_rate -r 1**

11g = 6 Mbps = wl **2g_rate -r 6**

11n20M = MCS0 = wl **2g_rate -h 0 -b 20**

11n40M = MCS0 = wl **2g_rate -h 0 -b 40**

11a = 6 Mbps = wl **5g_rate -r 6**

11n20M = MCS0 = wl **5g_rate -h 0 -b 20**

11n40M = MCS0 = wl **5g_rate -h 0 -b 40**

VHT80 = MCS0, 1 stream = wl **5g_rate -v 0x1 -b 80**

9.1 Regulatory WL command sequence used for testing

During regulatory certification process, the parameters (Power Output, Spectral Density, Occupied Bandwidth, Tx conducted Spurious/Harmonic emissions, Tx Radiated Spurious/Harmonic emissions, and FCC/Taiwan/Canada Tx Radiated Bandedge emissions) that are measured are mentioned along with the WL commands based on the modulation scheme, TX chains, and the frequency band of operation.

To understand the usage of individual WL commands, see the [WL Command Document](#) available on the [Infineon Developer Community](#).

For transmit/receive testing, example scripts can be found in: <WICED-SDK>\43xxx_Wi-Fi\libraries\test\wl_tool\scripts. Additional examples for wl commands for different testing conditions are also provided in this document in Section 4.

Overview of regulatory CLM process

10 Overview of regulatory CLM process

The end goal of the Infineon regulatory process is to provide Infineon customers with optimized, compliant Wi-Fi solutions, delivered in a timely and efficient manner. To summarize the overall regulatory approval process flow that is documented in this application note, see the following figure:

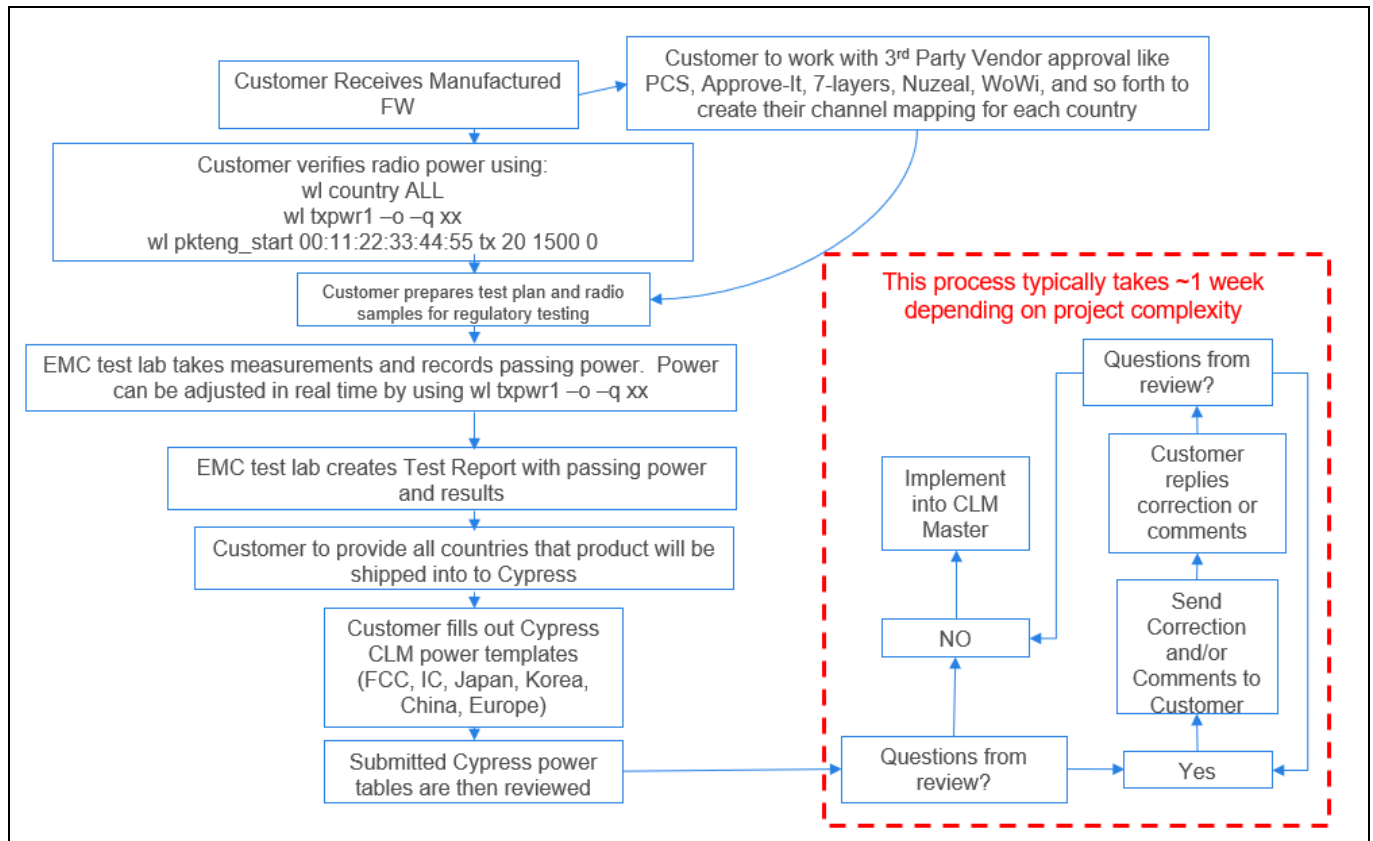


Figure 14 Regulatory CLM process flow

Revision history

Revision history

Document revision	Date	Description of changes
**	2018-11-16	New application note
*A	2022-06-10	Updated for 802.11ax and Wi-Fi 6E Migrated to Infineon template
*B	2023-03-24	Added information for Wi-Fi 6E Added Restriction changes to Japan and China test requirements. Added FCC Contigen based protocol testing for 6E.

Trademarks

All referenced product or service names and trademarks are the property of their respective owners.

Edition 2023-03-24

Published by

Infineon Technologies AG

81726 Munich, Germany

**© 2023 Infineon Technologies AG.
All Rights Reserved.**

Do you have a question about this document?

Email: erratum@infineon.com

Document reference

002-25347 Rev. *B

Important notice

The information contained in this application note is given as a hint for the implementation of the product only and shall in no event be regarded as a description or warranty of a certain functionality, condition or quality of the product. Before implementation of the product, the recipient of this application note must verify any function and other technical information given herein in the real application. Infineon Technologies hereby disclaims any and all warranties and liabilities of any kind (including without limitation warranties of non-infringement of intellectual property rights of any third party) with respect to any and all information given in this application note.

The data contained in this document is exclusively intended for technically trained staff. It is the responsibility of customer's technical departments to evaluate the suitability of the product for the intended application and the completeness of the product information given in this document with respect to such application.

Warnings

Due to technical requirements products may contain dangerous substances. For information on the types in question please contact your nearest Infineon Technologies office.

Except as otherwise explicitly approved by Infineon Technologies in a written document signed by authorized representatives of Infineon Technologies, Infineon Technologies' products may not be used in any applications where a failure of the product or any consequences of the use thereof can reasonably be expected to result in personal injury.